PROBLEMS FROM THE LITERATURE SETTLED BY TRANSFERRED METHODS: ROOK WALKS, HCM LAWS, HYPOCYCLOID DIVISION POINTS, MIXTURE LIKELIHOODS, POLYGAMMA WRONSKIANS AND COMPOSITE SEQUENCES

KEIO

Abstract. We take up twenty-five conjectures, questions and guesses stated in the literature by other authors and settle sixteen of them as posed (one conditionally on an earlier, not yet refereed manuscript of this series, and one by combining known results); for the remaining nine we prove partial results whose scope is stated precisely. The proofs transfer a small set of methods developed in earlier manuscripts of this series, sometimes in combination. Highlights: (i) for every d the numbers of diagonal rook paths in d dimensions satisfy a linear recurrence of order at most d with polynomial coefficients, produced by an explicit algorithm, which answers a challenge of Kauers and Zeilberger; we also prove their conjectured large-d asymptotics and the guessed recurrences for $d \le 7$ (Problem 17 of Chen and Kauers for $4 \le d \le 7$); (ii) the Gompertz critical-point conjecture of Polyanskiy and Wu is false, and mixtures of two analytic strongly log-concave densities can have arbitrarily many modes; (iii) Bosch's conjecture that $\sqrt{\gamma_t \gamma_s}$ is hyperbolically completely monotone fails exactly when $|t - s| > 1/2$, the condition $c \le a + b$ is necessary for $\Gamma_c \times \mathbf{B}_{a,b}$ to be HCM, as Bosch and Simon believed, and $\mathrm{HM} \cap \mathrm{GGC} \neq \mathrm{HCM}$; (iv) contrary to a speculation of Mani and Rubinstein-Salzedo, individual division points of the tricuspoid can be constructible when $n \nmid 6$ (the first case is $n = 32$), and we give the complete criterion; (v) several guesses and degree questions of Qi and coauthors on Wronskians of polygamma products are settled; (vi) assuming estimates from an earlier manuscript of this series, the binomial nonparametric MLE attains the minimax Wasserstein-p rate for every $p \ge 1$; (vii) every real biquadratic field contains infinitely many independent Pisot numbers all of whose integer parts of powers are composite and $\equiv 3, 4 \pmod 6$, and Šiurys's conjecture on composite k-step Fibonacci-like sequences holds for all $3 \le k \le 130$ and for a set of k of lower density at least $3/4$. The partial results also cover a conjecture and a problem of Faust et al. on Floquet isospectrality (the conjecture is proved for every period with a divisor in $[5, 69]$, and a structure theorem holds for all period lattices) and two questions of Erchenko, Jakobson and Tsypin on spectral volumes of weighted Laplacians. Each problem is quoted from its source and each outcome is labelled.

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1. Introduction

1.1. **What this paper does.** An earlier review of the manuscripts of this series singled out ten results, each of which answered a problem posed by other authors. Here we ask how far the *methods* behind those results reach. For each method we searched the literature for explicitly stated conjectures, questions and guesses, checked that they were still open as of October 2026, discarded those already treated in the series, and attacked the rest. This paper records what came out: complete solutions where we found them, counterexamples where the expectation was wrong, and partial results where neither was within reach.

The sections are independent. Each begins with the problem quoted from its source, followed by the main theorem and its proof. Labels and notation are local to each section.

1.2. **The transferred methods.** We refer to the ten methods by number; the earlier manuscripts contain the original applications.

- M1. *Singularity budgets for recurrences*: a recurrence of order r forces an ODE with at most r nonzero singular points, and critical values of a real Laurent polynomial on a torus produce genuine singularities (Franel's conjecture [1, 10]).
- M2. *Likelihood transport for mixtures*: optimality of the NPMLE, transport–entropy inequalities and Chebyshev–Bernstein bounds (binomial mixtures [6]).
- M3. *De Rham and $\mathcal{D}$-module counts* for minimal orders (banana maximal cuts [8]).
- M4. *Perturbative Floquet rigidity*: coordinate averaging, induction on dimension and eigenvalue perturbation [10].
- M5. *Spectral cones of weighted Laplacians*: norm obstructions, first-contact and double-contact arguments [3].
- M6. *Tower composition* for Ewald points [12].
- M7. *Local monodromy and irreducibility* (isotropic Fuchs squeeze) [11].
- M8. *Galois and transcendence classifications of division points and quantiles* [5].
- M9. *Laplace densities and complete monotonicity*: Bernstein's theorem turns complete monotonicity into the sign of a density, and a vanishing integral forces a sign change [9, Part VIII].
- M10. *Finite-field return and trace congruences* for composite values of recurrences [2].

Section 2 combines M1, M3 and M7; Section 3 combines M8 and M9; Section 4 combines M2 with Fourier windows and degenerate critical points; Section 5 combines M9 with a likelihood-ratio ordering; Section 6 uses M10 with Bang–Zsigmondy primitive divisors; Section 7 uses M4 with interval certification; Section 8 uses M5. Method M6 did not lead to any problem that we could treat.

1.3. **Outcomes.** Table 1 lists every problem we treat. *Closed* means that the statement asked in the source is proved or disproved as posed; *partial* means that only the stated part is proved. Some closures are easy once the problem is seen from the right angle, and we say so in the table and in the text.

Table 1: Problems treated and outcomes.

	Problem	Source	Outcome
1	Order-d recurrence for d-dimensional diagonal rook paths, every d	Kauers–Zeilberger [15, §2]	**Closed**: order $\leq d$ for all d, explicit algorithm (Thm. 2.8)
2	Large-d asymptotics of rook walks	Kauers–Zeilberger [15, §8]	**Closed** (Thm. 2.14)

	Problem	Source	Outcome
3	Prove the guessed recurrences for $d = 4, \dots, 12$	Chen–Kauers [16, Problem 17]	*Partial*: proved for $d \le 7$, computer-assisted exact arithmetic (Prop. 2.12)
4	Minimal order is exactly d	Kauers–Zeilberger (empirical)	*Partial*: $d \le 3$ (Prop. 2.18)
5	Constructible individual division points of the tricuspoid	Mani–Rubinstein-Salzedo [24, Rem. 5.7]	**Closed**: all three speculations fail; complete criterion (Thm. 3.3)
6	Division of the drawn lemniscate	[24, Rem. 1.4]	**Closed** by combining known results (Videla; Baek et al.; Cox–Shurman) (Thm. 3.7)
7	Is $\mathrm{HM} \cap \mathrm{GGC} = \mathrm{HCM}$?	Jedidi–Simon [29, Rem. 4.1(b)]	**Closed**: no; an elementary example (Thm. 3.12)
8	$\sqrt{\gamma_t \gamma_s}$ is HCM for all s, t; n-fold version	Bosch [30, Rem. 1]	**Closed**: false; HCM iff $\lvert t - s \rvert \le 1/2$; n-fold version fails for every $n \ge 2$ (Thm. 3.15, Prop. 3.16)
9	$c \le a + b$ is necessary for $\Gamma_c \times \mathbf{B}_{a,b}$ to be HCM	Bosch–Simon [32]	**Closed**: true (Thm. 3.18)
10	At most $O(a)$ critical points of Gompertz convolutions	Polyanskiy–Wu [42, §5.2]	**Closed**: false, $\Omega(a^2)$ (Thm. 4.8)
11	Is the $\Omega(k^2)$ mode count tight?	Polyanskiy–Wu [42, §5.5]	**Closed** as posed: unbounded already for $k = 2$ (Thm. 4.12)
12	Wasserstein-p guarantees for the binomial NPMLE	Vinayak–Kong–Valiant–Kakade [41, §5]	**Closed** conditionally on four estimates of [6] (Thm. 4.2)
13	Remark 17 guess and Ω-problem	Qi [54]	**Closed**; easy via a splitting identity (Thm. 5.9)
14	Remark 18 guess and Ω-problem	Qi [54]	**Closed** (guess); threshold: one half closed, the other characterized (Thm. 5.12)
15	Remark 4 guess and Λ-problem	Yin–Han–Qi [55]	**Closed** (guess); threshold: one half closed, the other characterized (Thm. 5.15)
16	Remark 2 sign guesses	Yin–Han–Qi [55]	**Closed**, elementary (Prop. 5.18)
17	CM degrees of H_2, H_3	Qi–Agarwal [56, §5.4]	**Closed** (Thm. 5.22)
18	CM degrees of the function (5.2)	Qi–Agarwal [56, §5.5]	*Partial*: closed for $\mu \ge 4$ and $\lambda = 0$; for $\lambda < 0$ non-integer in $(0, 1)$, values not in closed form (Thm. 5.23)
19	Sharp threshold and lower bound in (3.1)	Qi [59, Rem. 4.1]	*Partial* (Prop. 5.25)
20	Infinitely many biquadratic Pisot α with $\lfloor \alpha^n \rfloor \equiv 3, 4 \pmod 6$	Hahn–Ismailescu–Kim–Kim [66, §5]	**Closed**, in a stronger field-by-field form; the literal statement is easy (Thm. 6.2)
21	Composite k-step Fibonacci-like sequences for every $k > 2$	Šiurys [67]	*Partial*: all $3 \le k \le 130$, explicit residue classes, lower density $\ge 3/4$ (Thm. 6.12)

	Problem	Source	Outcome
22	Nonzero conjugate-reflection-symmetric potentials Floquet isospectral to 0 for every period $q > 3$	Faust et al. [78, Conj. 2]	*Partial*: every q that is even (known) or has a divisor in $[5, 69]$; exact for $q = 5, 9$, interval certificates for odd $q \le 69$; open e.g. for primes $q > 69$ (Cor. 7.8)
23	Classify the period lattices $\Gamma \subset \mathbb{Z}^d$ admitting nonzero potentials Floquet isospectral to 0	Faust et al. [78, Problem 1]	*Partial*: structure theorem for all lattices; rigidity for all $\Gamma \supseteq 2\mathbb{Z}^d$ or $3\mathbb{Z}^d$ and an infinite family of non-rectangular lattices; reduction to a core statement; planar lattices of index ≤ 12 (Thm. 7.13, Cor. 7.14, Prop. 7.17)
24	Does adding an edge increase the spectral volume $P(G)$?	Erchenko–Jakobson–Tsypin [89]	*Partial*: connected graphs on at most 6 vertices; two infinite families (Thms. 8.13, 8.8, 8.11)
25	Does suspension never decrease $P(G)$?	Erchenko–Jakobson–Tsypin [89]	*Partial*: graphs on at most 4 vertices; $P(G \vee K_1) \ge \gamma_n P(G)$ in general (Thms. 8.19, 8.17)

1.4. **Verification.** Every computation used in a proof is exact (rational or algebraic arithmetic, certified interval arithmetic, or finite certificates checked by an independent verifier); numerical experiments that only illustrate a statement are labelled as such. The scripts named in the text are collected in the supplementary archive that accompanies this paper; each prints the checks it performs and was run under `python3 -I`.

1.5. **Scope.** Several closures are modest, and we flag them: rows 13, 16 and 20 are easy once the right identity or example is found, row 6 combines known results, and row 12 depends on the binomial-mixture theorem of [6], which has not been refereed. The most substantial results are those in rows 1, 2, 5 and 8–12 of Table 1 and, among the partial results, rows 3 and 21–23. Statements about the literature are as of October 2026, and a missed recent preprint cannot be ruled out.

2. Recurrences of order at most d for diagonal rook paths in every dimension

2.1. **The problems and the status of this section.** For $d \ge 1$ let

$$R_d(x_1, \dots, x_d) = \frac{1}{1 - \sum_{i=1}^{d} \frac{x_i}{1-x_i}} = \sum_{m \in \mathbb{N}^d} \rho_d(m)\, x^m, \qquad a_n^{(d)} = \rho_d(n, n, \dots, n).$$

Here $\rho_d(m)$ is the number of ways a rook can move from the origin to m in $\mathbb{N}^d$ when every move increases exactly one coordinate by a positive integer; thus $a_n^{(2)}$ is OEIS A051708, $a_n^{(3)}$ is A144045, and the table $a_n^{(d)}$ is A181731 [21]. The sequence $(a_n^{(d)})_n$ is P-recursive by Lipshitz's theorem on diagonals [18]. Kauers and Zeilberger computed (by guessing) recurrences for $2 \le d \le 12$ and formulated the following challenges.

Problem 2.1 (Kauers–Zeilberger [15, §2])**.** "The recurrences for d-dimensional rook walks turn out (empirically) to have order d (for $2 \le d \le 12$). It is very possible that it won't be too hard to prove this sharp upper bound in general, or even a weaker—but still realistic—one. This may enable one to give a 'soft' proof that the empirically 'guessed' recurrences are indeed rigorously proved."

Problem 2.2 (Chen–Kauers [16, Problem 17])**.** "For $d = 4, \dots, 12$, prove recurrence equations for the diagonals of the rational series $1/(1 - \sum_{i=1}^{d} \frac{x_i}{1-x_i})$ conjectured in [Kauers–Zeilberger]." (The source adds: "For $d = 1, 2$, the problem is easy. For $d = 3$, it was solved in" [17].)

Problem 2.3 (Kauers–Zeilberger [15, §8])**.** "Let $w_n(d)$ be the number of ways a rook can positively walk from $[0^d]$ to $[n^d]$." "The asymptotic formulas for the individual n (for $1 \le n \le 4$) lead one to conjecture that the leading asymptotics for $w_n(d)$ as $d \to \infty$ is $e^{n-1}\frac{(nd)!}{n!^d}\big(1 + O(\frac{1}{d})\big)$." "We leave the rigorous proof of this as yet another challenge to the reader."

Note that $w_n(d) = a_n^{(d)}$.

Status. Theorem 2.8 proves, for *every* $d \ge 1$, that $a_n^{(d)}$ satisfies a nonzero linear recurrence of order at most d with polynomial coefficients, valid for all $n \ge 1$, and gives a finite algorithm producing it. This proves the upper bound asked for in Problem 2.1 and supplies rigorously proved recurrences of order at most d for every d. Together with an explicit degree bound (Lemma 2.11) it yields exactly the "soft proof" anticipated in Problem 2.1: Proposition 2.12 proves, for $2 \le d \le 7$, the unique recurrence of order d and of the coefficient degree reported in [15, §5] that fits the initial terms. This settles Problem 2.2 for $d = 4, 5, 6, 7$, with one caveat: we have not compared coefficient lists with the published files [23]; the identification with the recurrences of [15], which were fitted to initial terms, rests on this uniqueness. For $8 \le d \le 12$ Problem 2.2 remains open: rigorous recurrences of order at most d are proved to exist and are computable, and for every $d \le 12$ their coefficient degree, computed modulo a large prime, equals the degree reported in [15, §5] (Remark 2.10), but the specific recurrences of [15] are not proved here for $d \ge 8$. Theorem 2.14 proves the conjectured asymptotics of Problem 2.3 (an explicit second term is recorded, on numerical evidence only, in Remark 2.15). The *sharpness* of the bound (that no recurrence of order $< d$ exists) is proved here only for $d \le 3$ (Proposition 2.18); for $d \ge 4$ it remains open. We reduce it to a statement about local monodromy (Lemma 2.16, Conjecture 2.17), which holds for the differential operators found by guessing for $d \le 5$; this is evidence only, since these operators are not certified to be minimal.

2.2. **A Laguerre integral for the diagonal.** For $n \ge 1$ put

$$\ell_n(s) = \sum_{k=1}^{n} \binom{n-1}{k-1} \frac{s^k}{k!} = L_n^{(-1)}(-s), \tag{1}$$

where $L_n^{(\alpha)}$ is the Laguerre polynomial. (Integrals of products of Laguerre polynomials against e^{-s} have a classical combinatorial meaning, cf. Even–Gillis [22]; we only use the elementary identities below.)

Lemma 2.4. *For all $d \ge 1$ and $n \ge 1$,*

$$a_n^{(d)} = \sum_{k \in \{1,\dots,n\}^d} \frac{(k_1 + \cdots + k_d)!}{k_1! \cdots k_d!} \prod_{i=1}^{d} \binom{n-1}{k_i - 1} = \int_0^\infty e^{-s}\, \ell_n(s)^d \, ds.$$

Proof. Put $u_i = x_i/(1 - x_i)$. For $|x_i|$ small, $R_d = \sum_{K \ge 0} (u_1 + \cdots + u_d)^K$, and the multinomial theorem gives $R_d = \sum_{k \in \mathbb{N}^d} \frac{|k|!}{k_1! \cdots k_d!} \prod_i u_i^{k_i}$, an absolutely convergent expansion near 0. For $n \ge 1$ we have $[x^n]u^k = [x^{n-k}](1 - x)^{-k} = \binom{n-1}{k-1}$ if $1 \le k \le n$, and $[x^n]u^k = 0$ if $k = 0$ or $k > n$. Extracting the coefficient of $x_1^n \cdots x_d^n$ gives the finite sum. Finally insert $|k|! = \int_0^\infty e^{-s} s^{|k|} ds$ and factor the finite sum over k: $\sum_k \frac{|k|!}{\prod k_i!} \prod_i \binom{n-1}{k_i-1} = \int_0^\infty e^{-s} \prod_i \big(\sum_{k_i=1}^{n} \binom{n-1}{k_i-1} \frac{s^{k_i}}{k_i!}\big) ds$. □

Lemma 2.5. *For $n \ge 1$:*

(a) $\ell_n(0) = 0$, $\ell_n'(0) = 1$;

(b) $s\,\ell_n'' + s\,\ell_n' - n\,\ell_n = 0$;
(c) $(n+1)\ell_{n+1} = (n+s)\ell_n + s\ell_n'$ *and* $\ell_{n+1}' = \ell_n + \ell_n'$.

Proof. Write $\ell_n = \sum_k c_k s^k$, $c_k = \binom{n-1}{k-1}/k!$. (a) is clear. (b) The coefficient of s^k in $s\ell''+s\ell'-n\ell$ is $(k+1)kc_{k+1}+(k-n)c_k = \big(k\binom{n-1}{k} - (n-k)\binom{n-1}{k-1}\big)/k! = 0$. (c) For $1 \le k \le n$ the coefficient of s^k in $(n+s)\ell_n + s\ell_n'$ is $(n+k)c_k + c_{k-1} = \binom{n-1}{k-1}\frac{(n+k)(n-k+1)+k(k-1)}{(n-k+1)k!} = \binom{n-1}{k-1}\frac{n(n+1)}{(n-k+1)k!} = (n+1)\binom{n}{k-1}\frac{1}{k!}$, which is the coefficient of s^k in $(n+1)\ell_{n+1}$; for $k = n+1$ both sides have coefficient $1/n!$. The second identity is Pascal's rule for the coefficient of s^{k-1}: $\binom{n}{k-1} = \binom{n-1}{k-2} + \binom{n-1}{k-1}$. □

2.3. **Integration by parts and a reduction to d integrals.** Fix $d \ge 1$. For $n \ge 1$, $0 \le i \le d$ and integers $k \ge i - d$ put

$$\mathcal{J}_n(i,k) = \int_0^\infty e^{-s} s^k\, \ell_n(s)^{d-i} \ell_n'(s)^i\, ds.$$

The integral converges: by Lemma 2.5(a) the integrand is $O(s^{k+d-i})$ at 0, and $k+d-i \ge 0$. Note $\mathcal{J}_n(0,0) = a_n^{(d)}$.

Lemma 2.6 (integration by parts)**.** *For $n \ge 1$, $0 \le i \le d$ and $k \ge i - d + 1$,*

$$(1+i)\mathcal{J}_n(i,k) = k\,\mathcal{J}_n(i,k-1) + (d-i)\,\mathcal{J}_n(i+1,k) + i\,n\,\mathcal{J}_n(i-1,k-1), \tag{2}$$

where a term whose coefficient vanishes ($i = d$, resp. $i = 0$) is omitted.

Proof. Let $m_i = \ell^{d-i}\ell'^i$ ($\ell = \ell_n$) and $g = e^{-s}s^k m_i$. Since $k+d-i \ge 1$, $g(0) = 0$; also $g(s) \to 0$ as $s \to \infty$. Hence $\int_0^\infty g'\,ds = 0$. By Lemma 2.5(b), $\ell'' = n\ell/s - \ell'$, so $m_i' = (d-i)m_{i+1} + i\ell^{d-i}\ell'^{i-1}\ell'' = (d-i)m_{i+1} + \frac{in}{s}m_{i-1} - im_i$, and $g' = e^{-s}\big(-s^k m_i + ks^{k-1}m_i + (d-i)s^k m_{i+1} + ins^{k-1}m_{i-1} - is^k m_i\big)$. Each of the four integrals obtained by splitting converges (the exponents of s at 0 are $k+d-i$, $k-1+d-i$, $k+d-i-1$, $k+d-i$, all ≥ 0), which gives (2). □

Lemma 2.7 (reduction)**.** *There are polynomials $\gamma_b^{(i,k)} \in \mathbb{Q}[n]$, for $0 \le i \le d$, $k \ge 0$ and $1-d \le b \le 0$, depending only on d, i, k, b, such that for every integer $n \ge 1$*

$$\mathcal{J}_n(i,k) = \sum_{b=1-d}^{0} \gamma_b^{(i,k)}(n)\,\mathcal{J}_n(0,b).$$

Proof. All identities below hold for every integer $n \ge 1$, with coefficients that are polynomials in n independent of n; they are obtained from (2) by linear elimination in which we only divide by the nonzero integers $d-i+1$ and $d+1$. "Span" means $\mathbb{Q}[n]$-linear span.

Step 1 (lowering i). We claim: for $0 \le i \le d$ and $k \ge i-d$,

$$\begin{aligned} \mathcal{J}_n(i,k) &= \sum_{b=k-i}^{k} \lambda_b^{(i,k)}(n)\,\mathcal{J}_n(0,b), \\ \lambda_k^{(i,k)} &= \binom{d}{i}^{-1}, \qquad \lambda_{k-i}^{(i,k)} = (-1)^i \binom{k}{i}\binom{d}{i}^{-1} \end{aligned} \tag{3}$$

(for $i = 0$ both formulas read $\lambda_k^{(0,k)} = 1$). This is trivial for $i = 0$. For $i \ge 1$, relation (2) with $(i-1,k)$ in place of (i,k) is valid because $k \ge i-d = (i-1)-d+1$, and reads

$$(d-i+1)\mathcal{J}_n(i,k) = i\,\mathcal{J}_n(i-1,k) - k\,\mathcal{J}_n(i-1,k-1) - (i-1)n\,\mathcal{J}_n(i-2,k-1).$$

By induction the three terms on the right lie in the span of $\mathcal{J}_n(0,b)$ for $b \in [k-i+1,k]$, $[k-i,k-1]$ and $[k-i+1,k-1]$ respectively. The index $b = k$ occurs only in the first term and $b = k-i$

only in the second, which gives $\lambda_k^{(i,k)} = \frac{i}{d-i+1}\lambda_k^{(i-1,k)}$ and $\lambda_{k-i}^{(i,k)} = \frac{-k}{d-i+1}\lambda_{k-i}^{(i-1,k-1)}$, and hence the formulas in (3). In particular $\lambda_k^{(d,k)} = 1$.

Step 2 (a recurrence in k). Let $k \geq 1$. Relation (2) with $i = d$ (valid since $k \geq 1$) gives $(1+d)\mathcal{J}_n(d,k) = k\mathcal{J}_n(d,k-1) + dn\mathcal{J}_n(d-1,k-1)$. Insert (3): the left side is $(1+d)\mathcal{J}_n(0,k)$ plus a combination of $\mathcal{J}_n(0,b)$, $k-d \leq b \leq k-1$; on the right, $\mathcal{J}_n(d,k-1)$ involves $b \in [k-1-d, k-1]$, the coefficient of $\mathcal{J}_n(0,k-1-d)$ being $(-1)^d\binom{k-1}{d}$, and $\mathcal{J}_n(d-1,k-1)$ involves $b \in [k-d, k-1]$. Therefore

$$\mathcal{J}_n(0,k) = \sum_{b=k-d-1}^{k-1} \mu_b^{(k)}(n)\mathcal{J}_n(0,b), \qquad \mu_{k-d-1}^{(k)} = \frac{(-1)^d k}{d+1}\binom{k-1}{d}, \tag{4}$$

and $\mu_{k-d-1}^{(k)} = 0$ when $1 \leq k \leq d$.

Step 3. By induction on k: $\mathcal{J}_n(0,k)$ lies in the span B of $\mathcal{J}_n(0,1-d), \ldots, \mathcal{J}_n(0,0)$ for every $k \geq 1-d$. This is trivial for $k \leq 0$. For $1 \leq k \leq d$, (4) only involves indices b in $[k-d, k-1] \subset [1-d, k-1]$; for $k \geq d+1$ it involves indices b in $[k-d-1, k-1] \subset [0, k-1]$.

Step 4. Let $k \geq 0$. By (3), $\mathcal{J}_n(i,k)$ is a combination of $\mathcal{J}_n(0,b)$ with $b \geq k-i \geq -d$, and $b = -d$ can only occur for $(i,k) = (d,0)$, where its coefficient is $(-1)^d\binom{0}{d} = 0$. Hence only $b \geq 1-d$ occur, and Step 3 finishes the proof. □

2.4. **The upper bound.**

Theorem 2.8. *For every $d \geq 1$ there are polynomials $c_0, \ldots, c_d \in \mathbb{Z}[n]$, not all zero, such that*

$$\sum_{j=0}^{d} c_j(n)\, a_{n+j}^{(d)} = 0 \qquad \text{for all integers } n \geq 1.$$

In particular the minimal order of a recurrence with polynomial coefficients satisfied by $(a_n^{(d)})_{n\geq 0}$ is at most d. The polynomials c_j are produced by the finite procedure in the proof.

Proof. Iterating Lemma 2.5(c), for each $j \geq 0$ there are $\alpha_j, \beta_j \in \mathbb{Q}(n)[s]$, of degree at most j in s and with denominators dividing $(n+1)(n+2)\cdots(n+j)$, such that $\ell_{n+j} = \alpha_j\ell_n + \beta_j\ell'_n$ for all $n \geq 1$ (start with $\alpha_0 = 1, \beta_0 = 0$ and use $(\ell_{m+1}, \ell'_{m+1}) = (\frac{m+s}{m+1}\ell_m + \frac{s}{m+1}\ell'_m,\ \ell_m + \ell'_m)$ with $m = n, \ldots, n+j-1$). By Lemma 2.4, for $0 \leq j \leq d$ and $n \geq 1$,

$$a_{n+j}^{(d)} = \int_0^\infty e^{-s}(\alpha_j\ell_n + \beta_j\ell'_n)^d\, ds = \sum_{i=0}^{d}\binom{d}{i}\sum_{k\geq 0}[s^k]\big(\alpha_j^{d-i}\beta_j^i\big)\,\mathcal{J}_n(i,k),$$

a finite sum. By Lemma 2.7, $a_{n+j}^{(d)} = \sum_{b=1-d}^{0} V_{j,b}(n)\mathcal{J}_n(0,b)$ with $V_{j,b} \in \mathbb{Q}(n)$ whose denominators divide $\big((n+1)\cdots(n+d)\big)^d$; these identities hold for all $n \geq 1$. The $(d+1)\times d$ matrix $V = (V_{j,b})$ over the field $\mathbb{Q}(n)$ has a nonzero left kernel vector; multiplying it by a suitable nonzero element of $\mathbb{Q}(n)$ we get $c \in \mathbb{Z}[n]^{d+1}\backslash\{0\}$ with $\sum_j c_j V_{j,b} = 0$ in $\mathbb{Q}(n)$ for every b. Evaluating at an integer $n \geq 1$ (where all denominators of V are nonzero), $\sum_j c_j(n)a_{n+j}^{(d)} = \sum_b\big(\sum_j c_j(n)V_{j,b}(n)\big)\mathcal{J}_n(0,b) = 0$. □

Remark 2.9 (what the method is). Heuristically (this is not used anywhere), the count d is a de Rham cohomology count for the connection $e^{-s}\otimes\mathrm{Sym}^d(\text{Laguerre})$ on $\mathbb{G}_m$: the Laguerre connection of Lemma 2.5(b) is regular at $s = 0$ with a single unipotent Jordan block and has exponential factors $1, e^{-s}$ at ∞, so $e^{-s}\mathrm{Sym}^d$ has irregularity $d+1$ at ∞ and H^1 has dimension $d+1$; the integrals start at the boundary point $s = 0$, and Step 4 of the proof shows that the class of $\mathcal{J}_n(0,-d)$ never contributes. The shift $n \mapsto n+1$ acts on the remaining d-dimensional space, which bounds the order. This is the mechanism of the banana upper bound of [8] (method M3),

transported from a Laplace variable to the discrete variable n. The proof above is the elementary version of this count and does not use $\mathcal{D}$-module theory.

Remark 2.10 (computations). The script `rook_verify.py` checks Lemmas 2.4–2.6 exactly on many instances, and that for $2 \le d \le 7$ and $n = 1, \dots, 12$ the left kernel of $V(n)$ is one-dimensional, the coefficients $c_0(n), c_d(n)$ are nonzero (so the relation has order exactly d), and the relation annihilates the exact rook numbers. For $2 \le d \le 5$ the constructive relation is proportional, at $n = 1, \dots, 40$, to the minimal-degree order-d recurrence recomputed by guessing (degrees $1, 4, 9, 18$, as in [15, §5]). The script `rook_degrees_modp.py` computes the constructive recurrence modulo $p = 2^{61} - 1$ by evaluation and interpolation, without using the numbers $a_n^{(d)}$; for every $2 \le d \le 12$ its minimal coefficient degree is $1, 4, 9, 18, 31, 50, 75, 108, 149, 200, 261$, exactly the degrees listed in [15, §5]. For $8 \le d \le 12$ this is evidence, not proof, that the constructive recurrence *is* the one of [15]: if the recurrence of [15] is the only one of its order and degree fitting the initial terms used to guess it, and if the degree of the constructive recurrence over $\mathbb{Q}$ equals its degree modulo p, then the two agree up to a constant factor. For $d \le 7$ the identification is proved in Proposition 2.12 below.

2.5. **A soft proof of the guessed recurrences for $d \le 7$.** Kauers and Zeilberger hoped that an upper bound would yield a "soft" proof of the guessed recurrences. The proof of Theorem 2.8 does this once one bounds degrees. Let $P_j(n) = (n+1)(n+2)\cdots(n+j)$ ($P_0 = 1$), and let $V_{j,b} \in \mathbb{Q}(n)$ ($0 \le j \le d$, $1 - d \le b \le 0$) be the rational functions produced in that proof from the polynomials of Lemma 2.7, so that $a_{n+j}^{(d)} = \sum_b V_{j,b}(n)\mathcal{J}_n(0,b)$ for $n \ge 1$.

Lemma 2.11 (degree bound)**.** *Put $e = \lceil d/2 \rceil$. Then $P_j^d V_{j,b}$ is a polynomial in n of degree at most $\lfloor d/2 \rfloor + e\,dj$. Consequently, for every $K = (K_0, \dots, K_d) \in \mathbb{Q}[n]^{d+1}$ with $\deg K_j \le D$, each*

$$N_b(n) = P_d(n)^d \sum_{j=0}^{d} K_j(n) V_{j,b}(n) \qquad (1 - d \le b \le 0)$$

is a polynomial of degree at most $B_d(D) = D + \lfloor d/2 \rfloor + e\,d^2$.

Proof. We follow degrees in n through the proof of Lemma 2.7. In Step 1, $\lambda_b^{(i,k)}$ has degree $\le \lfloor i/2 \rfloor$, by induction on i: the recursion multiplies by n only in the term $(i-1)n\mathcal{J}_n(i-2,k-1)$, and $1 + \lfloor (i-2)/2 \rfloor = \lfloor i/2 \rfloor$. In Step 2 the coefficients come from $\mathcal{J}_n(d,k)$, $k\mathcal{J}_n(d,k-1)$ and $dn\mathcal{J}_n(d-1,k-1)$, so $\mu_b^{(k)}$ has degree $\le \max(\lfloor d/2 \rfloor, 1 + \lfloor (d-1)/2 \rfloor) = e$. By Step 3 and induction on k, $\mathcal{J}_n(0,k) = \sum_b \gamma_b^{(0,k)} \mathcal{J}_n(0,b)$ with $\deg \gamma_b^{(0,k)} \le ek$ for $k \ge 1$ (and $\gamma_b^{(0,k)}$ constant for $k \le 0$); with Step 4, $\deg \gamma_b^{(i,k)} \le \lfloor i/2 \rfloor + ek$ for $k \ge 0$. Next, by induction on j with Lemma 2.5(c), $P_j \ell_{n+j} = A_j \ell_n + B_j \ell'_n$ and $P_j \ell'_{n+j} = C_j \ell_n + D_j \ell'_n$ with $A_j, B_j, C_j, D_j \in \mathbb{Q}[n,s]$ of total degree $\le j$ (start with $A_0 = D_0 = 1$, $B_0 = C_0 = 0$; then $A_{j+1} = (n+j+s)A_j + sC_j$, $B_{j+1} = (n+j+s)B_j + sD_j$, $C_{j+1} = (n+j+1)(A_j + C_j)$, $D_{j+1} = (n+j+1)(B_j + D_j)$). Hence $P_j^d V_{j,b} = \sum_{i=0}^{d} \binom{d}{i} \sum_{k=0}^{dj} [s^k]\big(A_j^{d-i} B_j^i\big)\, \gamma_b^{(i,k)}$, where $[s^k](A_j^{d-i}B_j^i)$ has degree $\le dj - k$ in n; the total is $\le \max_{i,k}(dj - k + \lfloor i/2 \rfloor + ek) \le \lfloor d/2 \rfloor + e\,dj$ because $e \ge 1$ and $k \le dj$. Finally $N_b = \sum_j K_j\,(P_d/P_j)^d\, P_j^d V_{j,b}$ has degree at most $D + \max_j\big(d(d-j) + \lfloor d/2 \rfloor + e\,dj\big) = D + \lfloor d/2 \rfloor + e\,d^2$. $\square$

Proposition 2.12 (the guessed recurrences for $d \le 7$)**.** *Let $2 \le d \le 7$, $D_d = 1, 4, 9, 18, 31, 50$ and $N_d = 48, 63, 94, 159, 270, 455$ respectively. Up to a constant factor there is exactly one nonzero $K \in \mathbb{Q}[n]^{d+1}$ with $\deg K_j \le D_d$ and $\sum_{j=0}^{d} K_j(n) a_{n+j}^{(d)} = 0$ for $1 \le n \le N_d - d$, and this K satisfies $\sum_j K_j(n) a_{n+j}^{(d)} = 0$ for* all *$n \ge 0$. In particular, for $d = 4, 5, 6, 7$ the recurrences of order d and coefficient degree 9, 18, 31, 50 conjectured in* [15] *are proved, provided they are satisfied by the*

terms $a_1^{(d)},\dots,a_{N_d}^{(d)}$ (any recurrence of that order and degree satisfied by these terms equals K up to a constant factor; we have not compared coefficients with the published files [23]*).*

Proof. This is a computer-assisted proof (script `rook_softproof.py`); all arithmetic below is exact except the rank computation modulo a prime, which is only used as a lower bound for the rank over $\mathbb{Q}$. The homogeneous linear system (equations $1 \le n \le N_d - d$) for the coefficients of K has a one-dimensional solution space modulo a prime p; since a nonzero minor modulo p comes from a nonzero integer minor, the solution space over $\mathbb{Q}$ has dimension at most one. A nonzero solution K is obtained by Chinese remaindering and rational reconstruction (five primes for $d \le 5$, seven for $d = 6$, twelve for $d = 7$; the integer coefficients of the primitive K have at most $4, 18, 42, 104, 175, 317$ bits) and is checked to satisfy the equations for $0 \le n \le N_d - d$ exactly, so the dimension is one. By Lemma 2.11, each N_b is a polynomial of degree at most $B_d(D_d) = 6, 23, 43, 95, 142, 249$ for $d = 2,\dots,7$. The script evaluates the reduction of Lemma 2.7 and the shift of Theorem 2.8 at $n = n_0$ with exact rational arithmetic (evaluation commutes with these operations, which only divide by nonzero integers and by $P_j(n_0) \ne 0$) and checks $\sum_j K_j(n_0)V_{j,b}(n_0) = 0$ for $n_0 = 1,\dots,B_d(D_d)+1$ and all b. Hence $N_b \equiv 0$, and for every $n \ge 1$, $\sum_j K_j(n)a_{n+j}^{(d)} = \sum_b \big(\sum_j K_j(n)V_{j,b}(n)\big)\mathcal{J}_n(0,b) = 0$; the case $n = 0$ was checked directly above. □

Remark 2.13. The same procedure applies verbatim to any d and any candidate recurrence. For $8 \le d \le 12$ it needs the candidate (recomputed by guessing, or taken from the files [23]) and $B_d(D_d)+1 = 336, 518, 655, 932, 1132$ exact evaluations of the reduction; we did not carry this out for $d \ge 8$.

2.6. **The large-dimension asymptotics.**

Theorem 2.14. *For every fixed $n \ge 1$, $w_n(d) = a_n^{(d)} = e^{n-1}\dfrac{(nd)!}{n!^d}\big(1 + O(\frac{1}{d})\big)$ as $d \to \infty$.*

Proof. For $n = 1$, $a_1^{(d)} = d!$ exactly. Let $n \ge 2$. By (1), $\ell_n(s) = \frac{s^n}{n!}P(1/s)$ with $P(u) = \sum_{j=0}^{n-1}\binom{n-1}{j}\frac{n!}{(n-j)!}u^j$, so $P(0) = 1$, $P'(0) = n(n-1)$, P has nonnegative coefficients, and $1 \le P(u) \le (1+nu)^{n-1}$ for $u \ge 0$ (because $n!/(n-j)! \le n^j$). Put $N = nd$ and let S have the Gamma density $e^{-s}s^N/N!$ on $(0,\infty)$. Lemma 2.4 gives

$$\frac{n!^d}{(nd)!}a_n^{(d)} = \mathbb{E}\big[P(1/S)^d\big].$$

Bulk. On $[0,1]$ the function $\log P$ is smooth with $\log P(u) = n(n-1)u + r(u)$, $|r(u)| \le C_n u^2$. On the event $S \ge N/2$ (and $d \ge 4/n$, so $1/S \le 1$) we get $d\log P(1/S) = (n-1)N/S + \epsilon$ with $|\epsilon| \le 4C_n/(n^2 d)$. Write $S = N(1+X)$; then $\mathbb{E}X = 1/N$, $\mathbb{E}X^2 = (N+2)/N^2$, and on $X \ge -1/2$ the function $\phi(X) = \exp\big((n-1)(\frac{1}{1+X} - 1)\big)$ satisfies $|\phi(X) - 1 + (n-1)X| \le K_n X^2$. Since $\mathbb{P}(S < N/2) \le e^{-cN}$ and $\mathbb{E}[|X|; S < N/2] \le e^{-cN}$ for some $c > 0$ (Chernoff bound for the Gamma law), $\mathbb{E}[e^{(n-1)N/S}; S \ge N/2] = e^{n-1}\big(1 + O(1/N)\big)$, and therefore $\mathbb{E}[P(1/S)^d; S \ge N/2] = e^{n-1}\big(1 + O(1/d)\big)$.

Small S. Using $P(1/s)^d \le (1 + n/s)^{(n-1)d}$ and $N - (n-1)d = d$,

$$\begin{aligned}\mathbb{E}[P(1/S)^d; S < N/2] &\le \frac{1}{N!}\int_0^{N/2} e^{-s}s^d(s+n)^{(n-1)d}ds \\ &\le \frac{e^n}{N!}\int_n^{N/2+n} e^{-v}v^N dv \le e^n\,\mathbb{P}(S \le N/2 + n),\end{aligned}$$

which is $O(e^{-c'N})$. Adding the two parts proves the theorem. □

Remark 2.15 (second term; numerical only). Expanding one order further ($\log P(u) = n(n-1)u - n(n-1)^2u^2 + O(u^3)$ and $\mathrm{Var}(S/N) \sim 1/N$) suggests $a_n^{(d)} = e^{n-1}\frac{(nd)!}{n!^d}\big(1 - \frac{(n-1)^2}{2nd} + O(d^{-2})\big)$. Exact values for $n = 2, 3, 4$ and $d \le 80$ agree (`rook_verify.py`, part 6: $d(\mathrm{ratio} - 1) \to -\frac{1}{4}, -\frac{2}{3}, -\frac{9}{8}$). We did not write out the error analysis of this second term. Combinatorially, $a_n^{(d)} = \sum_w 2^{A(w)}$ over the arrangements w of the multiset $\{1^n, \dots, d^n\}$, where $A(w)$ counts adjacent equal letters (split each run into a composition); this explains the factor $e^{n-1} = \mathbb{E}[2^{\mathrm{Poisson}(n-1)}]$.

2.7. **Lower bounds: what is proved and what remains.** The following refinement of the singularity budget used in [1, 10] (Franel order, method M1) counts local monodromy with multiplicity.

Lemma 2.16 (index budget). *Let (a_n) satisfy a nonzero recurrence $\sum_{j=0}^{r} p_j(n)a_{n+j} = 0$ ($n \ge n_0$, $p_j \in \mathbb{C}[n]$), let $F(t) = \sum a_n t^n$ have positive radius of convergence, let V be the solution space (at a base point) of the minimal differential operator of F, and for $p \in \mathbb{C}^\times$ let h_p be the dimension of the space of local solutions at p (of that operator, continued to a punctured disc at p) that extend holomorphically to p. Then*

$$\sum_{p \in \mathbb{C}^\times} \big(\dim V - h_p\big) \le r, \qquad \text{and} \qquad \dim V - h_p \ge \mathrm{rk}(T_p - 1),$$

where T_p is the local monodromy at p.

Proof. As in the Franel argument of [1], with $\vartheta = t\,d/dt$ the operator $L_0 = \sum_j t^{r-j}p_j(\vartheta - j)$ satisfies $L_0F = R$ with R a polynomial, so $L = \partial_t^{1+\deg R}L_0$ annihilates F. Write $L = \sum_{k \le D'} a_k\partial_t^k$ with polynomial coefficients; its leading coefficient is $t^D H(t)$ with $\deg H \le r$ (the leading coefficient of L_0 is $t^D H(t)$, $H = \sum_j c_j t^{r-j}$, c_j the top coefficient of p_j, and left multiplication by ∂_t^e does not change it). Fix $p \in \mathbb{C}^\times$ and let $\mu_p = \mathrm{ord}_p H$. By Malgrange's index theorem [19], the index of L acting on convergent power series at p equals $D' - \mathrm{ord}_p a_{D'} = D' - \mu_p$ (the index on formal power series is $\max_k(k - \mathrm{ord}_p a_k)$; the difference is the irregularity, which we do not need); since the dimension of the kernel is at least the index, L has at least $D' - \mu_p$ linearly independent solutions holomorphic at p. The minimal operator divides L on the right, so near p its solution space V is contained in the D'-dimensional solution space of L; therefore $h_p \ge \dim V + (D' - \mu_p) - D'$, i.e. $\dim V - h_p \le \mu_p$. Summing over p gives $\sum_p(\dim V - h_p) \le \deg H \le r$. Holomorphic solutions are fixed by T_p, which gives the second inequality. □

For the rook diagonal $F_d(t) = \sum_n a_n^{(d)}t^n$, Lemma 2.16 reduces sharpness to local monodromy. Put $t_m = \big(\frac{m+1}{d+1-m}\big)^{d-2m}$ for $0 \le m < d/2$; these are the critical values of $\prod x_i$ on the polar divisor at the points with $d - m$ coordinates equal to $r_m = \frac{m+1}{d+1-m}$ and m equal to $1/r_m$ (Lagrange multipliers).

Conjecture 2.17 (sharpness; Kauers–Zeilberger's empirical statement). *For every $d \ge 1$, $a_n^{(d)}$ satisfies no recurrence of order $< d$. More precisely, the minimal operator of F_d has nontrivial local monodromy at each t_m ($0 \le m < d/2$) and $\mathrm{rk}(T_1 - 1) = \lfloor d/2 \rfloor$ at $t = 1$, so that Lemma 2.16 gives order $\ge \lceil d/2 \rceil + \lfloor d/2 \rfloor = d$; and the characteristic polynomial of the order-d recurrence is $(T-1)^{\lfloor d/2 \rfloor}\prod_{0 \le m < d/2}(T - t_m^{-1})$.*

Proposition 2.18. *Conjecture 2.17 (first sentence) holds for $d \le 3$. Hence for $d \le 3$ the minimal order is exactly d.*

Proof. A recurrence of order 0 is impossible since $a_n^{(d)} \ne 0$; this settles $d = 1$ ($a_n^{(1)} = 2^{n-1}$ for $n \ge 1$). By Lemma 2.16 it suffices, for $d = 2, 3$, to exhibit d distinct points of $\mathbb{C}^\times$ at which some continuation of F_d is not holomorphic.

$d=2$: $F_2=\frac12\big(1+\sqrt{(1-t)/(1-9t)}\big)$, which has branch points at $t=1/9$ and $t=1$. (This is classical: $F_2(t)$ is the sum of the residues of $R_2(x,t/x)/x=\frac{(1-x)(x-t)}{x(-2x^2+(1+3t)x-2t)}$ at its poles that tend to 0 with t, namely $\frac12$ at $x=0$ and $\frac12\sqrt{(1-t)/(1-9t)}$ at the small root of $2x^2-(1+3t)x+2t$.)

$d=3$: by [17] (van Hoeij's formula), $F_3(t)=1+\int_0^t g(w)\,dw$ with $g(w)=\frac{6\,G(z(w))}{(1-4w)(1-64w)}$, $G={}_2F_1(\frac13,\frac23;2;\cdot)$ and $z(w)=\frac{27w(2-3w)}{(1-4w)^3}$; one checks $1-z(w)=\frac{(1-w)^2(1-64w)}{(1-4w)^3}$. At $w=1/4$ every branch of g is analytic (the local solutions of the hypergeometric equation at $z=\infty$ are $z^{-1/3}$ and $z^{-2/3}$ times power series in $1/z$, and z^{-1} has a triple zero at $w=1/4$), so the only possible singularities of branches of F_3 in $\mathbb{C}^\times$ are $1/64$, $2/3$, 1. The preimage $z^{-1}([1,\infty))$ is the segment $[\frac{1}{64},\frac14]$ together with two arcs from $\frac14$ to 1 (near $w=1$, $1-z\sim\frac73(w-1)^2$, so the arcs arrive vertically), which bound a lens containing $(\frac14,1)$; the complementary component containing 0 is unbounded and contains $(1,\infty)$. On it $G\circ z$ stays on the principal branch. (i) As $w\uparrow\frac{1}{64}$, $z\uparrow1$ and $G(z)\to G(1)=\frac{\Gamma(2)\Gamma(1)}{\Gamma(5/3)\Gamma(4/3)}\neq0$; g has a simple pole and F_3 a logarithmic singularity at $1/64$. (ii) Since $c-a-b=1$, near $z=1$ the principal branch is $G=A(z)+B(1-z)\log(1-z)\,h(z)$ with A,h analytic, $h(1)=1$ and $B=\Gamma(2)/(\Gamma(\frac13)\Gamma(\frac23))\neq0$ [20, 15.8.10]. Approaching $w=1$ along $(1,1+\delta)$ inside the unbounded component, $z\uparrow1$ and g contains $\frac73B\,(w-1)^2\log((w-1)^2)$ times a nonvanishing analytic factor, so this continuation of F_3 is singular at 1. (iii) Crossing one of the arcs once replaces G by $G+\kappa y_1$, $\kappa=\pm2\pi iB\neq0$, where $y_1=(1-z)h(z)$ is the exponent-1 solution at $z=1$. Inside the lens we may follow the real axis to $w\to\frac23$, where z has a simple zero. The solutions analytic at $z=0$ form the line $\mathbb{C}G$ (the exponents there are $0,-1$), and y_1 is not a multiple of G on $(0,1)$ because $y_1(1)=0\neq G(1)$; hence y_1 has a nonzero pole part βz^{-1} at 0, and β does not depend on the side from which 0 is approached. Thus g acquires a simple pole at $\frac23$ and this continuation of F_3 has a logarithmic singularity there. Three points give order ≥3. □

Remark 2.19 (evidence for Conjecture 2.17). For $d\leq5$ we reconstructed (by guessing, exactly over $\mathbb{Q}$) differential operators annihilating F_d that are minimal among those of small degree; they have orders $2,3,5,8$ ($d=2,\dots,5$; order 11 for $d=6$, found modulo a prime), and for $d\leq4$ they annihilate constants. Their exact local analysis (`explore_t1.py`, `explore_jordan.py`) gives: at each t_m a pseudo-reflection (exponents $0,\dots,r-2$ and one exponent in $\frac12+\mathbb{Z}$ for even d, a logarithm for odd d); at $t=1$, $\dim\ker(T_1-1)=r-\lfloor d/2\rfloor$ (for $d=2$ the eigenvalue -1 occurs once), with unipotent Jordan blocks $(2,1)$, $(3,1,1)$, $(2,2,1,1,1,1)$ for $d=3,4,5$; and the leading coefficient vanishes to order exactly 1 at each t_m and $\lfloor d/2\rfloor$ at $t=1$. The recurrences of Proposition 2.12 (which are proved) have characteristic polynomial exactly $(T-1)^{\lfloor d/2\rfloor}\prod_m(T-t_m^{-1})$ up to a constant, for $2\leq d\leq6$ (`rook_charpoly.py`), and no recurrence of order $<d$ with coefficient degree up to 20 ($d=4$), resp. 35 ($d=5$), fits the data. The differential operators are not certified as minimal, so all this is evidence, not proof. What is missing for a proof is global: one must show that the continuations of F_d realize the singular local solutions at every t_m and a $\lfloor d/2\rfloor$-dimensional image of T_1-1 (for odd d the Jordan blocks at $t=1$ have size 2, so no single branch suffices). For $d=4$ it would suffice to know that the guessed operator $M_4=L_1\partial_t$ (order 5) annihilates F_4, which can be proved like Proposition 2.12, and that L_1 is irreducible. Indeed, the derivatives of the continuations of F_4 would then span the solution space of L_1; L_1 has $3,3,2$ linearly independent holomorphic solutions at $\frac{1}{625},\frac14,1$ (`explore_sharp4b.py`), and the argument of Lemma 2.16 would give order $\geq1+1+2=4$. The exponents of L_1 are $\{0,0,-1,-1\}$ at 0, $\{0,1,2,-\frac12\}$ at $\frac{1}{625}$ and at $\frac14$, $\{0,1,1,3\}$ at 1, $\{1,1,2,2\}$ at ∞ (solutions $\sim t^{-e}$), and $\{0,1,2,4\}$ at four apparent singularities. The Fuchs relation therefore does not exclude right factors of orders $1,2,3$, so the cheap irreducibility test is inconclusive, and a factorization algorithm or a certified monodromy computation would be needed.

3. Division points, drawn curves, and hyperbolic complete monotonicity

This section settles five explicitly posed questions by other authors.

(1) The speculation of Mani and Rubinstein-Salzedo on individual constructible division points of the tricuspoid is false; we give the complete classification (Theorem 3.3).
(2) Their question on the lemniscate *with the curve drawn* is answered by combining known results: a drawn lemniscate gives exactly the origami field (as for any drawn non-circular conic, by Videla and Baek et al.), and the admissible n are those of the Cox–Shurman origami theorem (Theorem 3.7).
(3) The identification $\mathrm{HM} \cap \mathrm{GGC} = \mathrm{HCM}$ asked about by Jedidi and Simon fails (Theorem 3.12).
(4) Bosch's conjecture that $\sqrt{\mathbf{\Gamma}_t \mathbf{\Gamma}_s}$ is always HCM is false; in fact it is HCM if and only if $|t-s| \le 1/2$ (Theorem 3.15); the n-fold version fails for every $n \ge 2$ (Proposition 3.16).
(5) The belief of Bosch and Simon that $a+b \ge c$ is necessary for $\mathbf{\Gamma}_c \times \mathbf{B}_{a,b}$ to be HCM is true (Theorem 3.18).

Items 1–2 are division-point dichotomies in the spirit of the algebraic/transcendental division classifications of method M8, carried out in the constructible (two-power) setting. Items 3–5 use method M9's mechanism: a Bernstein (positive Laplace) representation forces sign conditions, here in the form of a zero-free and Pick-type obstruction (Lemma 3.9), which sharpens the zero-free continuation argument used for McKay laws in [6].

Throughout, "constructible" means constructible with straightedge and compass from the points 0 and 1 of $\mathbb{C} = \mathbb{R}^2$, unless a drawn curve is explicitly allowed.

3.1. **Individual division points of the tricuspoid.**

Problem 3.1 (Mani and Rubinstein-Salzedo [24, Remark 5.7]; Remark 4.3 of the arXiv version)**.** "One may be tempted to wonder which, if any, of the individual n-division points are constructible when $n \nmid 6$. It appears that no n-division points other than those contained among the 6-division points are constructible. The x-coordinates of n-division points are roots of the polynomials $f_r(x)$, where r is a rational number ≥ 3 whose numerator divides n. It appears that, except in the cases already discussed, $f_r(x)$ is indeed irreducible, and further, that the field $\mathbb{Q}(\alpha)$ obtained by adjoining a root of $f_r(x)$ is totally ramified at 3."

In [24] the tricuspoid (deltoid) is $x = 2\cos\varphi + \cos 2\varphi$, $y = 2\sin\varphi - \sin 2\varphi$, i.e.

$$z(\varphi) = 2e^{i\varphi} + e^{-2i\varphi}, \qquad 0 \le \varphi < 2\pi,$$

the curve is not drawn, and the n-division points are taken relative to the cusp $z(0) = 3$: they are the points $P(k/n)$, $0 \le k < n$, where $P(r)$ denotes the point at arc length $16r$ from 3 in the direction of increasing φ (the total length is 16). Their main theorem for this curve [24, Theorem 5.1] says that *all* n-division points are constructible if and only if $n \mid 6$. Put $T_3(X) = 4X^3 - 3X$ and $\omega = e^{2\pi i/3}$.

Lemma 3.2. *$|z'(\varphi)| = 4|\sin(3\varphi/2)|$; for $0 \le \varphi \le 2\pi/3$ the arc length from $z(0)$ is $s(\varphi) = \frac{8}{3}\big(1 - \cos\frac{3\varphi}{2}\big)$, with $s(2\pi/3) = 16/3$; and $z(\varphi + 2\pi/3) = \omega z(\varphi)$.*

Proof. $|z'|^2 = |2ie^{i\varphi} - 2ie^{-2i\varphi}|^2 = 8 - 8\cos 3\varphi = 16\sin^2(3\varphi/2)$, and $\sin(3\varphi/2) \ge 0$ on $[0, 2\pi/3]$; integrate. Finally $e^{-2i(\varphi+2\pi/3)} = e^{-2i\varphi}e^{-4\pi i/3} = \omega e^{-2i\varphi}$. □

Theorem 3.3 (Constructible points of the tricuspoid)**.** *Let $r \in \mathbb{Q} \cap [0,1)$ and write $r = j/3 + \rho$ with $j \in \{0,1,2\}$ and $\rho \in [0, 1/3)$. The following are equivalent:*

(i) *$P(r)$ is constructible;*
(ii) *the cubic $T_3(X) - (1 - 6\rho)$ has a rational root;*

(iii) $\rho = \frac{1}{6}(1-u)(1+2u)^2$ *for some* $u \in \mathbb{Q} \cap [-1,1]$.

If these fail, the abscissa of $P(r)\,\omega^{-j}$ *generates a cubic field.*

Proof. (ii)⇔(iii) since $1-T_3(u) = (1-u)(1+2u)^2$, and a rational root u of $T_3(X) = 1-6\rho \in (-1,1]$ lies in $[-1,1]$.

By Lemma 3.2, $P(r) = \omega^j z(\varphi)$ where $\varphi \in [0, 2\pi/3)$ is defined by $\cos(3\varphi/2) = \tau := 1-6\rho$. Put $\theta = \varphi/2 \in [0,\pi/3)$, $c = \cos\theta$, $s = \sin\theta \geq 0$. Then $T_3(c) = \cos 3\theta = \tau$ and

$$x := \operatorname{Re} z(\varphi) = 2\cos 2\theta + \cos 4\theta = 8c^4 - 4c^2 - 1, \qquad y := \operatorname{Im} z(\varphi) = 2\sin 2\theta - \sin 4\theta = 8cs^3.$$

Multiplication by $\omega^{\pm 1}$ preserves constructibility ($\cos\frac{2\pi}{3} = -\frac{1}{2}$, $\sin\frac{2\pi}{3} = \frac{\sqrt{3}}{2}$), so (i) holds iff $z(\varphi)$ is constructible.

(ii)⇒(i): if $g(X) := T_3(X) - \tau$ has a rational root, all its roots lie in a field of degree ≤ 2 over $\mathbb{Q}$; in particular $[\mathbb{Q}(c):\mathbb{Q}] \leq 2$, $s = \sqrt{1-c^2}$, and $x, y \in \mathbb{Q}(c,s)$, a two-step quadratic tower.

(i)⇒(ii): suppose g is irreducible over $\mathbb{Q}$. Then $[\mathbb{Q}(c):\mathbb{Q}] = 3$ and $x \in \mathbb{Q}(c)$, so $[\mathbb{Q}(x):\mathbb{Q}] \in \{1,3\}$. Modulo g we have $8c^4 = 2c\cdot 4c^3 = 6c^2 + 2\tau c$, hence $x = 2c^2 + 2\tau c - 1$. If $x = x_0 \in \mathbb{Q}$, then c would be a root of $2X^2 + 2\tau X - 1 - x_0 \in \mathbb{Q}[X]$, contradicting $[\mathbb{Q}(c):\mathbb{Q}] = 3$. So $[\mathbb{Q}(x):\mathbb{Q}] = 3$, x is not constructible, and neither is $z(\varphi)$. A reducible cubic over $\mathbb{Q}$ has a rational root. □

The minimal polynomial of x is the characteristic polynomial of $2c^2 + 2\tau c - 1$ on $\mathbb{Q}[c]/(g)$, namely

$$F_\tau(x) = 4x^3 - (24\tau^2 + 3)x + (1 - 20\tau^2 - 8\tau^4). \tag{5}$$

For $\rho = 1/n$ one checks $n^4 F_{1-6/n}(x) = 4n^4x^3 - (27n^4 - 288n^3 + 864n^2)x - (27n^4 - 432n^3 + 2448n^2 - 6912n + 10368)$, which is exactly the cubic $f_n(x)$ printed in [24, Section 5] (Section 4 of the arXiv version); the polynomial f_r of Problem 3.1 is $F_{1-6/r}$ up to a constant.

Corollary 3.4 (All three speculations in Problem 3.1 fail)**.** (1) *For* $n = 32$ *the division points* $P(k/32)$ *with* $k \in \{9, 13, 19, 23\}$ *are constructible, and none of them is a* 6*-division point. For instance, with* $u = 1/4$, $T_3(1/4) = -11/16 = 1 - 6\cdot\frac{9}{32}$, *and*

$$P\left(\tfrac{9}{32}\right) = \left(\frac{39 - 45\sqrt{5}}{64},\ \sqrt{\frac{3969 + 1755\sqrt{5}}{2048}}\right).$$

No $n < 32$ *has a constructible* n*-division point outside the* 6*-division points, but infinitely many* n *do: every* $u \in \mathbb{Q} \cap (-1,1)$ *with* $(1-u)(1+2u)^2 \notin \mathbb{Z}$ *gives a constructible point* $P(\rho)$, $\rho = \frac{1}{6}(1-u)(1+2u)^2 \in (0, \frac{1}{3}) \setminus \{\frac{1}{6}\}$, *and these* ρ *have unbounded denominators.*

(2) $f_{32/9}$ *is reducible:* $8192\,F_{-11/16}(x) = (32x+39)(1024x^2 - 1248x - 2151)$.

(3) *For* $r = 28/9$ *(that is* $\rho = 9/28$, $\tau = -13/14$*),* f_r *is irreducible over* $\mathbb{Q}$ *and* 3 *is unramified in* $\mathbb{Q}(\alpha)$.

Proof. (1) For $k = 9, 13, 19, 23$ one has $\rho = 9/32,\ 7/96,\ 25/96,\ 5/96$ and $1 - 6\rho = T_3(u)$ with $u = 1/4,\ -3/4,\ 3/4,\ -1/4$; apply Theorem 3.3. The explicit point: the roots of $T_3(X) + 11/16$ are $1/4$ and $(-1 \pm 3\sqrt{5})/8$; the root in $[1/2, 1]$ is $c = (3\sqrt{5} - 1)/8$, and $x = 2c^2 + 2\tau c - 1$, $y^2 = 64c^2(1-c^2)^3$ give the displayed values. Minimality of 32 and the remaining claims of (1)–(3) are finite computations, carried out by `R7_verify_tricuspoid.py` (rational-root test, factorization of F_τ, and a PARI computation of the decomposition of 3 in the cubic field for (3)). □

Remark 3.5. The script also checks Theorem 3.3 against an independent criterion for all k/n with $n \leq 60$: the order of the Galois group of the minimal polynomial of $z = x + iy$ is a power of 2 exactly when (ii) holds; it confirms $s(\varphi) = 4.5$ for $P(9/32)$ by 10^{-40}-accurate quadrature, and re-derives [24, Theorem 5.1] for $n \leq 2000$ (for $n \nmid 6$ the first division point is not constructible).

3.2. **The lemniscate with the curve drawn.**

Problem 3.6 ([24, Remark 1.4])**.** "For curves other than the circle, there are actually two possible problems regarding the constructibility of n-division points. First, for a closed curve C, we can ask whether, given the curve C already drawn on the plane, it is possible to construct the n-division points using a compass and a straightedge. However, we can also ask whether it is possible to construct the n-division points of C without C ever being drawn. [...] Abel's Theorem is an answer to a problem of the second type. To the best of our knowledge, the corresponding question with a drawn lemniscate has not yet been answered."

Let $L = \{(x^2+y^2)^2 = x^2 - y^2\}$, with node $O = 0$ and vertex $A = 1$. Let $\mathcal{K}_L \subset \mathbb{C}$ be the set of points constructible from O, A when, in addition to intersecting lines and circles, one may intersect a constructed line or circle with the drawn curve L. Let $\mathcal{O}$ be the field of origami numbers; by Alperin [27, Theorem 5.2] it is the smallest subfield of $\mathbb{C}$ closed under square roots, cube roots and complex conjugation, and $z \in \mathcal{O}$ iff z lies in a tower of extensions of degrees 2 and 3 over $\mathbb{Q}$ [28, Section 10.3].

Theorem 3.7. *Let $\mathcal{K}_L$ be the set of points constructible with straightedge and compass from O and A when intersections of constructed lines and circles with the drawn lemniscate L are also allowed. Then $\mathcal{K}_L = \mathcal{O}$. Consequently the n-division points of the lemniscate (relative to O, or to any point of $L \cap \mathcal{K}_L$) are constructible with straightedge and compass from the drawn lemniscate if and only if*

$$n = 2^a 3^b p_1 \cdots p_r, \qquad a, b \geq 0,$$

where $p_1, \ldots, p_r$ are distinct Pierpont primes ($p_i = 2^u 3^v + 1$) with $p_i = 7$ or $p_i \equiv 1 \pmod 4$. For instance $n = 7, 9, 13, 21, 37$ are possible with the drawn curve but not without it (Abel), while $n = 11, 19, 23, 25$ remain impossible.

Proof. Inversion. The map $\iota(z) = 1/\bar{z}$ is a straightedge-and-compass operation on points, maps lines and circles to lines and circles (constructibly), and maps $L \setminus \{0\}$ onto the hyperbola $H = \{X^2 - Y^2 = 1\}$: if $x + iy = \iota(X + iY)$ then $x^2 + y^2 = (X^2 + Y^2)^{-1}$ and $x^2 - y^2 = (X^2 - Y^2)(X^2 + Y^2)^{-2}$. Hence intersecting a constructed circle Γ with H can be done by intersecting $\iota(\Gamma)$ with L and applying ι.

Real cubics. Let $p(w) = w^3 + \beta_2 w^2 + \beta_1 w + \beta_0$ with real $\beta_i \in \mathcal{K}_L$, $\beta_0 \neq 0$, and let w_0 be a real root. Put $q(w) = (w - m)p(w)$ with $m = -\operatorname{sign}\beta_0$; then $q = w^4 + a_3 w^3 + a_2 w^2 + a_1 w + a_0$ with $a_0 = |\beta_0| > 0$. With $\lambda = (4a_0)^{-1/4} \in \mathcal{K}_L$ the substitution $u = \lambda w$ turns $q = 0$ into $u^4 + Du^3 + Fu^2 + \frac{E}{2}u + \frac{1}{4} = 0$, where $D = a_3\lambda$, $F = a_2\lambda^2$, $E = 2a_1\lambda^3$. In the rotated coordinates $u = (X + Y)/\sqrt{2}$, $v = (X - Y)/\sqrt{2}$ the hyperbola is $uv = \frac{1}{2}$, and substituting $v = 1/(2u)$ shows that the circle $u^2 + v^2 + Du + Ev + F = 0$ meets it exactly at the points $(u, 1/(2u))$ with u a real root of this quartic. As λw_0 is such a root, the circle is real, it is constructible, and the intersection point is in $\mathcal{K}_L$. Hence $w_0 \in \mathcal{K}_L$.

Lower bound. $\mathcal{K}_L$ is a subfield of $\mathbb{C}$ closed under conjugation and square roots (classical). If $z = \varrho e^{i\vartheta} \in \mathcal{K}_L$, then $\varrho^{1/3}$ (root of $w^3 - \varrho$) and $\cos(\vartheta/3)$ (root of $4w^3 - 3w - \cos\vartheta$) lie in $\mathcal{K}_L$, hence so do $\sin(\vartheta/3)$ and $z^{1/3}$. By Alperin's characterization, $\mathcal{O} \subseteq \mathcal{K}_L$.

Upper bound. Suppose all points constructed so far have coordinates in a subfield $K \subset \mathcal{O}$ (with $i \in K$). Intersections of lines and circles give coordinates in a quadratic extension of K. For a line $\alpha x + \beta y + \gamma = 0$ over K, eliminating y from the equation of L gives a quartic in x. For a circle $x^2 + y^2 + Dx + Ey + F = 0$, points of L on it satisfy $(Dx + Ey + F)^2 = x^2 - y^2$, so they are intersections of the circle with a conic, and their abscissae are roots of a polynomial of degree ≤ 4 over K (Bézout); then y satisfies a quadratic over $K(x)$. Roots of quartics are obtained by Ferrari's formulas with square and cube roots, so the new coordinates lie in $\mathcal{O}$. By induction $\mathcal{K}_L \subseteq \mathcal{O}$.

Division points. By the first part, a point is constructible from the drawn lemniscate iff it is origami constructible. Cox and Shurman proved that the lemniscate can be divided into n arcs of equal length by origami iff n has the displayed form [25, Lemniscate Theorem, part (2)]. The lemniscatic addition formula $\mathrm{sl}(u+v) = (\mathrm{sl}\,u\,\mathrm{sl}'v + \mathrm{sl}\,v\,\mathrm{sl}'u)/(1+\mathrm{sl}^2u\,\mathrm{sl}^2v)$, together with $x = r\sqrt{(1+r^2)/2}$, $y = r\sqrt{(1-r^2)/2}$ for the point of L at polar radius r, shows that changing the base point to another constructible point of L does not change constructibility. □

Remark 3.8. The arithmetic behind the theorem is that, for odd n, the Galois group over $\mathbb{Q}(i)$ of the n-division value $\mathrm{sl}(2\varpi/n)$ is $(\mathbb{Z}[i]/n)^\times$ [26, Corollary 4.8] (powers of 2 only add 2-power factors), whose order $\prod_{p^e\|n} |(\mathbb{Z}[i]/p^e)^\times|$ is 3-smooth exactly for the displayed n. `R7_verify_drawn_lemniscate.py` checks this equivalence for $n \le 2\cdot 10^4$, confirms for odd $n \le 13$ by factoring the division polynomials of $y^2 = x^3 - x$ over $\mathbb{Q}(i)$ that the division value $\wp(2\varpi/n) = \mathrm{sl}(2\varpi/n)^{-2}$ has degree $|(\mathbb{Z}[i]/n)^\times|/2$, and carries out the constructions of $2^{1/3}$ and $\cos(\pi/9)$ with the drawn lemniscate numerically (error $< 10^{-59}$). The theorem uses the cited theorems as inputs. The identification $\mathcal{K}_L = \mathcal{O}$ is an instance of the known fact that straightedge, compass and a fixed drawn non-circular conic construct exactly Alperin's field [39, 40], because inversion maps L to the hyperbola $X^2 - Y^2 = 1$; the answer to Problem 3.6 is thus a combination of known results, which we record because the question was left open in [24].

3.3. **A zero-free and Pick obstruction for HCM functions.** A function $f : (0,\infty) \to (0,\infty)$ is *HCM* if for each $u > 0$ the function $H_u(w) = f(uv)f(u/v)$, $w = v + v^{-1}$, is completely monotone (CM) on $(2,\infty)$, and *HM* if each H_u is nonincreasing there [29, 34]. A law on $(0,\infty)$ is GGC if it is a weak limit of finite sums of independent gamma variables. Put $\Omega = \mathbb{C} \setminus (-\infty, 0]$.

Lemma 3.9. *Let $f > 0$ on $(0,\infty)$ extend analytically to Ω, and suppose f is HCM. Then*

(1) *f has no zero in Ω;*
(2) *$\psi_f(z) := -zf'(z)/f(z)$ satisfies $\operatorname{Im}\psi_f(z) \ge 0$ for $\operatorname{Im} z > 0$.*

Proof. Fix $u > 0$. The map $v \mapsto w = v + v^{-1}$ sends Ω two-to-one onto $D := \mathbb{C} \setminus (-\infty, -2]$ (it sends $(-\infty, 0)$ onto $(-\infty, -2]$). The function $G(v) = f(uv)f(u/v)$ is analytic on Ω and invariant under $v \mapsto 1/v$, so it descends to an analytic function H_u on D (at $w = 2$ write $v = e^t$: G is even in t and $w - 2 = t^2 + O(t^4)$).

By Bernstein's theorem $H_u(w) = \int_{[0,\infty)} e^{-wt}\mu(dt)$ for $w > 2$, with $\mu \ge 0$, $\mu \ne 0$. Let σ be the abscissa of convergence. By Landau's theorem [36, Chapter II, Theorem 5b] the real point σ, if finite, is a singular point of the function defined by the integral. That function agrees with H_u on $\{\operatorname{Re} w > \max(\sigma, -2)\}$, and H_u is analytic on $D \supset (-2,\infty)$; hence $\sigma \le -2$, and for every $w \in (-2,\infty)$

$$H_u(w) = \int e^{-wt}\mu(dt) > 0, \qquad H_u'(w) = -\int te^{-wt}\mu(dt) \le 0.$$

For $\phi \in (0,\pi)$, $w = 2\cos\phi \in (-2,2)$ corresponds to $v = e^{i\phi}$, and since $f(\bar z) = \overline{f(z)}$,

$$H_u(2\cos\phi) = f(ue^{i\phi})f(ue^{-i\phi}) = |f(ue^{i\phi})|^2.$$

Positivity gives (1) (every $z \in \Omega \setminus (0,\infty)$ is $ue^{\pm i\phi}$). Differentiating in ϕ, with $z = ue^{i\phi}$,

$$2|f(z)|^2 \operatorname{Im}\psi_f(z) = \frac{d}{d\phi}|f(ue^{i\phi})|^2 = -2\sin\phi\, H_u'(2\cos\phi) \ \ge 0,$$

because $\frac{d}{d\phi}\log|f(ue^{i\phi})| = \operatorname{Re}\big(izf'(z)/f(z)\big) = \operatorname{Im}\psi_f(z)$. This is (2). □

Remark 3.10. The lemma also follows from Bondesson's representation $f(x) = Cx^{\beta-1}h_1(x)h_2(1/x)$ with $-(\log h_i)'$ Stieltjes [34, (5.2.3), Theorem 5.3.1], recalled as equation (7) of [35]; the proof above uses only Bernstein's and Landau's theorems. Part (1) is the zero-free obstruction used for McKay laws in [6]; part (2) is stronger and is what decides the boundary cases below.

3.4. $\mathrm{HM} \cap \mathrm{GGC} \neq \mathrm{HCM}$.

Problem 3.11 (Jedidi and Simon [29, Remark 4.1(b)])**.** "From the above equivalence, one can ask if the general identification $\mathrm{HM} \cap \mathrm{GGC} = \mathrm{HCM}$ is true or not, and we could not find any counterexample. Such an identification would show Bondesson's conjecture in view of the main result of [22]." ([22] refers to the source's bibliography.)

Theorem 3.12. *Let $0 < \lambda_1 < \lambda_2$ and let $X = E_1/\lambda_1 + E_2/\lambda_2$ with E_1, E_2 independent standard exponential variables. Then X is GGC, its density is HM, and its density is not HCM. Hence* $\mathrm{HM} \cap \mathrm{GGC} \neq \mathrm{HCM}$.

Proof. X is a sum of two independent gamma variables, hence GGC (Thorin measure $\delta_{\lambda_1} + \delta_{\lambda_2}$). Its density is $f(x) = \frac{\lambda_1\lambda_2}{\delta} e^{-\lambda_1 x}(1 - e^{-\delta x})$, $\delta = \lambda_2 - \lambda_1$.

HM: write $u = e^{\sigma}$, $v = e^{t}$ and $\Phi(y) = \log f(e^y)$; then $\log H_u = \Phi(\sigma + t) + \Phi(\sigma - t)$, and $w = 2\cosh t$ increases with $t \geq 0$, so H_u is nonincreasing for every u iff Φ' is nonincreasing, i.e. iff $x \mapsto x f'(x)/f(x)$ is nonincreasing. Here

$$\frac{x f'(x)}{f(x)} = -\lambda_1 x + \frac{\delta x}{e^{\delta x} - 1},$$

and both terms decrease, since $\frac{d}{dt}\frac{t}{e^t - 1} = -\frac{te^t - e^t + 1}{(e^t - 1)^2} < 0$ for $t > 0$.

Not HCM: f is entire and vanishes at $z = 2\pi i/\delta \in \Omega$, contradicting Lemma 3.9(1). (Concretely, for $\lambda_1 = 1, \lambda_2 = 2$ and $u = 2\pi$ one gets $H_u(0) = |f(2\pi i)|^2 = 0$, and the 16th Taylor coefficient of $H_{2\pi}$ at $w = 2.2$ has the wrong sign for complete monotonicity; see `R7_verify_hcm.py`.) □

Remark 3.13. The example is elementary; we found no earlier statement that it answers the question of [29]. The same zero argument shows that no sum of two independent gamma variables with different rates is HCM (the density is $x^{\beta_1+\beta_2-1}e^{-\lambda_2 x}M(\beta_1, \beta_1 + \beta_2, \delta x)$, and Kummer's M is the Laplace transform of a non-degenerate measure on $[0, 1]$, so it has infinitely many zeros, all non-real); such sums are always GGC, and they are HM in parameter ranges that include the exponential case above.

3.5. **Square roots of gamma products.** Let $\mathbf{\Gamma}_t$ denote a gamma variable with density $x^{t-1}e^{-x}/\Gamma(t)$; products are of independent copies.

Problem 3.14 (Bosch [30, Remark 1])**.** (a) "It is hence natural to raise the [following] Conjecture. For all $s, t > 0$, the independent product $\sqrt{\gamma_t \gamma_s}$ is HCM." (b) "[...] we believe that the latter should be true for all independent products of the type" $(\gamma_{t_1} \times \cdots \times \gamma_{t_n})^{1/n}$, $t_i > 0$.

For $|t - s| < 1/2$, (a) follows from the argument sketched by Bosch in [30, Remark 1(a)]; see also [31, Section 3.3]. For $|t - s| = 1/2$ it follows from the duplication formula, which gives $\sqrt{\mathbf{\Gamma}_t \mathbf{\Gamma}_{t+1/2}} \overset{d}{=} \mathbf{\Gamma}_{2t}/2$, a gamma law.

Theorem 3.15. *For $s, t > 0$ the density of $\sqrt{\mathbf{\Gamma}_t \mathbf{\Gamma}_s}$ is HCM if and only if $|t - s| \leq 1/2$. In particular Bosch's conjecture fails, e.g. for $\sqrt{\mathbf{\Gamma}_1 \mathbf{\Gamma}_2}$ and for $\sqrt{\mathbf{\Gamma}_3 \mathbf{\Gamma}_{1/2}}$, whose density is $\frac{1}{4}(4x^2 + 6x + 3)e^{-2x}$.*

Proof. The density of $\mathbf{\Gamma}_t \mathbf{\Gamma}_s$ is $\frac{2}{\Gamma(t)\Gamma(s)} y^{(t+s)/2-1} K_{t-s}(2\sqrt{y})$, so that of $Y = \sqrt{\mathbf{\Gamma}_t \mathbf{\Gamma}_s}$ is

$$f(x) = \frac{4}{\Gamma(t)\Gamma(s)} x^{t+s-1} K_{\nu}(2x), \qquad \nu = |t - s|$$

($K_{-\nu} = K_{\nu}$). For $(t, s) = (3, \frac{1}{2})$, $K_{5/2}(\zeta) = \sqrt{\pi/(2\zeta)}\, e^{-\zeta}(1 + 3\zeta^{-1} + 3\zeta^{-2})$ gives the displayed density. Sufficiency is the case discussed before the theorem; we prove necessity. Let $\nu > 1/2$.

The function f is analytic on Ω and positive on $(0,\infty)$. From $K_\nu'(\zeta) = -K_{\nu-1}(\zeta) - \frac{\nu}{\zeta}K_\nu(\zeta)$ [37, 10.29.2],

$$\psi_f(z) = \nu - (t+s-1) + R_\nu(2z), \qquad R_\nu(\zeta) := \frac{\zeta K_{\nu-1}(\zeta)}{K_\nu(\zeta)}.$$

We use the convergent expansions $K_\mu(\zeta) = \frac{1}{2}\Gamma(\mu)(\zeta/2)^{-\mu}(1+O(|\zeta|^{\min(2,2\mu)}))$ for $\mu > 0$, $\mu \notin \mathbb{Z}$, the analogous statement with logarithmic corrections for $\mu \in \mathbb{Z}_{>0}$, and $K_0(\zeta) = -\log(\zeta/2) - \gamma + O(|\zeta|^2|\log\zeta|)$ [37, 10.27.4, 10.31.1, 10.31.2], all uniform for $0 < \arg\zeta < \pi$ with principal branches. Hence, as $\zeta \to 0$ in the upper half-plane:

$$R_\nu(\zeta) = \begin{cases} \dfrac{2\Gamma(1-\nu)}{\Gamma(\nu)}\Big(\dfrac{\zeta}{2}\Big)^{2\nu}(1+o(1)), & \frac{1}{2} < \nu < 1, \\ \zeta^2\big(-\log(\zeta/2)-\gamma\big)(1+o(1)), & \nu = 1, \\ \dfrac{\zeta^2}{2(\nu-1)}(1+o(1)), & \nu > 1. \end{cases}$$

Take $\zeta = \varrho e^{i\theta}$ with $\varrho \to 0$ and fixed θ: for $\frac{1}{2} < \nu < 1$ choose $\theta \in (\pi/(2\nu), \pi)$, so that $\sin(2\nu\theta) < 0$; for $\nu = 1$ choose $\theta \in (3\pi/4, \pi)$, so that $\sin 2\theta < 0 < \cos 2\theta$ and $\operatorname{Im}[\zeta^2(-\log(\zeta/2)-\gamma)] = \varrho^2[\sin 2\theta\,(\log(2/\varrho)-\gamma) - \theta\cos 2\theta] \le -|\sin 2\theta|\varrho^2(\log(2/\varrho)-\gamma)$; for $\nu > 1$ choose $\theta \in (\pi/2, \pi)$. In each case $\operatorname{Im} R_\nu(\zeta) < 0$ for small ϱ (the relative errors are $o(1)$ while the main terms have imaginary parts of the same order as their moduli). Then $\operatorname{Im}\psi_f(\zeta/2) < 0$ with $\operatorname{Im}(\zeta/2) > 0$, contradicting Lemma 3.9(2). □

Proposition 3.16. *For every $n \ge 2$ there are $t_1, \dots, t_n > 0$ such that $(\mathbf{\Gamma}_{t_1} \cdots \mathbf{\Gamma}_{t_n})^{1/n}$ is not HCM. Explicitly:*

1. *for $n \in \{2,3,4\}$ and any $t > 0$, with $t_1 = t+3$ and $t_j = t + (j-1)/n$ $(2 \le j \le n)$, the density is $Cx^{nt}e^{-nx}Q_n(x)$ with $Q_n(x) = x^2 + \frac{3(n-1)}{n}x + \frac{(n-1)(2n-1)}{n^2}$, which has non-real zeros;*
2. *for every $n \ge 2$, $t_1 = 1$ and $t_j = 1 + \frac{j+1}{2n}$ $(2 \le j \le n)$ violate Lemma 3.9(2).*

Proof. (1) Let $g(x) = x^{nt-1}e^{-nx}$ and let $\mathcal{D}h = -(xh)'$, so that $\int_0^\infty x^s \mathcal{D}h\,dx = s\int_0^\infty x^s h\,dx$. By Gauss's multiplication formula the Mellin transform of $(\mathbf{\Gamma}_{t+3}\prod_{j=1}^{n-1}\mathbf{\Gamma}_{t+j/n})^{1/n}$ is a constant times $(t+\frac{s}{n})(t+1+\frac{s}{n})(t+2+\frac{s}{n})\,\Gamma(nt+s)n^{-s}$, i.e. the law has density proportional to $T_tT_{t+1}T_{t+2}\,g$ with $T_c = c + \mathcal{D}/n$. Since $T_c[x^\alpha e^{-nx}] = (x + c - \frac{\alpha+1}{n})x^\alpha e^{-nx}$, a direct computation gives $T_tT_{t+1}T_{t+2}\,g = x^{nt}e^{-nx}Q_n(x)$. The discriminant of Q_n is $(n-1)(n-5)/n^2 < 0$ for $n \le 4$, so f has zeros in Ω; apply Lemma 3.9(1).

(2) All t_j lie in $[1,2)$ and are distinct, so the poles of the Mellin transform $M(s) = \prod_j \Gamma(t_j + s/n)/\Gamma(t_j)$ are simple and the density of Y is the convergent residue series (Meijer G-function expansion)

$$f_Y(x) = \frac{n}{\prod_i \Gamma(t_i)} \sum_{j=1}^{n} \sum_{k \ge 0} \frac{(-1)^k}{k!} \prod_{i \ne j} \Gamma(t_i - t_j - k)\, x^{n(t_j+k)-1},$$

valid on Ω with principal powers (the terms are $O(|x|^{nk}/(k!)^n)$ in k, cf. [37, 16.17.2]). Since $nt_1 - 1 = n-1$, $nt_2 - 1 = n - 1 + \frac{3}{2}$, $nt_j - 1 \ge n+1$ for $j \ge 3$, and $n \ge 2$,

$$f_Y(x) = \kappa\, x^{n-1}\Big(1 + \frac{c_2}{c_1}x^{3/2} + O(|x|^2)\Big), \qquad \kappa > 0, \quad c_j := \prod_{i \ne j}\Gamma(t_i - t_j);$$

here $c_1 > 0$ and $c_2 < 0$ (only the factor $\Gamma(t_1 - t_2) = \Gamma(-\frac{3}{2n})$ is negative). Differentiating termwise, $\psi_{f_Y}(z) = -(n-1) - \frac{3}{2}\frac{c_2}{c_1}z^{3/2} + O(|z|^2)$, so for $z = \varrho e^{i\theta}$ with $\theta \in (2\pi/3, \pi)$, $\operatorname{Im}\psi_{f_Y}(z) = \frac{3}{2}\big|\frac{c_2}{c_1}\big|\varrho^{3/2}\sin\frac{3\theta}{2} + O(\varrho^2) < 0$ for small ϱ. □

The cases $t_j = j/n$ singled out by Bosch for stable laws are not affected: there consecutive gaps are $1/n$ and the leading correction exponent is 1.

3.6. **Gamma–beta products.** Let $\mathbf{B}_{a,b}$ denote a beta variable with density $x^{a-1}(1-x)^{b-1}/B(a,b)$ on $(0,1)$.

Problem 3.17 (Bosch and Simon [32, after Lemma 1])**.** "A perusal of the proof of Lemma 1 in [12] shows also that the product $\Gamma_c \times \mathbf{B}_{a,b}$ has a HCM density as soon as $a+b \geq c$. We believe that this latter condition is necessary." (In the source, [12] is [33].)

Theorem 3.18. *For $a, b, c > 0$ the law of $\mathbf{\Gamma}_c \times \mathbf{B}_{a,b}$ has an HCM density if and only if $c \leq a+b$.*

Proof. Density. With $y = 1/(1+t)$ in $f(x) = \int_0^1 y^{-1} g_c(x/y)\beta_{a,b}(y)\,dy$ one finds $f(x) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(c)} x^{c-1}e^{-x}U(b, c-a+1, x)$, and Kummer's transformation $U(\mathsf{a},\mathsf{b},x) = x^{1-\mathsf{b}}U(\mathsf{a}-\mathsf{b}+1, 2-\mathsf{b}, x)$ [37, 13.2.40] gives

$$f(x) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(c)}\, x^{a-1}e^{-x}\, U(-\alpha, 1-\delta, x), \qquad \alpha = c-a-b, \quad \delta = c-a.$$

Sufficiency ($c \leq a+b$). If $c \leq a$ this is Lemma 1 of [32] (Simon's lemma [33], extended to $c \leq \min a_i$ there by weak closure). If $a < c \leq a+b$, write $\mathbf{B}_{a,b} \overset{d}{=} \mathbf{B}_{a,c-a}\mathbf{B}_{c,a+b-c}$ (independent factors; $\mathbf{B}_{c,0} := 1$) and use $\mathbf{\Gamma}_c\mathbf{B}_{a,c-a} \overset{d}{=} \mathbf{\Gamma}_a$: then $\mathbf{\Gamma}_c\mathbf{B}_{a,b} \overset{d}{=} \mathbf{\Gamma}_a\mathbf{B}_{c,a+b-c}$ with $a < c$, which is HCM by the same lemma.

Necessity. Let $c > a+b$, i.e. $\alpha > 0$, and suppose f were HCM. Put $U(z) = U(-\alpha, 1-\delta, z)$, analytic on Ω; $U > 0$ on $(0,\infty)$ because $f > 0$. Then

$$\psi_f(z) = (1-a) + z - \frac{zU'(z)}{U(z)}.$$

We use: (i) $U(z) = z^{\alpha}(1 + O(|z|^{-1}))$ as $z \to \infty$ with $0 \leq \arg z \leq \pi$ [37, 13.2.6, 13.7.3]; (ii) $U(z) \to \Gamma(\delta)/\Gamma(b) > 0$ as $z \to 0$ uniformly in $0 \leq \arg z \leq \pi$ (the limiting forms [37, 13.2.20–13.2.22] for second parameter $1-\delta < 1$, with $\Gamma(1-(1-\delta)) = \Gamma(\delta)$ and $-\alpha - (1-\delta) + 1 = b$); (iii) U continues analytically across $(-\infty, 0)$ from above; write $W(x) = U(-x+i0)$ for $x > 0$.

By Lemma 3.9, U has no zero in the open upper half-plane and $\operatorname{Im}\psi_f \geq 0$ there. W has no zero either: if $W(x_0) = 0$ with order $m \geq 1$, then near $z_0 = -x_0$ (from above) $\psi_f(z) = -\frac{mz_0}{z-z_0} + O(1)$, and $\operatorname{Im}\psi_f(-x_0 + i\eta) = -mx_0/\eta + O(1) \to -\infty$, a contradiction. Hence ψ_f is continuous up to $(-\infty, 0)$ from above, and since $\frac{d}{dx}\log W(x) = -U'/U$ at $z = -x + i0$,

$$0 \leq \operatorname{Im}\psi_f(-x+i0) = \operatorname{Im}\Big(-x\frac{W'(x)}{W(x)}\Big) = -x\,\frac{d}{dx}\arg W(x),$$

so a continuous branch of $\arg W$ is nonincreasing on $(0,\infty)$.

Now apply the argument principle to U on the half-annulus $\{\varepsilon < |z| < R,\ 0 < \arg z < \pi\}$, whose boundary carries no zero of U for small ε and large R, by (i), (ii) and the above. Along $[\varepsilon, R]$ the argument of U is constant; along $|z| = R$ it increases by $\alpha\pi + O(1/R)$ by (i); along $|z| = \varepsilon$ it changes by $o(1)$ by (ii); along the upper side of the cut, traversed from $-R$ to $-\varepsilon$, it changes by $\arg W(\varepsilon) - \arg W(R)$. Since there are no zeros inside, the total is 0, whence

$$\arg W(R) - \arg W(\varepsilon) = \alpha\pi + o(1) \qquad (\varepsilon \to 0,\ R \to \infty).$$

This is positive for small ε and large R, contradicting the monotonicity of $\arg W$. Hence f is not HCM. □

Remark 3.19. For $c = a+b+1$ the density is proportional to $x^{a-1}(x+b)e^{-x}$ and the obstruction is visible directly: $\psi_f(z) = (1-a) + z - 1 + \frac{b}{z+b}$ has $\operatorname{Im}\psi_f(-b+i\eta) = \eta - b/\eta < 0$ for $0 < \eta < \sqrt{b}$.

The script `R7_verify_hcm.py` checks the density formula by Mellin moments and, for several (a, b, c) with $c > a + b$, the identity $\arg W(R) - \arg W(\varepsilon) \approx \pi(\alpha - 2N_+)$, N_+ the number of zeros of U in the upper half-plane: both alternatives occur ($N_+ = 0$ with $\arg W$ increasing, e.g. $(a, b, c) = (0.7, 0.4, 1.6)$; $N_+ = 1$, e.g. $(2, 0.5, 4.3)$, where f itself vanishes in Ω), and each excludes HCM. It also checks $\operatorname{Im} \psi_f \geq 0$ on a grid when $c \leq a + b$.

3.7. **Verification, scope, and an unresolved target.** The scripts `R7_verify_tricuspoid.py`, `R7_verify_drawn_lemniscate.py`, `R7_verify_hcm.py` and `R7_verify_lorch_muldoon.py` (run with `python3 -I`) print PASS/FAIL for each check described above. They are independent of the proofs: Galois-group orders, exact factorizations, quadrature of arc length, Mellin moments, explicit Taylor coefficients of H_u, and argument tracking of U.

Theorems 3.7 and 3.15 (sufficiency) and 3.18 (sufficiency) rest on the cited results of Alperin, Cox–Shurman, Videla, Baek et al., Bosch, and Bosch–Simon; the remaining statements are proved here, the minimality of $n = 32$ and the irreducibility and ramification claims of Corollary 3.4 by exact computer algebra.

Remark 3.20 (Lorch–Muldoon, Section 2.3(b),(c): not settled). For $f = x^{-\nu} J_\nu(x)$ and $c = 2\nu + 1$, Lorch and Muldoon [38, Section 2.3] conjectured that the resultant $r_n(\nu)$ of the two coefficient polynomials of $f^{(n)}$ has exactly the rational roots $-m/2$ $(1 \leq m \leq n - 1)$, together with 0 for $n = 4$ and 1 for $n = 5$. Since r_n is monic in c ([14]), rational roots are half-integers. `R7_verify_lorch_muldoon.py` verifies (b),(c) for all $n \leq 30$ (Lorch–Muldoon: $n \leq 13$). Computations show that for $6 \leq n \leq 30$ the non-linear part of r_n is $Q_n(c)^2$ with Q_n irreducible of degree $\binom{\lfloor n/2 \rfloor}{2}$, and (for $|c| \leq 41$, $n \leq 26$) that $Q_n(c)$ is odd for even c; we have no proof for all n.

4. Mixtures and populations of parameters: W_p recovery, self-regularization, and a moment-matching constant

This section treats four questions about mixture estimation that were posed by other authors and that we could not find settled, either in the literature or in the earlier parts of this programme. Their status is as follows.

(A) Vinayak, Kong, Valiant and Kakade asked for Wasserstein-p guarantees for the maximum likelihood estimator of a population of coin biases [41, Sec. 5]. Theorem 4.2 gives the full minimax rate for every fixed $p \in [1, \infty)$, for every number of tosses t and population size N, and shows that every nonparametric maximum likelihood estimator (NPMLE) attains it. Relative to $p = 1$ [6, Part VI], only the parametric term changes, from $N^{-1/2}$ to $N^{-1/(2p)}$. *Closed, conditionally* on the W_1 theorem and three auxiliary estimates of [6, Part VI], recalled as (F1)–(F4) in Lemma 4.3; that manuscript is not yet refereed.

(B) Polyanskiy and Wu conjectured that a Gompertz location mixture with mixing law on $[-a, a]$ has $O(a)$ critical points [42, Sec. 5.2]. Theorem 4.8 shows that it can have at least $a(a - 4\log 2)/(2\pi^2) - 2$ local maxima. *Disproved.*

(C) Polyanskiy and Wu asked whether their $\Omega(k^2)$ lower bound for the number of modes of a mixture of k analytic strongly log-concave densities is tight [42, Sec. 5.5]. Theorem 4.12 shows that it is not. Already for $k = 2$, and with all second derivatives of the log-densities confined to $[-3/2, -1/2]$, the number of modes is unbounded. *Answered (negatively).*

(D) Musco, Musco, Rosenblatt and Singh asked whether the constant in their Chebyshev moment-matching bound $W_1 \leq 36/k + \Gamma$ can be improved "at least to $2\pi/k + \Gamma$, and possibly further" [43, Sec. 1.2 and 3.1]. Theorem 4.15 shows that $\pi/(k + 1) + \Gamma$ holds and that the constant π is asymptotically optimal. *Closed, but by classical approximation theory*: the upper bound is a direct application of Favard's 1937 theorem. We include it for completeness and do not count it as a new mathematical result.

Items (B) and (C) are short constructive disproofs. Item (A) is the main result. It reuses the likelihood-transport, Hellinger and Chebyshev–Bernstein estimates of [6, Part VI]. The new observation is that smoothing the fitted and true mixing laws at polynomial degree at most $t/2$ makes the effect of the unobservable components negligible (at most 2^{-m} after polynomial approximation). The W_1 distance between the smoothed laws is then purely statistical, and $W_p^p \le W_1$ on $[0,1]$ converts its parametric size into $N^{-1/(2p)}$ (Remark 4.6). Subsection 4.5 lists candidate questions that turned out to be already settled by others.

4.1. **Wasserstein-p minimax rates for the binomial NPMLE.**

Problem 4.1 (Vinayak, Kong, Valiant and Kakade [41, Sec. 5])**.** "Another direction of interest is to extend the analysis to provide guarantees in general Wasserstein-p norms."

Here "the analysis" is the minimax analysis, in Wasserstein-1 distance, of the nonparametric MLE of the population of coin biases. That analysis was completed for every (N,t) in [6, Part VI, Theorem p06:bm:full]. We answer the question for every finite p.

Setting. Let G be a probability measure on $[0,1]$. We observe N independent counts $X_1,\dots,X_N$, where $X_i \mid \theta_i \sim \mathrm{Bin}(t,\theta_i)$ and $\theta_i \sim G$ are independent. Put $B_j^t(x) = \binom{t}{j}x^j(1-x)^{t-j}$ and $p_H(j) = \int B_j^t \,\mathrm{d}H$ for $0 \le j \le t$. Let q be the empirical law of the counts. An NPMLE is any maximizer $\widehat{G}$ of $H \mapsto \sum_j q_j \log p_H(j)$ over all probability measures on $[0,1]$. Write $\widehat{p} = p_{\widehat{G}}$, $L_N = \log(eN)$,

$$r_{N,t} = \min\Big\{1, \frac{1}{t} + \frac{1}{\sqrt{tL_N}} + \frac{1}{\sqrt{N}}\Big\}, \qquad \rho_p(N,t) = \min\Big\{1, \frac{1}{t} + \frac{1}{\sqrt{tL_N}} + N^{-1/(2p)}\Big\}.$$

On the count grid we use the normalized transport distance $W_{1,t}$, with cost $|j-l|/t$, and the Hellinger distance $h(u,v)^2 = \sum_j(\sqrt{u_j}-\sqrt{v_j})^2$.

Theorem 4.2 (W_p-minimax optimality of the binomial NPMLE)**.** *Assume the facts (F1)–(F4) of Lemma 4.3, taken from* [6, Part VI]*. For every $p \in [1,\infty)$ there are constants $0 < c_p \le C_p < \infty$ such that, for all integers $N,t \ge 1$,*

$$c_p\,\rho_p(N,t) \;\le\; \inf_{\widetilde{G}} \sup_{G} \mathbb{E}_G W_p(\widetilde{G},G) \;\le\; \sup_{G} \mathbb{E}_G W_p(\widehat{G},G) \;\le\; C_p\,\rho_p(N,t). \tag{6}$$

The infimum is over all (possibly randomized) estimators. The upper bound holds for every measurable choice of an NPMLE.

For $p=1$ the theorem is [6, Part VI]. For $p>1$ the deconvolution terms $1/t + (tL_N)^{-1/2}$ are unchanged, and the parametric term becomes $N^{-1/(2p)}$. This term dominates once $t \gtrsim N^{1/p}/L_N$. As $p \to \infty$ it tends to 1, which is consistent with W_∞: there the minimax risk is bounded below by a positive constant for all N and t. Indeed δ_0 and $(1-\varepsilon)\delta_0 + \varepsilon\delta_1$ are at W_∞-distance one, and they cannot be told apart when $\varepsilon \ll 1/N$.

4.1.1. *Facts recalled from Part VI of* [6].

Lemma 4.3 (recalled)**.** *There is an absolute constant $C_0 \ge 1$ with the following properties.*

(F1) $\sup_G \mathbb{E}_G W_1(\widehat{G},G) \le C_0 r_{N,t}$ *for all $N,t \ge 1$ and every NPMLE. Also*

$$\inf_{\widetilde{G}} \sup_{G} \mathbb{E}_G W_1(\widetilde{G},G) \ge C_0^{-1} r_{N,t}.$$

([6, Part VI, Theorem p06:bm:full].)

(F2) *If $P = \sum_{j=0}^t b_j B_j^t$ is a polynomial of degree $k \le t/2$, then $\max_j |b_j| \le C_0 e^{2k^2/t}\|P\|_\infty$. If moreover $k \ge 1$, then $\max_{j<t} t|b_{j+1}-b_j| \le C_0 e^{4k^2/t}\|P'\|_\infty$. Here and below $\|\cdot\|_\infty$ is the supremum on $[0,1]$.* ([6, Part VI, Lemma p06:bm:approx]; the second bound is the first one applied to $P' = t\sum_{j<t}(b_{j+1}-b_j)B_j^{t-1}$.)

(F3) $\mathbb{E}_G W_{1,t}(\widehat{p}, p_G) \le C_0 N^{-1/2}$ *for all* N, t *and* G. ([6, Part VI, Lemma p06:bm:transport].)

(F4) *If* $1 \le t \le N$, *then* $\mathbb{E}_G h(p_G, \widehat{p}) \le C_0 t^{1/4} L_N^{3/4} N^{-1/2}$. ([6, Part VI, Corollary p06:bm:hellinger].)

We also use two elementary dualities on the count grid. Let u, v be probability vectors on $\{0, \dots, t\}$ and let $b \in \mathbb{R}^{t+1}$. Then

$$\Big|\sum_j b_j(u_j - v_j)\Big| \le \Big(\max_{j<t} t|b_{j+1} - b_j|\Big)\, W_{1,t}(u, v), \qquad \Big|\sum_j b_j(u_j - v_j)\Big| \le 2 \max_j |b_j|\, h(u, v). \tag{7}$$

The first inequality is Kantorovich–Rubinstein duality. The second follows from $\|u - v\|_1 \le h(u,v)\big(\sum_j (\sqrt{u_j} + \sqrt{v_j})^2\big)^{1/2} \le 2h(u, v)$.

4.1.2. *Reflected Jackson smoothings.* Let $\mathrm{fold}(z) = \mathrm{dist}(z, 2\mathbb{Z})$. This map sends $\mathbb{R}$ onto $[0, 1]$, it is 1-Lipschitz, and it is the identity on $[0, 1]$. For a real random variable U and a probability measure H on $[0, 1]$, let $S_U H$ be the law of $\mathrm{fold}(Y + U)$, where $Y \sim H$ is independent of U. Then S_U is a Markov operator, and $|\,\mathrm{fold}(Y + U) - Y| \le |U|$. Hence

$$W_q(H, S_U H) \le (\mathbb{E}|U|^q)^{1/q} \qquad (q \ge 1). \tag{8}$$

For integers $n \ge 1$ and $\rho \ge 2$ let

$$J_{n,\rho}(u) = c_{n,\rho}\Big(\frac{\sin(n\pi u/2)}{\sin(\pi u/2)}\Big)^{2\rho}, \qquad u \in [-1, 1],$$

normalized to be a probability density on $[-1, 1]$, and let $U_{n,\rho}$ have this density. Write $S_{n,\rho} = S_{U_{n,\rho}}$ and $\Lambda = \rho(n - 1)$.

Lemma 4.4. *Let* $U = U_{n,\rho}$ *and* $S = S_{n,\rho}$.

(i) *For* $0 < v \le 1$, $\mathbb{P}(|U| > v) \le (\pi/2)^{2\rho}(nv)^{1-2\rho}$. *Consequently, for* $1 \le q < 2\rho - 1$,

$$\mathbb{E}|U|^q \le \Big(1 + \frac{q(\pi/2)^{2\rho}}{2\rho - 1 - q}\Big) n^{-q}.$$

(ii) *For bounded measurable* f *on* $[0, 1]$, *the adjoint* $S^* f(y) = \mathbb{E} f(\mathrm{fold}(y + U))$ *is the restriction to* $[0, 1]$ *of a trigonometric polynomial* $\sum_{|l| \le \Lambda} c_l e^{i\pi l y}$ *with* $|c_l| \le \sup |f|$. *If* f *is* 1*-Lipschitz, then so is* $S^* f$, *and* $\|S^* f - f\|_\infty \le \mathbb{E}|U|$.

(iii) *Let* $g = \sum_{|l| \le \Lambda} c_l e^{i\pi l y}$ *and* $D \ge 9\Lambda + 1$. *Let* $\mathcal{T}_D[g]$ *be the polynomial obtained by replacing each* $e^{i\pi l y}$ *by its Taylor polynomial of degree* D *at* $y = 1/2$. *Then on* $[0, 1]$,

$$\|g - \mathcal{T}_D[g]\|_\infty \le (2\Lambda + 1) \max_l |c_l|\, 2^{-D-1}, \qquad \|g' - \mathcal{T}_D[g]'\|_\infty \le (2\Lambda + 1)\pi\Lambda \max_l |c_l|\, 2^{-D}.$$

If g *is real-valued, so is* $\mathcal{T}_D[g]$.

Proof. (i) For $|u| \le 1$ we have $|\sin(\pi u/2)| \ge |u|$, so $J_{n,\rho}(u) \le c_{n,\rho}|u|^{-2\rho}$. For $|u| \le 1/n$, concavity of sin on $[0, \pi/2]$ gives $|\sin(n\pi u/2)| \ge n|u|$. Also $|\sin(\pi u/2)| \le \pi|u|/2$, so the ratio is at least $2n/\pi$. Hence $c_{n,\rho}^{-1} \ge (2/n)(2n/\pi)^{2\rho}$. Therefore $\mathbb{P}(|U| > v) \le 2c_{n,\rho} \int_v^\infty u^{-2\rho}\, \mathrm{d}u \le (\pi/2)^{2\rho}(nv)^{1-2\rho}/(2\rho - 1)$. Then

$$\mathbb{E}|U|^q = \int_0^1 q v^{q-1} \mathbb{P}(|U| > v)\, \mathrm{d}v \le n^{-q} + q(\pi/2)^{2\rho} n^{1-2\rho} \int_{1/n}^\infty v^{q-2\rho}\, \mathrm{d}v,$$

which is the stated bound.

(ii) Fejér's identity $(\sin(n\pi u/2)/\sin(\pi u/2))^2 = \sum_{|l|<n}(n - |l|)e^{i\pi l u}$ shows that $J_{n,\rho}$ is a nonnegative even trigonometric polynomial of degree Λ in πu. Its coefficients $\hat{\jmath}_l = \frac{1}{2}\int_{-1}^1 J_{n,\rho}(u) e^{-i\pi l u}\, \mathrm{d}u$ satisfy $|\hat{\jmath}_l| \le \frac{1}{2}$. With $\tilde{f} = f \circ \mathrm{fold}$, which is even and 2-periodic, we get $S^* f(y) = \int_{-1}^1 \tilde{f}(y + u) J_{n,\rho}(u)\, \mathrm{d}u = \sum_l \hat{\jmath}_l e^{i\pi l y} \int_{-1}^1 \tilde{f}(s) e^{-i\pi l s}\, \mathrm{d}s$, using that $J_{n,\rho}$ is even. This proves the first claim. The

map $y \mapsto f(\mathrm{fold}(y+u))$ is 1-Lipschitz for each u, and so is its average. Finally $|S^*f(y) - f(y)| \le \mathbb{E}|\,\mathrm{fold}(y+U) - y| \le \mathbb{E}|U|$.

(iii) For real z, $|e^{iz} - \sum_{d\le D}(iz)^d/d!| \le |z|^{D+1}/(D+1)!$. Here $|z| = \pi|l||y - \frac12| \le \pi\Lambda/2$. Also $(D+1)! \ge ((D+1)/e)^{D+1}$ and $D+1 \ge e\pi\Lambda$, so the remainder is at most 2^{-D-1}. The derivative of the remainder is $i\pi l$ times the remainder of order $D-1$, which gives the second bound. If g is real, then $c_{-l} = \overline{c_l}$, the terms of indices $\pm l$ are complex conjugates, and so $\mathcal{T}_D[g]$ is real. □

4.1.3. *Smoothed W_1 errors of the NPMLE.* The next proposition is the two-scale inversion of [6, Part VI, Proposition p06:bm:twoscale], applied to smoothed measures. For these, the test functions are polynomials up to an exponentially small error, so the Jackson term C/k of that proposition disappears. This is the only place where the binomial data enter.

Proposition 4.5. *Let $1 \le t \le N$ and let $\rho \ge 2$ and $n \ge 1$, $n_s \ge 1$, $m \ge 1$ be integers. Put $\Lambda = \rho(n-1)$, $\Lambda_s = \rho(n_s - 1)$, $D = \lceil 9\Lambda\rceil + m + 1$ and $D_s = \lceil 9\Lambda_s\rceil + m + 1$, and assume $\max(D, D_s) \le t/2$. Then for every G and every NPMLE,*

$$W_1(S_{n,\rho}\widehat{G}, S_{n,\rho}G) \le 2C_0 e^{4D_s^2/t}\, W_{1,t}(\widehat{p}, p_G) + 2C_0 e^{2\max(D,D_s)^2/t}\big(\mathbb{E}|U_{n_s,\rho}| + 2^{-m}\big)h(\widehat{p}, p_G) + 2^{-m}.$$

Proof. Write $S = S_{n,\rho}$ and $S_s = S_{n_s,\rho}$. By Kantorovich–Rubinstein duality, $W_1(S\widehat{G}, SG) = \sup_f \int S^*f\,\mathrm{d}(\widehat{G} - G)$ over 1-Lipschitz f on $[0,1]$. Adding a constant to f does not change the integral, so we may take $f(0) = 0$; then $|f| \le 1$.

Put $g = S^*f$ and $g_s = S_s^*g$. By Lemma 4.4(ii), g is 1-Lipschitz, and it is a real trigonometric polynomial of degree Λ with coefficients of modulus at most 1. Similarly g_s is 1-Lipschitz, and it is a real trigonometric polynomial of degree at most Λ_s with coefficients of modulus at most 1. Moreover $\|g - g_s\|_\infty \le \mathbb{E}|U_{n_s,\rho}|$. Let $P = \mathcal{T}_D[g]$ and $Q = \mathcal{T}_{D_s}[g_s]$. These are real polynomials of degree at most D and D_s. Since $9\Lambda + m + 1 \le D$, Lemma 4.4(iii) gives $\|g - P\|_\infty \le (2\Lambda + 1)2^{-9\Lambda - m - 2} \le 2^{-m-2}$. In the same way $\|g_s - Q\|_\infty \le 2^{-m-2}$ and $\|Q'\|_\infty \le 1 + 2^{-m} \le 2$.

Every polynomial of degree at most t is a combination of the B_j^t, and $\int B_j^t\,\mathrm{d}(\widehat{G} - G) = \widehat{p}_j - p_G(j)$. We write

$$\int g\,\mathrm{d}(\widehat{G} - G) = \int (g - P)\,\mathrm{d}(\widehat{G} - G) + \int Q\,\mathrm{d}(\widehat{G} - G) + \int (P - Q)\,\mathrm{d}(\widehat{G} - G).$$

The first term is at most $2\|g - P\|_\infty \le 2^{-m-1}$. By (F2) and the first inequality of (7), the second term is at most $C_0 e^{4D_s^2/t} \cdot 2 \cdot W_{1,t}(\widehat{p}, p_G)$. For the third term, $\|P - Q\|_\infty \le \|g - g_s\|_\infty + 2^{-m-1}$. By (F2) and the second inequality of (7), the third term is therefore at most $2C_0 e^{2\max(D,D_s)^2/t}(\mathbb{E}|U_{n_s,\rho}| + 2^{-m})h(\widehat{p}, p_G)$. These bounds hold uniformly in f, and also for $-f$. □

4.1.4. *Proof of Theorem 4.2.*

Proof of the lower bound. Since $W_p \ge W_1$, (F1) gives $\inf\sup \mathbb{E}W_p \ge C_0^{-1} r_{N,t}$. For the parametric term, let $\delta = 1/(4\sqrt{N})$ and $G_\pm = (\frac12 \pm \delta)\delta_0 + (\frac12 \mp \delta)\delta_1$. Under $G_\pm$ every count equals 0 or t. So the sample is a vector of N independent $\{0, t\}$-valued labels with probabilities $\frac12 \pm \delta$. The Bhattacharyya coefficient of one label pair is at least $1 - 4\delta^2$. For the product laws it is therefore at least $(1-4\delta^2)^N \ge 1 - 4N\delta^2 = \frac34$, and their total variation is at most $\sqrt{1 - (3/4)^2} < 0.7$. Moreover $W_p(G_+, G_-) = (2\delta)^{1/p}$. Le Cam's two-point bound gives $\inf\sup \mathbb{E}W_p \ge \frac12(1 - 0.7)(2\delta)^{1/p} \ge c\,N^{-1/(2p)}$. Finally $\rho_p \le r_{N,t} + N^{-1/(2p)}$, and the maximum of two numbers is at least half their sum. □

Proof of the upper bound. Fix $p > 1$; the case $p = 1$ is (F1). Constants C may depend on p and change from line to line. We use two elementary facts. First, (8) holds. Second, $W_p^p(\mu, \nu) \le W_1(\mu, \nu)$ for probability measures on $[0,1]$: use a W_1-optimal coupling and $|x - y|^p \le |x - y|$.

Since $W_p \le 1$ and ρ_p is bounded below by a positive constant when t is bounded, we may assume $t \ge T_0(p)$ for a suitable $T_0(p)$.

Case $t > N$. For a law H on $[0,1]$, let $H^{\natural}$ be the law of X/t, where $X \mid \theta \sim \mathrm{Bin}(t,\theta)$ and $\theta \sim H$. Coupling θ with X/t, and using $\mathbb{E}|X/t-\theta|^q \le C_q t^{-q/2}$ for an even $q \ge p$, gives $W_p(H, H^{\natural}) \le Ct^{-1/2}$. Moreover $W_p^p(\widehat{G}^{\natural}, G^{\natural}) \le W_1(\widehat{G}^{\natural}, G^{\natural}) = W_{1,t}(\widehat{p}, p_G)$. By Jensen's inequality and (F3),

$$\mathbb{E}W_p(\widehat{G}, G) \le 2Ct^{-1/2} + (\mathbb{E}W_{1,t}(\widehat{p}, p_G))^{1/p} \le 2CN^{-1/2} + C_0^{1/p} N^{-1/(2p)} \le C\rho_p(N,t).$$

Case $T_0(p) \le t \le N$. Fix

$$\rho = \lceil p \rceil + 1, \qquad \gamma = \frac{1}{4(2p+1)}, \qquad c_1 = \min\Big\{\frac{1}{36\rho}, \frac{\sqrt{\gamma}}{18\rho}\Big\},$$

and set $k_0 = \lfloor c_1 \min\{t, \sqrt{tL_N}\} \rfloor$. For $t \ge T_0(p)$ we have $k_0 \ge c_1\sqrt{t}/2 \ge 1$. Let $m = \lceil p \log_2(k_0+1) \rceil + 2$ and $n_s = \lfloor \sqrt{t}/(40\rho) \rfloor$. Let $S = S_{k_0,\rho}$, so that $\Lambda = \rho(k_0-1)$ and $D = \lceil 9\Lambda \rceil + m + 1 \le 9\rho k_0 + m + 2$. By the triangle inequality, (8), $W_p^p \le W_1$ for the smoothed laws, and Jensen's inequality,

$$\mathbb{E}W_p(\widehat{G}, G) \le 2\big(\mathbb{E}|U_{k_0,\rho}|^p\big)^{1/p} + \big(\mathbb{E}W_1(S\widehat{G}, SG)\big)^{1/p}. \tag{9}$$

The smoothing cost. Since $2\rho - 1 \ge 2p+1 > p$, Lemma 4.4(i) gives $(\mathbb{E}|U_{k_0,\rho}|^p)^{1/p} \le C/k_0$.

Parameters. We have $D_s \le 9\rho n_s + m + 2 \le \sqrt{t}/4$ for $t \ge T_0(p)$, so $e^{4D_s^2/t} \le e^{1/4}$. Also $D \le t/2$, and

$$\frac{2D^2}{t} \le \frac{4(9\rho c_1)^2 \min\{t, \sqrt{tL_N}\}^2}{t} + \frac{4(m+2)^2}{t} \le \gamma L_N + 1.$$

So $e^{2\max(D,D_s)^2/t} \le e\,(eN)^{\gamma}$. By Lemma 4.4(i), $\mathbb{E}|U_{n_s,\rho}| \le C/\sqrt{t}$. Moreover $2^{-m} \le (k_0+1)^{-p}/4 \le k_0^{-1} \le C/\sqrt{t}$, because $k_0 \ge c_1\sqrt{t}/2$.

The smoothed W_1 error. Proposition 4.5, (F3) and (F4) give

$$\begin{aligned}\mathbb{E}W_1(S\widehat{G}, SG) &\le C\Big[N^{-1/2} + (eN)^{\gamma}\frac{1}{\sqrt{t}} \cdot \frac{t^{1/4}L_N^{3/4}}{\sqrt{N}} + k_0^{-p}\Big] \\ &= C\big[N^{-1/2} + (eN)^{\gamma} L_N^{3/4} t^{-1/4} N^{-1/2} + k_0^{-p}\big].\end{aligned}$$

If $t \ge (eN)^{4\gamma} L_N^3$, the middle term is at most $N^{-1/2}$. Otherwise $k_0 \le c_1\sqrt{tL_N} \le c_1 (eN)^{2\gamma} L_N^2$. Since $\gamma(1+2p) = 1/4$, the middle term divided by k_0^{-p} is then at most

$$e^{1/2} c_1^p\, (eN)^{\gamma(1+2p)-1/2} L_N^{2p+3/4} = e^{1/2} c_1^p (eN)^{-1/4} L_N^{2p+3/4} \le C.$$

Hence $\mathbb{E}W_1(S\widehat{G}, SG) \le C(k_0^{-p} + N^{-1/2})$.

Inserting both bounds into (9) gives

$$\mathbb{E}W_p(\widehat{G}, G) \le C\Big(\frac{1}{k_0} + N^{-1/(2p)}\Big), \qquad \frac{1}{k_0} \le \frac{2}{c_1}\max\Big\{\frac{1}{t}, \frac{1}{\sqrt{tL_N}}\Big\}.$$

Together with $W_p \le 1$, this proves the upper bound in (6). □

Remark 4.6. (1) *Mechanism.* The moment-limited error of a mixture estimate consists of components of polynomial degree above t. These are invisible to the data, and after smoothing at degree at most $t/2$ their effect is at most 2^{-m}. So $W_1(S\widehat{G}, SG)$ is purely statistical, and the inequality $W_p^p \le W_1$ on the unit interval turns its parametric size $N^{-1/2}$ into $N^{-1/(2p)}$. The smoothing itself costs only $(\mathbb{E}|U|^p)^{1/p} \asymp 1/k_0$ in W_p. Applying $W_p^p \le W_1$ directly to $(\widehat{G}, G)$ would only give the weaker rate $(1/k_0 + N^{-1/2})^{1/p}$.

(2) The proof uses only (F1)–(F4) and deterministic facts, so it applies to every estimator for which analogues of (F1), (F3) and (F4) hold. This includes approximate NPMLEs within N^{-1} of the optimal log-likelihood [6, Part VI].

(3) A Monte Carlo check (script `R5_wp_npmle_sim.py`, with the NPMLE computed by EM on a grid of 2001 points) is consistent with (6). For $G = \frac{1}{2}(\delta_{0.2} + \delta_{0.8})$, $t = 4000$ and $N = 100, \dots, 6400$, the fitted log–log slopes of $\mathbb{E}W_p$ in N are -0.54, -0.28 and -0.18 for $p = 1, 2, 3$. The predictions are $-1/2$, $-1/4$ and $-1/6$, and the dependence of the slope on p is the new parametric term $N^{-1/(2p)}$. For uniform G, $N = 20000$ and $t = 10, 40, 160, 640$, the ratios $\mathbb{E}W_p/\mathbb{E}W_1$ lie between 1.2 and 1.45 for $p = 2, 3$. So for this smooth G the dependence on t is that of the case $p = 1$, and the worst-case term $N^{-1/(2p)}$ is not visible. The products $\mathbb{E}W_p \cdot \sqrt{tL_N}$ lie between 0.18 and 0.50. They rise at $t = 640$, where $N^{-1/2}$ is already about $0.6\,(tL_N)^{-1/2}$. These experiments use an EM approximation of the NPMLE (KKT gaps up to $1.8 \cdot 10^{-2}$) and only illustrate the theorem.

(4) As in [41] and [6], the estimation target is the mixing law G. If the target is instead the empirical law of $\theta_1, \dots, \theta_N$, the upper bound is unchanged, by the triangle inequality and $\mathbb{E}W_p(G, N^{-1}\sum_i \delta_{\theta_i}) \le CN^{-1/(2p)}$. For that target, however, the two-point lower bound of order $N^{-1/(2p)}$ need not hold.

4.2. **The Gompertz critical-point conjecture of Polyanskiy and Wu is false.** Polyanskiy and Wu [42] prove that the NPMLE of a Gaussian location mixture self-regularizes: for subgaussian data it has $O(\log n)$ atoms. For a mixture of exponential distributions they pose the open question of whether the NPMLE has $\Theta(\log n)$ atoms with high probability when the mixing law is supported in $[1, 2]$ [42, Sec. 5.2, eq. (37)]. They reduce the upper bound to an analytic conjecture.

Conjecture 4.7 (Polyanskiy and Wu [42, Sec. 5.2])**.** *"The crucial $O(\log n)$ upper bound would follow from the following analytic* conjecture*: For any distribution π on $[-a, a]$ the convolution $(\pi * h)(x) \triangleq \int h(x-y)\pi(dy)$ has at most $O(a)$ critical points, where $h(x) = e^{-e^x + x}$ is the density of a Gompertz distribution."*

Theorem 4.8. *Let $h(x) = e^{x - e^x}$. For every $a \ge 4$ let $\omega_a = (a - 4\log 2)/\pi$ and let π_a be the probability measure on $[-a, a]$ with density proportional to $1 + \sin(\omega_a y)$. Then $\pi_a * h$ has at least*

$$\Big\lfloor \frac{a(a - 4\log 2)}{2\pi^2} \Big\rfloor - 1$$

strict local maxima in $(-a/2, a/2)$, and each of them is a critical point. In particular Conjecture 4.7 is false: no bound of the form Ca can hold for all large a.

For the Gaussian kernel the analogous question had a similar history. A conjectured $O(a)$ bound was refuted by Kashyap and Krishnapur [48], and [42, Sec. 5.4] gives a short proof of an $\Omega(a^2)$ lower bound. The disproof is a direct corollary of the lemma Polyanskiy and Wu use to show that the $O(a^2)$ bound for Gaussian kernels is sharp [42, Lemma 9], which already applies to arbitrary continuous densities, combined with $|\Gamma(1 - i\omega)|^2 = \pi\omega/\sinh(\pi\omega)$: the Gompertz kernel has an exponentially decaying Fourier transform and an exponential left tail, and these balance to give quadratic growth. For completeness we include a short proof of the lemma in the form we use.

Lemma 4.9 (Sinusoidal windows)**.** *Let h be a bounded continuous probability density on $\mathbb{R}$ with a bounded continuous derivative. Let H be its distribution function and $\hat{h}(\omega) = \int e^{-i\omega s} h(s)\,\mathrm{d}s$. Let $a, \omega > 0$ and suppose*

$$A := |\hat{h}(\omega)| > 2T, \qquad T := H(-a/2) + 1 - H(a/2).$$

Let π have density proportional to $(1 + \sin(\omega y))\mathbf{1}_{[-a,a]}(y)$. Then $\pi * h$ is C^1 and has at least $\lfloor a\omega/(2\pi) \rfloor - 1$ local maxima in $(-a/2, a/2)$, each of them a critical point.

Proof. Let Z be the normalizing constant, so that $Z\,(\pi * h)(x) = g(x) - \Delta(x)$ with $g(x) = \int_{\mathbb{R}}(1+\sin\omega y)h(x-y)\,\mathrm{d}y$ and $\Delta(x) = \int_{|y|>a}(1+\sin\omega y)h(x-y)\,\mathrm{d}y$. Write $\hat{h}(\omega) = Ae^{i\beta}$. Then $\int \sin(\omega y)h(x-y)\,\mathrm{d}y = \mathrm{Im}\big(e^{i\omega x}\hat{h}(\omega)\big)$, so $g(x) = 1 + A\sin(\omega x+\beta)$. For $|x| \le a/2$,

$$0 \le \Delta(x) \le 2\mathbb{P}_{S\sim h}(|x-S|>a) = 2\big[H(x-a) + 1 - H(x+a)\big] \le 2T.$$

Let $u_m = (\pi/2 - \beta + 2\pi m)/\omega$ and $v_m = u_m + \pi/\omega$. If $u_m \in [-a/2, a/2]$, then $Z(\pi*h)(u_m) \ge 1 + A - 2T > 1$. If $v_m \in [-a/2, a/2]$, then $Z(\pi*h)(v_m) \le 1 - A < 1$. Consider an interval $[v_{m-1}, v_m] \subset [-a/2, a/2]$. It contains u_m, and the continuous function $\pi * h$ is larger at u_m than at both endpoints. So its maximum over the interval is attained at an interior point, which is a local maximum on $\mathbb{R}$. Distinct m give disjoint interiors. The points v_m have spacing $2\pi/\omega$, so at least $\lfloor a\omega/(2\pi)\rfloor - 1$ such intervals lie in $[-a/2, a/2]$. Differentiation under the integral shows that $\pi * h \in C^1$, so local maxima are critical points. □

Proof of Theorem 4.8. The density $h(x) = e^{x-e^x}$ is that of $\log E$ with $E \sim \mathrm{Exp}(1)$. It is smooth with bounded derivatives, and its distribution function is $H(x) = 1 - \exp(-e^x)$. Substituting $u = e^s$ gives $\hat{h}(\omega) = \int_0^\infty u^{-i\omega}e^{-u}\,\mathrm{d}u = \Gamma(1-i\omega)$. From $\Gamma(1+z)\Gamma(1-z) = \pi z/\sin(\pi z)$ with $z = i\omega$, and $\Gamma(1+i\omega) = \overline{\Gamma(1-i\omega)}$, we get

$$|\hat{h}(\omega)|^2 = \frac{\pi\omega}{\sinh(\pi\omega)} \ge 2\pi\omega\, e^{-\pi\omega}.$$

Next, $H(-a/2) = 1 - \exp(-e^{-a/2}) \le e^{-a/2}$ and $1 - H(a/2) = \exp(-e^{a/2}) \le e^{-a/2}$, so $T \le 2e^{-a/2}$. With $\omega = \omega_a$ we have $e^{-\pi\omega/2} = 4e^{-a/2}$. Hence $|\hat{h}(\omega_a)| \ge 4\sqrt{2\pi\omega_a}\,e^{-a/2} > 4e^{-a/2} \ge 2T$ whenever $2\pi\omega_a > 1$, which holds for $a \ge 4$. Lemma 4.9 gives at least $\lfloor a\omega_a/(2\pi)\rfloor - 1$ local maxima, as claimed. The function $\pi_a * h$ is real-analytic and nonconstant, so these local maxima are strict. If the number of critical points were at most Ca for all large a, we would need $a(a - 4\log 2)/(2\pi^2) - 2 \le Ca$ for all large a, which is impossible. □

Remark 4.10. (1) *Numerical confirmation.* The script `R5_gompertz_check.py` evaluates $\pi_a * h$ exactly through incomplete Gamma functions, in 60-digit arithmetic, and cross-checks the formula against direct quadrature. Counting local maxima in $(-a/2, a/2)$ on a fine grid gives $2, 6, 11, 26, 47, 110$ for $a = 8, 12, 16, 24, 32, 48$. The proved bounds are $1, 4, 9, 24, 46, 108$, and $a^2/(2\pi^2)$ takes the values $3, 7, 13, 29, 52, 117$.

(2) *A better constant.* The window can also be placed asymmetrically, using the double-exponential right tail of h. With $\omega \approx 2a/\pi$ the numerics give about a^2/π^2 local maxima (for example 137 at $a = 36$). We do not prove this refinement.

(3) *What remains of the open question.* Riemann sums converge uniformly to $\pi_a * h$, and the local maxima produced by Lemma 4.9 have a quantitative margin $A - 2T$. Hence the theorem also holds for finite positive combinations $\sum_i w_i h(x - y_i)$ with $y_i \in [-a, a]$. Through $\theta = e^{-x}$ and $x_i = e^{y_i}$, these are, up to reflection of h, exactly the gradient functions $\theta \mapsto \sum_i w_i \theta e^{-\theta x_i}$ whose critical points carry the atoms of the exponential-mixture NPMLE [42, eq. (35)]. So the deterministic route proposed in [42] cannot give $O(\log n)$: for suitable positive weights it allows $\Omega((\log(x_{\max}/x_{\min}))^2)$ critical points. The open question itself is not resolved by this. Its upper bound would have to use the special NPMLE weights $w_i \propto 1/\widehat{f}(x_i)$ or the randomness of the data. We also do not know whether $O(a^2)$ is the correct upper bound for Gompertz mixtures.

4.3. **Mixtures of two analytic strongly log-concave densities can have arbitrarily many modes.**

Problem 4.11 (Polyanskiy and Wu [42, Sec. 5.5])**.** The authors observe that mixtures of two unimodal, or even log-concave, densities can have infinitely many modes, and that C^∞ strongly log-concave examples with flat pieces exist. They then ask: given a convex combination of k

analytic densities that are strongly log-concave, how many modes can the mixture have? Their Corollary 11 gives $\Omega(k^2)$ modes, and they write: "Whether this is tight is an open question."

A density f is c-strongly log-concave if $(\log f)'' \le -c < 0$.

Theorem 4.12. *Let $\varepsilon \in (0,1)$ and $\omega > 0$, and define*

$$f_\pm(x) = Z^{-1} \exp\Big(-\frac{(x \mp 1)^2}{2} + \frac{\varepsilon}{\omega^2}\cos(\omega x)\Big), \qquad m = \tfrac{1}{2}(f_+ + f_-).$$

Both functions use the same constant Z. Then:

(i) *f_+ and f_- are entire probability densities with $-1-\varepsilon \le (\log f_\pm)'' \le -1+\varepsilon$. So they are $(1-\varepsilon)$-strongly log-concave and $(1+\varepsilon)$-log-smooth, uniformly in ω.*

(ii) *The mixture m has at least $\frac{1}{\pi}\big(\frac{3\varepsilon}{2}\big)^{1/3}\omega^{2/3} - \frac{5}{2}$ strict local maxima.*

Consequently, for every $k \ge 2$ and every M there is a mixture of k distinct analytic, $\frac{1}{2}$-strongly log-concave densities with at least M modes. The $\Omega(k^2)$ bound is therefore not tight: no function of k bounds the number of modes.

Proof. (i) The substitution $x \mapsto -x$ and the evenness of cos show that the two normalizing integrals coincide. The exponents are entire functions of x and are bounded above by $-(x \mp 1)^2/2 + \varepsilon/\omega^2$, so $f_\pm$ are integrable entire functions. Directly, $(\log f_\pm)'' = -1 - \varepsilon\cos(\omega x)$.

(ii) Because the factor $e^{(\varepsilon/\omega^2)\cos\omega x}$ is common to both components,

$$\log m(x) = \text{const} + \frac{\varepsilon}{\omega^2}\cos(\omega x) - \frac{x^2}{2} + \log\cosh x, \qquad (\log m)'(x) = -\frac{\varepsilon}{\omega}\sin(\omega x) - (x - \tanh x).$$

For $x \ge 0$ we have $0 \le x - \tanh x \le x^3/3$. Indeed, the function $\tanh x - x + x^3/3$ vanishes at 0 and has derivative $x^2 - \tanh^2 x \ge 0$. Let $X = (3\varepsilon/(2\omega))^{1/3}$, so that $X^3/3 = \varepsilon/(2\omega)$, and let $x_j = (\pi/2 + j\pi)/\omega$ for $j \ge 0$. If $x_j \le X$, then $\sin(\omega x_j) = (-1)^j$ and $0 \le x_j - \tanh x_j \le \varepsilon/(2\omega)$. Hence $(\log m)'(x_j)$ is negative for even j and positive for odd j. On each interval $[x_{2i+1}, x_{2i+2}] \subset (0, X]$, the function m increases at the left end and decreases at the right end, so it has an interior local maximum. These intervals are disjoint. Let $J = \#\{j \ge 0 : x_j \le X\}$. Then $J \ge \omega X/\pi - \frac{1}{2}$, and $(0, X]$ contains at least $\lfloor (J-1)/2 \rfloor$ such maxima. Since m is even, the same holds on $[-X, 0)$. Hence there are at least $J - 2 \ge \omega X/\pi - \frac{5}{2}$ local maxima. The function m is real-analytic and nonconstant, so its critical points are isolated and its local maxima are strict.

For $k = 2$ the mixture m itself, with $\varepsilon = \frac{1}{2}$ and ω large, already has at least M modes. For $k \ge 3$, take $\varepsilon = \frac{1}{2}$, choose ω so large that m has at least M strict local maxima in $[-1, 1]$, and let $\delta \in (0,1)$. Consider $(1-\delta)m + \frac{\delta}{k-2}\sum_{i=3}^{k}\phi(\cdot - 10i)$, where ϕ is the standard normal density. This is a mixture of k distinct analytic $\frac{1}{2}$-strongly log-concave densities. On $[-1, 1]$, the function m is bounded below and the added term is $O(\delta)$ in C^1. At the finitely many points x_j used above, the strict sign conditions on m' therefore persist for small δ, and so do the M local maxima. □

Remark 4.13. (1) The proof locates the extra modes near the degenerate critical point $x = 0$ of the equal mixture of $N(\pm 1, 1)$, at which $(\log m)'(x) \sim -x^3/3$. Analytic wiggles of amplitude ε/ω^2 in the log-densities change the second derivatives by at most ε, but they create about $\omega^{2/3}$ modes. The script `R5_modes_check.py` and an independent high-precision recount give $9, 37, 169, 785$ strict local maxima for $\omega = 10^2, 10^3, 10^4, 10^5$ with $\varepsilon = \frac{1}{2}$ (an odd number, as it must be: m is even and $(\log m)''(0) = -\varepsilon < 0$).

(2) The construction needs $|(\log f)'''|$ of order $\varepsilon\omega$. If one also imposes a uniform bound on the third derivative of $\log f_i$, the question becomes different, and we leave it open. The components in the construction behind [42, Cor. 11] are of a special kind: Gaussian convolutions of laws on consecutive intervals of length $1/4$. If all components are Gaussian convolutions of laws supported

in a common interval of length $O(k)$, the mixture is a Gaussian location mixture, and the $O(a^2)$ bound of [42, Sec. 5.4] shows that $\Theta(k^2)$ is the correct order in that structured class.

4.4. **The optimal constant in Chebyshev moment matching.**

Problem 4.14 (Musco, Musco, Rosenblatt and Singh [43])**.** For distributions p, q on $[-1, 1]$ whose Chebyshev moments satisfy $\sum_{j=1}^{k} j^{-2}(\mathbb{E}_p T_j - \mathbb{E}_q T_j)^2 \le \Gamma^2$, the authors prove $W_1(p, q) \le 36/k + \Gamma$ and write: "we believe the constants can be improved, at least to $\frac{2\pi}{k} + \Gamma$, and possibly further" [43, footnote 3 and Sec. 3.1]. The 36 is twice the constant 18 in their Jackson-type Fact 12. Citing [44, App. C.2], they add that "strong numerical evidence suggests that this 18 can be improved to π".

Theorem 4.15. *Let $k \ge 1$.*

(a) *(Favard's damping.) Let $\lambda_j = \frac{j\pi}{2(k+1)} \cot \frac{j\pi}{2(k+1)}$ for $1 \le j \le k$ and $\lambda_0 = 1$. Then $1 = \lambda_0 > \lambda_1 > \cdots > \lambda_k > 0$. For every ℓ-Lipschitz f on $[-1, 1]$ with Chebyshev series $\sum_j a_j T_j$, the damped truncation $f_k = \sum_{j \le k} \lambda_j a_j T_j$ satisfies $\|f - f_k\|_\infty \le \pi\ell/(2(k + 1))$.*

(b) *Under the hypothesis of Problem 4.14, $W_1(p, q) \le \pi/(k + 1) + \Gamma$.*

(c) *The constant π in (b) cannot be lowered. For every $k \ge 4$ there are distributions p, q on $[-1, 1]$ with the same first k Chebyshev moments (so $\Gamma = 0$) and $W_1(p, q) \ge \frac{\pi}{k}(1 - 5k^{-1/2})$.*

(d) *(The damping of [44].) For the Jackson coefficients of [44] ($N = 4z$, coefficients $(g*g)*(g*g)$ with $g = \mathbf{1}_{2z+1}$, normalized), every 1-Lipschitz f on $[-1, 1]$ satisfies $\|f - \bar{f}_N\|_\infty \le \pi/N$, for every $z \ge 1$. The best constant $c_N = N \sup_f \|f - \bar{f}_N\|_\infty$ tends to $12 \log 2/\pi \approx 2.648$.*

Remark 4.16 (Status). Part (a) is Favard's theorem for W_∞^1 [45, 46] (see also [47, Ch. 4]), transported to $[-1, 1]$ by $x = \cos\theta$. Part (c) is the matching lower bound, which reduces to the classical asymptotics of best approximation of Lipschitz functions at the centre of the interval. So the answer, "the optimal constant is π", follows from known results, and we count this item as *settled by classical theory* rather than as a new theorem. The only new content is part (d), which confirms rigorously the numerical observation of [44], and the self-contained proofs below.

Proof. (a) Let $n = k$ and $N' = n+1$. For $s \in (0, \pi/2)$ the function $s \cot s$ is decreasing with values in $(0, 1)$, which gives the ordering of the λ_j. Let $B(\theta) = \pi - \theta = 2\sum_{j \ge 1} \sin(j\theta)/j$ on $(0, 2\pi)$, and let $\sigma(\theta) = 2\sum_{j=1}^{n} \lambda_j \sin(j\theta)/j$. A direct computation with the discrete sine transform shows that σ is the unique sine polynomial of degree n with $\sigma(j\pi/N') = B(j\pi/N')$ for $1 \le j \le 2n+1$. Indeed $\frac{2}{N'} \sum_{i=1}^{N'-1} (\pi - \frac{i\pi}{N'}) \sin \frac{ij\pi}{N'} = \frac{\pi}{N'} \cot \frac{j\pi}{2N'} = 2\lambda_j/j$. Put $\Theta = B - \sigma$ on $(0, 2\pi)$. It vanishes at the $2n+1$ nodes, while $\Theta(0^+) = \pi$ and $\Theta(2\pi^-) = -\pi$. Its derivative $-1 - \sigma'$ is a trigonometric polynomial of degree n, so it has at most $2n$ zeros per period. By Rolle's theorem these are exhausted between consecutive nodes. Hence Θ has no other zeros, they are all simple, and $\operatorname{sgn}\Theta = \operatorname{sgn}\sin(N'\theta)$. Since $\operatorname{sgn}\sin(N'\theta) = \frac{4}{\pi}\sum_{m \text{ odd}} \sin(mN'\theta)/m$ is orthogonal to σ,

$$\int_0^{2\pi} |\Theta| = \int_0^{2\pi} B(\theta) \operatorname{sgn}\sin(N'\theta)\, \mathrm{d}\theta = \sum_{m \text{ odd}} \frac{2}{mN'} \cdot \frac{4}{\pi m} \cdot \pi = \frac{\pi^2}{N'}.$$

Now let g be 2π-periodic with $|g'| \le \ell$ almost everywhere, and let $F_n g = \sum_j \lambda_j (a_j \cos j\theta + b_j \sin j\theta)$. Integration by parts gives $(g - F_n g)(0) = -\frac{1}{2\pi}\int_0^{2\pi} g'(s)\Theta(s)\,\mathrm{d}s$, so $\|g - F_n g\|_\infty \le \ell\pi/(2N')$ by translation invariance. For f Lipschitz on $[-1, 1]$, the function $g(\theta) = f(\cos\theta)$ satisfies $|g'| \le \ell$, and $F_n g(\theta) = f_k(\cos\theta)$.

(b) This is the proof of [43, Thm. 1] with Fact 12 replaced by (a). That proof uses Fact 12 only through $|\lambda_j| \le 1$ and the uniform error bound; together with their Lemma 13 it gives $W_1 \le \Gamma + 2\|f - f_k\|_\infty$.

(c) Let $D = \lceil \sqrt{k} \rceil$, $N' = k + 1 + 2D$, $\theta_j = j\pi/N'$, $x_j = \cos\theta_j$ $(0 \le j \le N')$, and let w_j be the Chebyshev–Lobatto weights ($1/N'$ inside, $1/(2N')$ at the ends). The Lobatto rule is exact

for polynomials of degree at most $2N'-1$ against the arcsine law, and $T_{N'}(x_j)=(-1)^j$. Hence the signed measure $\mu=\sum_j w_j(-1)^j(1-x_j^2)^D\delta_{x_j}$ annihilates every polynomial of degree at most $N'-1-2D=k$. So $\mu=\frac{\|\mu\|}{2}(p-q)$ for probability measures p,q with the same first k moments.

Let $s_j=\frac12\min(x_{j-1}-x_j,x_j-x_{j+1})$ for $0<j<N'$, and let f be the piecewise linear function with $f(x_j)=(-1)^j s_j$ and $f(x_0)=f(x_{N'})=0$. This f is 1-Lipschitz, and

$$W_1(p,q)\ge\frac{2}{\|\mu\|}\int f\,\mathrm{d}\mu=\frac{2\sum_j w_j\sin^{2D}\theta_j\,s_j}{\sum_j w_j\sin^{2D}\theta_j}.$$

Here

$$s_j=\sin\frac{\pi}{2N'}\Big(\sin\theta_j\cos\frac{\pi}{2N'}-|\cos\theta_j|\sin\frac{\pi}{2N'}\Big).$$

Lobatto exactness gives $\sum w_j\sin^{2D+2}\theta_j/\sum w_j\sin^{2D}\theta_j=(2D+1)/(2D+2)$, and $\sin\theta\ge\sin^2\theta$. Therefore $W_1(p,q)\ge\sin\frac{\pi}{N'}\cdot\frac{2D+1}{2D+2}-2\sin^2\frac{\pi}{2N'}$. For $k\ge4$ this is at least $\frac{\pi}{k}(1-5k^{-1/2})$, by elementary estimates with $N'\le k+3+2\sqrt{k}$.

(d) The BKM kernel is $K(\theta)=C_\nu^{-1}(\sin(\nu\theta/2)/\sin(\theta/2))^4$ with $\nu=2z+1$ and $C_\nu=\nu(2\nu^2+1)/3$. It has degree $N=2\nu-2$ and is nonnegative. For a nonnegative kernel the worst 1-Lipschitz f at $x_0=\cos\theta_0$ is $|x-x_0|$. Moreover $|\cos\theta_0-\cos(\theta_0-s)|\le2|\sin(s/2)|$. Hence

$$\sup_f\|f-\bar f_N\|_\infty\le\frac{1}{2\pi}\int_{-\pi}^{\pi}2|\sin\tfrac{s}{2}|K(s)\,\mathrm{d}s=\frac{4}{\pi C_\nu}I(\nu),\qquad I(\nu)=\int_0^{\pi/2}\frac{\sin^4(\nu u)}{\sin^3u}\,\mathrm{d}u.$$

On $(0,1]$ we have $\sin^{-3}u\le u^{-3}+u^{-1}$, because $\sin u\ge u(1-u^2/6)$ and $(1-u^2/6)^{-3}\le(1-u^2/2)^{-1}\le1+u^2$. On $[1,\pi/2]$ we have $\sin^{-3}u\le1.678$. Two integrations by parts and Frullani's integral give $\int_0^\infty\sin^4v\,v^{-3}\,\mathrm{d}v=\int_0^\infty(\cos2v-\cos4v)v^{-1}\,\mathrm{d}v=\log2$, since $(\sin^4)''(v)=2\cos2v-2\cos4v$. Also $\int_0^1\sin^4(\nu u)u^{-1}\,\mathrm{d}u\le\frac14+\log\nu$. Together these give $I(\nu)\le\nu^2\log2+\log\nu+1.21$. Since $(2\nu-2)/(\nu(2\nu^2+1))\le\nu^{-2}$,

$$N\sup_f\|f-\bar f_N\|_\infty\le\frac{12}{\pi}\Big(\log2+\frac{\log\nu+1.21}{\nu^2}\Big)\le3.08<\pi\qquad(\nu\ge5).$$

For $\nu=3$ ($N=4$), exact evaluation gives 1.807.

The displayed bound also gives $\limsup c_N\le12\log2/\pi$. For the reverse inequality, test $f=|x|$ at $\theta_0=\pi/2$. Since $|\sin s|\ge2|\sin\frac{s}{2}|-|s|^3/8$, we get $c_N\ge N\big(\frac{4}{\pi C_\nu}I(\nu)-\frac{1}{16\pi}\int_{-\pi}^{\pi}|s|^3K(s)\,\mathrm{d}s\big)$. Moreover $I(\nu)\ge\nu^2\log2-2/\pi^2$. The kernel satisfies $K(s)\le C\nu(\nu|s|)^{-4}$, so $\int|s|^3K=O(\nu^{-3}\log\nu)$. Hence $\liminf c_N\ge12\log2/\pi$. The script `R5_jackson_check.py` evaluates the sharp constant for $N\le192$: it increases from 1.635 to 2.620. □

4.5. **Candidates examined and found settled elsewhere.** The following explicitly posed questions in this area were checked and found to be already answered by other authors, so they were not pursued.

- Polyanskiy–Wu conjecture that the statistical degree of Gaussian mixtures with compactly supported mixing law is $\asymp\log n/\log\log n$ [42, Sec. 5.3]: resolved by Ma, Wu and Yang [49].
- Polyanskiy–Wu conjecture that the multivariate Gaussian NPMLE is unique [42, Sec. 5.6]: non-uniqueness for an equilateral triangle in $\mathbb{R}^2$ [50, Lemma 2], and uniqueness for almost every data set [51].
- Minimax rates for smooth binomial mixing densities with heterogeneous trials, stated as open in [52, Sec. 2.3, Sec. 6]: determined by Lee and Kennedy [53]. Adaptivity to the smoothness remains listed there as future work.

- The Vinayak et al. Chebyshev–Bernstein coefficient conjecture and the binomial NPMLE W_1 range, as well as Poisson-mixture rates, are treated in earlier parts of this programme [6]. The fixed-test "fingerprint" conjecture of [41, Sec. 5.1] remains open and is not addressed here.

5. Wronskians of polygamma products and two degree questions

This section settles several explicitly posed guesses and questions of F. Qi and coauthors on complete monotonicity (CM) of functions built from polygamma functions, answers one CM-degree question from the survey [56] and partly answers another. All proofs follow the pattern of the degree theorem for Stirling remainders ([9, Part VIII]): write the function as a Laplace transform, reduce CM to the sign of an explicit density (Bernstein's theorem plus uniqueness), and decide that sign. Monotonicity of ratios of Laplace transforms is standard; what we add for the guesses is a likelihood-ratio argument for convolution densities, driven by the log-concavity of the Bernoulli kernel $c(t) = t/(1-e^{-t})$ and of its derivative, and by the concavity of $tc'(t)/c(t)$.

Summary of outcomes.

- Qi, C. R. Math. 2022, Remark 17 [54]: guess *true*; the attached necessary-and-sufficient-condition problem is solved with explicit constants (Theorem 5.9). We point out (Remark 5.10) that the whole remark also follows from Qi's own Theorem 1 in [54] by a splitting identity, so this closure is easy rather than deep.
- Qi, C. R. Math. 2022, Remark 18 [54]: guess *true* (Theorem 5.12). Attached problem: one half solved explicitly ($\Omega \geq 1$); the other half is reduced to an explicit one-variable minimum $\Omega_* \in (0,1)$, computed numerically (*partial* for that half: no closed form).
- Yin–Han–Qi, Demonstr. Math. 2024, Remark 4 [55]: guess *true* (Theorem 5.15). Attached problem: one half solved ($\Lambda \leq 1$), the other half reduced to an explicit maximum $\Lambda^*_{m,n} > 1$ (*partial*: no closed form). As for Remark 17, the guess itself can also be obtained by splitting from a published majorization theorem (Remark 5.16(2)); our proof is independent of it.
- Yin–Han–Qi, Remark 2 [55]: both sign guesses *true*, with the complete sign pattern (Proposition 5.18; elementary).
- Qi–Agarwal 2019, Section 5.4 [56]: $\deg_{\mathrm{cm}} H_2 = 0$ and $\deg_{\mathrm{cm}} H_3 = 1$ (Theorem 5.22).
- Qi–Agarwal 2019, Section 5.5 [56]: the negative has degree exactly 1 for every $\mu \geq 4$; for $\lambda = 0$ the degree is 0; for every $\lambda < 0$ the degree is a non-integer in $(0,1)$, nonincreasing in λ, tending to 1 and to 0 at the two ends (Theorem 5.23). For $\lambda < 0$ the degree is therefore located in $(0,1)$ but not determined, so this question is answered only *partially*.
- Qi, Math. Inequal. Appl. 2021, Remark 4.1 [59]: *partial* only (Proposition 5.25).

5.1. **Notation and general tools.** A function $f\colon (0,\infty) \to \mathbb{R}$ is *completely monotonic* (CM) if $f \in C^\infty$ and $(-1)^k f^{(k)} \geq 0$ for all $k \geq 0$. Products, sums, nonnegative multiples and pointwise limits (with all derivatives) of CM functions are CM. Following [56], for a CM function $f \not\equiv 0$ with $f(\infty) = 0$ (the only case needed here; in general [56] subtracts $f(\infty)$) the *completely monotonic degree* is

$$\deg_{\mathrm{cm}} f = \sup\{\alpha \in \mathbb{R} :\ x^\alpha f(x) \text{ is CM on } (0,\infty)\}.$$

Lemma 5.1. *Let $f \not\equiv 0$ be CM. The set $A_f = \{\alpha :\ x^\alpha f$ is CM$\}$ is of the form $(-\infty, d]$ or $(-\infty,\infty)$, and $d = \deg_{\mathrm{cm}} f$ when it is finite. If $x^\alpha f(x) \to 0$ as $x \to 0^+$, then $\alpha \notin A_f$.*

Proof. If $\alpha \in A_f$ and $\beta < \alpha$, then $x^\beta f = x^{\beta-\alpha} \cdot x^\alpha f$ is a product of CM functions. If $\alpha_k \uparrow \alpha$ with $\alpha_k \in A_f$, then $x^{\alpha_k} f \to x^\alpha f$ locally uniformly on $(0,\infty)$ together with all derivatives, so the sign conditions pass to the limit. Finally, a CM function is nonnegative and nonincreasing; if it tends to 0 at 0^+ it vanishes identically, which is impossible for $x^\alpha f$. □

Lemma 5.2 (Laplace criterion)**.** *Let $\varphi\colon (0,\infty)\to\mathbb{R}$ be continuous with $\int_0^\infty |\varphi(t)|e^{-xt}\,dt<\infty$ for every $x>0$, and let $F(x)=\mathcal{L}\varphi(x)=\int_0^\infty \varphi(t)e^{-xt}dt$. Then F is CM on $(0,\infty)$ if and only if $\varphi\ge 0$ on $(0,\infty)$. If $\varphi\ge 0$ and $\varphi\not\equiv 0$, then $(-1)^kF^{(k)}>0$ for all k.*

Proof. If $\varphi\ge 0$, differentiation under the integral sign (dominated convergence) gives $(-1)^kF^{(k)}(x)=\int_0^\infty t^k\varphi(t)e^{-xt}dt$. Conversely, by Bernstein's theorem $F(x)=\int_{[0,\infty)}e^{-xt}d\mu(t)$ with a positive measure μ. For fixed $a>0$ the finite signed measures $e^{-at}\varphi(t)dt$ and $e^{-at}d\mu(t)$ have equal Laplace transforms on $[0,\infty)$, hence coincide [64, Ch. II, §6]. Thus $\varphi\,dt=\mu$ on $(0,\infty)$, so $\varphi\ge 0$ almost everywhere, and everywhere by continuity. □

Lemma 5.3 (Wronskian identity)**.** *Let $\alpha,\gamma\colon(0,\infty)\to(0,\infty)$ be continuous with $\int_0^\infty(1+t)(\alpha+\gamma)(t)e^{-xt}dt<\infty$ for all $x>0$, and let $A=\mathcal{L}\alpha$, $C=\mathcal{L}\gamma$. Put*

$$M_1(t)=\int_0^t\alpha(u)\gamma(t-u)\,u\,du,\qquad M_2(t)=\int_0^t\alpha(u)\gamma(t-u)(t-u)\,du.$$

Then $AC'=-\mathcal{L}M_2$, $A'C=-\mathcal{L}M_1$, so $AC'-A'C=\mathcal{L}[M_1-M_2]$, and

$$M_1(t)-M_2(t)=\int_0^{t/2}(t-2u)\big[\alpha(t-u)\gamma(u)-\alpha(u)\gamma(t-u)\big]du. \tag{10}$$

If α/γ is nondecreasing, then $M_1\ge M_2$ and $AC'-A'C$ is CM; if α/γ is strictly increasing, then $M_1>M_2$ on $(0,\infty)$.

Proof. $A'=-\mathcal{L}[t\alpha]$ and $C'=-\mathcal{L}[t\gamma]$, and the convolution theorem (Fubini, absolute convergence) gives $AC'=-\mathcal{L}[\alpha*(t\gamma)]=-\mathcal{L}M_2$ and $A'C=-\mathcal{L}[(t\alpha)*\gamma]=-\mathcal{L}M_1$. Splitting $[0,t]$ at $t/2$ and substituting $u\mapsto t-u$ on $[t/2,t]$ gives (10). For $0<u<t/2$ we have $t-u>u$, so monotonicity of α/γ gives $\alpha(t-u)\gamma(u)\ge\alpha(u)\gamma(t-u)$ (strictly in the strict case). Lemma 5.2 finishes the proof. □

Lemma 5.4 (likelihood ratio)**.** *Let ν_1,ν_2 be finite positive measures on an interval J, with $\nu_2=\rho\,\nu_1$ for a nondecreasing $\rho\ge 0$, and let h be bounded and nondecreasing on J. Then $\int h\,d\nu_2/\nu_2(J)\ge\int h\,d\nu_1/\nu_1(J)$. If ν_1 has a positive density and ρ, h are strictly increasing, the inequality is strict.*

Proof. $\iint[h(v)-h(w)][\rho(v)-\rho(w)]\,d\nu_1(v)\,d\nu_1(w)\ge 0$ (Chebyshev), which expands to $\nu_1(J)\int h\rho\,d\nu_1\ge\int h\,d\nu_1\int\rho\,d\nu_1$. □

The kernels. For $t>0$ put

$$c(t)=\frac{t}{1-e^{-t}},\qquad b(t)=c'(t)=\frac{e^t(e^t-1-t)}{(e^t-1)^2},$$

$$\ell(t)=\frac{tc'(t)}{c(t)}=1-\frac{t}{e^t-1}.$$

For $k\ge 1$ [65, 5.15.1]

$$|\psi^{(k)}(x)|=(-1)^{k+1}\psi^{(k)}(x)=\int_0^\infty t^{k-1}c(t)e^{-xt}dt, \tag{11}$$

and, integrating by parts in $x\psi'(x)=x\mathcal{L}[c](x)$ (note $c(0^+)=1$),

$$\Phi(x):=x\psi'(x)-1=\mathcal{L}[b](x),\qquad(-1)^k\Phi^{(k)}=\mathcal{L}[t^kb]. \tag{12}$$

Moreover $|\psi^{(k)}(x)|=k!\,Z_k(x)$ with the Hurwitz function $Z_k(x)=\zeta(k+1,x)=\sum_{j\ge0}(x+j)^{-k-1}$; here $Z_k'=-(k+1)Z_{k+1}$ and [65, 25.11.43]

$$Z_k(x)=x^{-k-1}+O(1)\ \ (x\to0^+),\qquad Z_k(x)=\frac{x^{-k}}{k}\big(1+O(x^{-1})\big)\ \ (x\to\infty). \tag{13}$$

Lemma 5.5. *On $(0,\infty)$:*

(a) *c is strictly log-concave:* $(\log c)''(t) = \dfrac{1}{4\sinh^2(t/2)} - \dfrac{1}{t^2} < 0$;

(b) *ℓ is strictly concave:* $\ell''(t) = -\dfrac{e^t[(t-2)e^t + t + 2]}{(e^t-1)^3}$ *and* $(t-2)e^t + t + 2 = \sum_{k\geq 3} \frac{(k-2)t^k}{k!} > 0$;

(c) *b is strictly log-concave, increasing, and* $\frac{1}{2} \leq b < 1$.

Proof. (a) $\log c = \log t - \log(1-e^{-t})$ and $\sinh y > y$ for $y > 0$. (b) $\ell = 1 - f$ with $f(t) = t/(e^t-1)$; differentiation gives the displayed f'', and the series follows from $e^t = \sum t^k/k!$. (c) Since $\log b = t + \log(e^t - 1 - t) - 2\log(e^t - 1)$,

$$(\log b)'' = -\frac{1 + (t-1)e^t}{(e^t-1-t)^2} + \frac{2e^t}{(e^t-1)^2} = -\frac{P(t)}{(e^t-1-t)^2(e^t-1)^2},$$

where

$$P(t) = (t-3)e^{3t} + (2t+7)e^{2t} - (2t^2+3t+5)e^t + 1 = \sum_{n\geq 1} c_n \frac{t^n}{n!},$$
$$c_n = 3^{n-1}(n-9) + 2^n(n+7) - (2n^2+n+5).$$

One computes $c_1 = \cdots = c_5 = 0$, $c_6 = 20$, $c_7 = 224$, $c_8 = 1512$; for $n \geq 9$, $3^{n-1}(n-9) \geq 0$ and $2^n(n+7) \geq n^2(n+7) > 2n^2 + n + 5$ (because $n^3 + 5n^2 - n - 5 = (n+5)(n^2-1) > 0$). Hence $P > 0$ on $(0,\infty)$. Finally $c = t + f$, so $b' = c'' = f'' > 0$ by (b): b increases from $b(0^+) = \frac{1}{2}$ to $b(\infty) = 1$; directly, $b < 1 \iff e^t(1-t) < 1$, which holds for $t > 0$. □

Lemma 5.6 (symmetrized Beta weights)**.** *For integers $a, b \geq 1$ let $\sigma_{a,b}(v) = v^{a-1}(1-v)^{b-1} + v^{b-1}(1-v)^{a-1}$, $0 < v < 1$. If $m, n, p, q \geq 1$, $m + n = p + q$ and $\max(p,q) > \max(m,n)$, then $\sigma_{m,n}/\sigma_{p,q}$ is strictly increasing on $(0, \frac{1}{2}]$, with value 1 at $v = \frac{1}{2}$.*

Proof. With $z = v/(1-v) \in (0,1]$ one has $\sigma_{a,b}(v) = (1-v)^{a+b-2} z^{\min(a,b)-1}(1 + z^{|a-b|})$. Put $d = |m-n|$ and $k = \min(m,n) - \min(p,q) \geq 1$; then $|p-q| = d + 2k$ and, with $z = e^{-\tau}$,

$$\frac{\sigma_{m,n}}{\sigma_{p,q}} = \frac{z^k(1+z^d)}{1+z^{d+2k}} = \frac{\cosh(d\tau/2)}{\cosh((d/2+k)\tau)}.$$

Its logarithmic τ-derivative is $\frac{d}{2}\tanh\frac{d\tau}{2} - (\frac{d}{2}+k)\tanh((\frac{d}{2}+k)\tau) < 0$ for $\tau > 0$, because $y \mapsto y\tanh(y\tau)$ is strictly increasing on $[0,\infty)$. As v increases to $\frac{1}{2}$, $\tau = \log\frac{1-v}{v}$ decreases to 0. □

Lemma 5.7. *For integers $r > s \geq 1$ the function $Z_{r+1}Z_s - Z_r Z_{s+1}$ is CM on $(0,\infty)$.*

Proof. By absolute convergence,

$$Z_{r+1}Z_s - Z_r Z_{s+1} = \sum_{i,j\geq 0} \frac{(x+j) - (x+i)}{(x+i)^{r+2}(x+j)^{s+2}},$$

and symmetrizing in (i,j) gives

$$Z_{r+1}Z_s - Z_r Z_{s+1} = \sum_{0\leq i<j} \frac{j-i}{(x+i)^{s+2}(x+j)^{s+2}}\Big[(x+i)^{-(r-s)} - (x+j)^{-(r-s)}\Big].$$

For $i < j$ the bracket equals $\mathcal{L}\big[t^{r-s-1}(e^{-it} - e^{-jt})\big]/(r-s-1)!$, so it is CM; each summand is a product of CM functions, and the series converges with all derivatives locally uniformly. □

5.2. **Qi's Remark 17: the three-polygamma Wronskian.**

Problem 5.8 (Qi [54, Remark 17])**.** "The decreasing property of $Q_{m,n}(x)$ in Theorem 11 implies that the inequality

$$\psi^{(m+n)}(x)[\psi^{(m)}(x)\psi^{(n)}(x)]' - \psi^{(m+n+1)}(x)\psi^{(m)}(x)\psi^{(n)}(x) > 0$$

... is valid on $(0,\infty)$ for $m,n\in\mathbb{N}$. We guess that, for $m,n\in\mathbb{N}$, the function

$$\psi^{(m+n)}(x)[\psi^{(m)}(x)\psi^{(n)}(x)]' - \psi^{(m+n+1)}(x)\psi^{(m)}(x)\psi^{(n)}(x)$$

should be completely monotonic in $x\in(0,\infty)$. Generally, one can discuss necessary and sufficient conditions on $\Omega_{m,n}\in\mathbb{R}$ such that the function

$$\psi^{(m+n)}(x)[\psi^{(m)}(x)\psi^{(n)}(x)]' - \Omega_{m,n}\psi^{(m+n+1)}(x)\psi^{(m)}(x)\psi^{(n)}(x)$$

and its opposite are respectively completely monotonic on $(0,\infty)$." Here $Q_{m,n}$ denotes the quotient $\psi^{(m+n)}/(\psi^{(m)}\psi^{(n)})$.

Theorem 5.9. *Let $m,n\in\mathbb{N}$, $M=m+n$, $\Omega\in\mathbb{R}$, and $H_\Omega = \psi^{(M)}[\psi^{(m)}\psi^{(n)}]' - \Omega\,\psi^{(M+1)}\psi^{(m)}\psi^{(n)}$. Then*

(i) *H_Ω is CM on $(0,\infty)$ if and only if $\Omega\le 1$; in particular Qi's guess ($\Omega=1$) holds;*

(ii) *$-H_\Omega$ is CM on $(0,\infty)$ if and only if $\Omega\ge\Omega^\sharp := \dfrac{M+2}{M+1}$.*

Proof. Since $|\psi^{(k)}| = k!Z_k$ and $Z_k' = -(k+1)Z_{k+1}$, a sign count gives

$$-H_\Omega = M!\,m!\,n!\Big[\Omega(M+1)Z_{M+1}Z_mZ_n - (m+1)Z_MZ_{m+1}Z_n - (n+1)Z_MZ_mZ_{n+1}\Big]. \tag{14}$$

Sufficiency in (i). By (11), $P := |\psi^{(m)}\psi^{(n)}| = \mathcal{L}\alpha$ and $B := |\psi^{(M)}| = \mathcal{L}\beta$ with $\alpha = a_m * a_n$, $\beta = a_M$, $a_k(t) = t^{k-1}c(t)$. Also $\psi^{(m)}\psi^{(n)} = (-1)^M P$, $\psi^{(M)} = (-1)^{M+1}B$ and $\psi^{(M+1)} = (-1)^{M+1}B'$, whence $H_1 = PB' - P'B$. By Lemma 5.3 (with $(A,C)=(P,B)$) it suffices that α/β be increasing. Now

$$\frac{\alpha(s)}{\beta(s)} = \int_0^1 v^{m-1}(1-v)^{n-1}e^{L_v(s)}dv,$$
$$L_v(s) = \log c(sv) + \log c(s(1-v)) - \log c(s),$$

and $L_v'(s) = v(\log c)'(sv) + (1-v)(\log c)'(s(1-v)) - (\log c)'(s) > 0$, because $(\log c)'$ is strictly decreasing (Lemma 5.5(a)) and sv, $s(1-v) < s$. So H_1 is CM (indeed strictly). For $\Omega<1$, $H_\Omega = H_1 + (1-\Omega)\,|\psi^{(M+1)}|\,|\psi^{(m)}\psi^{(n)}|$ is a sum of CM functions.

Necessity in (i). By (13) at infinity, the bracket in (14) equals

$$\Big(\frac{\Omega}{mn} - \frac{1}{Mn} - \frac{1}{Mm}\Big)x^{-2M-1} + O(x^{-2M-2}) = \frac{\Omega-1}{mn}\,x^{-2M-1} + O(x^{-2M-2}).$$

Hence $H_\Omega(x) = M!(m-1)!(n-1)!(1-\Omega)x^{-2M-1} + O(x^{-2M-2})$, which is negative for large x when $\Omega>1$.

Sufficiency in (ii). At $\Omega^\sharp$ write $M+2 = (m+1)+(n+1)$ in (14):

$$-H_{\Omega^\sharp} = M!\,m!\,n!\Big[(m+1)Z_n\big(Z_{M+1}Z_m - Z_MZ_{m+1}\big) + (n+1)Z_m\big(Z_{M+1}Z_n - Z_MZ_{n+1}\big)\Big].$$

Since $M>m$ and $M>n$, Lemma 5.7 shows that $-H_{\Omega^\sharp}$ is CM. For $\Omega>\Omega^\sharp$ add the CM function $(\Omega-\Omega^\sharp)(M+1)!\,m!\,n!\,Z_{M+1}Z_mZ_n$.

Necessity in (ii). By (13) at 0^+, the bracket in (14) is $[\Omega(M+1)-(M+2)]x^{-2M-4}(1+o(1))$, which is negative near 0 if $\Omega<\Omega^\sharp$. □

Remark 5.10. Remark 17 is in fact a consequence of Theorem 1 of [54] (that is, [58, Thm. 3.1]): for $p'>m'\ge n'>q'\ge 1$ with $m'+n'=p'+q'$, the function $|\psi^{(m')}\psi^{(n')}|-c\,|\psi^{(p')}\psi^{(q')}|$ is CM iff $c\le \frac{(m'-1)!(n'-1)!}{(p'-1)!(q'-1)!}$, and its negative is CM iff $c\ge \frac{m'!n'!}{p'!q'!}$. Indeed, splitting $|\psi^{(M+1)}\psi^{(m)}\psi^{(n)}|$ with weights $\frac{m}{M}+\frac{n}{M}=1$,

$$\begin{aligned}H_1 = |\psi^{(n)}|\Big(|\psi^{(M)}\psi^{(m+1)}|-\tfrac{m}{M}|\psi^{(M+1)}\psi^{(m)}|\Big)\\ +|\psi^{(m)}|\Big(|\psi^{(M)}\psi^{(n+1)}|-\tfrac{n}{M}|\psi^{(M+1)}\psi^{(n)}|\Big),\end{aligned}$$

and $\frac{m}{M}=\frac{(M-1)!\,m!}{M!\,(m-1)!}$ is exactly the threshold for the pair $(M+1,m)\succ(M,m+1)$. Likewise part (ii) at $\Omega^\sharp$ follows from the "negative" half of the same theorem (equivalently, from Lemma 5.7). We have not seen this observation in print; the two-term inequalities used are also contained in Z.-H. Yang's majorization theorem [61, Prop. 2]. The proof given above is independent of both.

5.3. **Qi's Remark 18: the four-polygamma Wronskian.**

Problem 5.11 (Qi [54, Remark 18])**.** "The decreasing property of $Q_{m,n;p,q}(x)$ in Theorem 12 implies that the inequality

$$[\psi^{(m)}(x)\psi^{(n)}(x)][\psi^{(p)}(x)\psi^{(q)}(x)]'-[\psi^{(m)}(x)\psi^{(n)}(x)]'[\psi^{(p)}(x)\psi^{(q)}(x)]>0$$

... is valid on $(0,\infty)$ for $(p,q)\succ(m,n)$. We guess that, for $(p,q)\succ(m,n)$, the function

$$[\psi^{(m)}(x)\psi^{(n)}(x)][\psi^{(p)}(x)\psi^{(q)}(x)]'-[\psi^{(m)}(x)\psi^{(n)}(x)]'[\psi^{(p)}(x)\psi^{(q)}(x)]$$

should be completely monotonic in $x\in(0,\infty)$. Generally, for $(p,q)\succ(m,n)$, one can discuss necessary and sufficient conditions on $\Omega_{m,n;p,q}\in\mathbb{R}$ such that the function

$$[\psi^{(m)}\psi^{(n)}][\psi^{(p)}\psi^{(q)}]'-\Omega_{m,n;p,q}[\psi^{(m)}\psi^{(n)}]'[\psi^{(p)}\psi^{(q)}]$$

and its opposite are respectively completely monotonic on $(0,\infty)$." Here $Q_{m,n;p,q}=\psi^{(m)}\psi^{(n)}/(\psi^{(p)}\psi^{(q)})$, $m,n,p,q\in\mathbb{N}$, and $(p,q)\succ(m,n)$ means $p+q=m+n$ and $\max(p,q)>\max(m,n)$.

Theorem 5.12. *Let $m,n,p,q\in\mathbb{N}$ with $(p,q)\succ(m,n)$, $M=m+n$, and*

$$W_\Omega=[\psi^{(m)}\psi^{(n)}][\psi^{(p)}\psi^{(q)}]'-\Omega[\psi^{(m)}\psi^{(n)}]'[\psi^{(p)}\psi^{(q)}].$$

Let $\alpha=a_m*a_n$, $\gamma=a_p*a_q$ ($a_k=t^{k-1}c$), and let M_1,M_2 be as in Lemma 5.3.

(i) *W_Ω is CM on $(0,\infty)$ if and only if $\Omega\ge 1$; in particular Qi's guess ($\Omega=1$) holds.*
(ii) *$-W_\Omega$ is CM on $(0,\infty)$ if and only if $\Omega\le\Omega_*$, where $\Omega_*=\Omega_*(m,n;p,q)=\min_{t>0}M_2(t)/M_1(t)$; the minimum exists and $0<\Omega_*<1$.*

Proof. Put $P=|\psi^{(m)}\psi^{(n)}|=\mathcal{L}\alpha$ and $C=|\psi^{(p)}\psi^{(q)}|=\mathcal{L}\gamma$. Both products carry the sign $(-1)^M$, so $W_\Omega=PC'-\Omega P'C=\mathcal{L}[\Omega M_1-M_2]$ (Lemma 5.3).

Step 1: α/γ is strictly increasing. With $K_s(v)=c(sv)c(s(1-v))$, which is symmetric under $v\mapsto 1-v$,

$$\frac{\alpha(s)}{\gamma(s)}=\frac{\int_0^1 v^{m-1}(1-v)^{n-1}K_s(v)dv}{\int_0^1 v^{p-1}(1-v)^{q-1}K_s(v)dv}=\frac{\int_0^{1/2}\sigma_{m,n}K_s\,dv}{\int_0^{1/2}\sigma_{p,q}K_s\,dv}=\frac{\int_J h\,d\nu_s}{\nu_s(J)},$$

where $J=(0,\frac12]$, $d\nu_s=\sigma_{p,q}K_s\,dv$ and $h=\sigma_{m,n}/\sigma_{p,q}$, which is bounded and strictly increasing on J by Lemma 5.6. For $s<s'$, $d\nu_{s'}/d\nu_s=K_{s'}/K_s=\exp\int_s^{s'}\partial_\sigma\log K_\sigma\,d\sigma$, and

$$\partial_\sigma\log K_\sigma(v)=\frac{\ell(\sigma v)+\ell(\sigma(1-v))}{\sigma},$$
$$\partial_v\big[\ell(\sigma v)+\ell(\sigma(1-v))\big]=\sigma\big[\ell'(\sigma v)-\ell'(\sigma(1-v))\big]>0$$

for $0<v<\frac12$, since ℓ' is strictly decreasing (Lemma 5.5(b)). So $K_{s'}/K_s$ is strictly increasing on J, and Lemma 5.4 gives the claim.

Step 2: proof of (i). By Step 1 and Lemma 5.3, $M_1>M_2$, so W_1 is CM. For $\Omega>1$, $W_\Omega=W_1+(\Omega-1)(-P')C$, and $-P'=\mathcal{L}[t\alpha]$ is CM. For $\Omega<1$: as $u\to0^+$, $\alpha(u)=u^{M-1}(\mathrm{B}(m,n)+o(1))$ and $\gamma(u)=u^{M-1}(\mathrm{B}(p,q)+o(1))$ (because $c(0^+)=1$), hence

$$\frac{M_2(t)}{M_1(t)}\to\frac{\int_0^1 v^{M-1}(1-v)^M dv}{\int_0^1 v^M(1-v)^{M-1}dv}=1\qquad(t\to0^+),$$

so $\Omega M_1-M_2<0$ for small t, and Lemma 5.2 shows that W_Ω is not CM.

Step 3: proof of (ii). By Lemma 5.2, $-W_\Omega$ is CM iff $M_2-\Omega M_1\ge0$, i.e. iff $\Omega\le\inf_{t>0}r(t)$ with $r=M_2/M_1$; r is continuous and $r<1$ (Step 2), and $r(0^+)=1$. As $t\to\infty$, write $M_i(t)=t^2\int_0^1\alpha(tv)\gamma(t(1-v))\,\omega_i(v)\,dv$ with $\omega_1(v)=v$, $\omega_2(v)=1-v$. Since $c(y)/y\to1$, one has $\alpha(u)/u^{M+1}\to\mathrm{B}(m+1,n+1)$ and $\gamma(u)/u^{M+1}\to\mathrm{B}(p+1,q+1)$ as $u\to\infty$; and the bound $\alpha(u)\le\mathrm{B}(m,n)u^{M-1}(1+u)^2$ (from $c(y)\le1+y$), with its analogue for γ, shows that for $t\ge1$ the functions $t^{-2M-2}\alpha(tv)\gamma(t(1-v))$ are dominated by $16\,\mathrm{B}(m,n)\,\mathrm{B}(p,q)\,v^{M-1}(1-v)^{M-1}$. Dominated convergence gives

$$r(t)\to\frac{\int_0^1 v^{M+1}(1-v)^{M+2}dv}{\int_0^1 v^{M+2}(1-v)^{M+1}dv}=1.$$

Hence $\inf r$ is attained at an interior point, and $0<\Omega_*<1$. □

Remark 5.13. Numerically (Gauss–Legendre quadrature of the densities, script `R2_verify_remark18.py`): $\Omega_*(2,2;3,1)\approx0.97612$, $\Omega_*(2,3;4,1)\approx0.97174$, $\Omega_*(2,4;5,1)\approx0.97207$, $\Omega_*(3,3;4,2)\approx0.99206$ and $\Omega_*(3,3;5,1)\approx0.96453$, each attained near $t\approx16$–19. We do not know a closed form, so the second half of the Ω-problem is answered only by this characterization. Unlike Remark 17, we found no way to reduce Theorem 5.12(i) to known two-term inequalities: any two of the four products in W_1 share two factors, and the pairwise splittings we tried, with the sharp two-term thresholds of [58], cannot be balanced.

5.4. Yin–Han–Qi's Remark 4: the function $\Phi(x)=x\psi'(x)-1$.

Problem 5.14 (Yin–Han–Qi [55, Remark 4])**.** With $\Phi(x)=x\psi'(x)-1$: "We guess that, for $m,n\in\{0\}\cup\mathbb{N}$, the function

$$\Phi^{(m+n+1)}(x)\big[\Phi^{(m)}(x)\Phi^{(n)}(x)\big]'-\Phi^{(m+n+2)}(x)\big[\Phi^{(m)}(x)\Phi^{(n)}(x)\big]$$

is completely monotonic in $x\in(0,\infty)$. One can also consider necessary and sufficient conditions on $\Lambda_{m,n}\in\mathbb{R}$ for $m,n\in\{0\}\cup\mathbb{N}$ such that the function

$$\Phi^{(m+n+1)}(x)\left[\Phi^{(m)}(x)\Phi^{(n)}(x)\right]'-\Lambda_{m,n}\Phi^{(m+n+2)}(x)\left[\Phi^{(m)}(x)\Phi^{(n)}(x)\right]$$

and its opposite are respectively completely monotonic on $(0,\infty)$."

Theorem 5.15. *Let $m,n\ge0$, $M=m+n$, $\Lambda\in\mathbb{R}$, and*

$$G_\Lambda=\Phi^{(M+1)}[\Phi^{(m)}\Phi^{(n)}]'-\Lambda\Phi^{(M+2)}\Phi^{(m)}\Phi^{(n)}.$$

Let $\alpha = (t^m b) * (t^n b)$, $\beta = t^{M+1} b$, *and*

$$N_1(t) = \int_0^t \alpha(u)\beta(t-u)\,u\,du, \qquad N_2(t) = \int_0^t \alpha(u)\beta(t-u)(t-u)\,du.$$

(i) G_Λ *is CM on* $(0,\infty)$ *if and only if* $\Lambda \le 1$*; in particular the guess* ($\Lambda = 1$) *holds.*
(ii) $-G_\Lambda$ *is CM on* $(0,\infty)$ *if and only if* $\Lambda \ge \Lambda^*_{m,n} := \max_{t>0} N_1(t)/N_2(t)$*; the maximum exists and* $1 < \Lambda^*_{m,n} < \infty$.

Proof. By (12), $P := (-1)^M \Phi^{(m)}\Phi^{(n)} = \mathcal{L}\alpha$ and $B := (-1)^{M+1}\Phi^{(M+1)} = \mathcal{L}\beta$, with $\Phi^{(M+2)} = (-1)^{M+1}B'$. Hence $G_\Lambda = \Lambda B'P - BP' = \mathcal{L}[N_1 - \Lambda N_2]$ (Lemma 5.3 with $(A,C) = (P,B)$).

(i) For $\Lambda = 1$ we need α/β increasing. Here

$$\frac{\alpha(s)}{\beta(s)} = \int_0^1 v^m (1-v)^n \,\frac{b(sv)\,b(s(1-v))}{b(s)}\,dv,$$

and for each v the logarithmic s-derivative of the integrand is $v(\log b)'(sv) + (1-v)(\log b)'(s(1-v)) - (\log b)'(s) > 0$ by Lemma 5.5(c). So $N_1 > N_2$ and G_1 is CM. For $\Lambda < 1$, $G_\Lambda = G_1 + (1-\Lambda)(-B')P$ with $-B' = \mathcal{L}[t\beta]$ CM. For $\Lambda > 1$: b is continuous on $[0,\infty)$ with $b(0) = \frac12$, so $\alpha(u) \sim \frac14 \mathrm{B}(m+1,n+1)u^{M+1}$ and $\beta(s) \sim \frac12 s^{M+1}$ at 0; thus $N_1/N_2 \to 1$ as $t \to 0^+$, and $N_1 - \Lambda N_2 < 0$ for small t. Lemma 5.2 applies.

(ii) $-G_\Lambda$ is CM iff $\Lambda N_2 - N_1 \ge 0$, i.e. iff $\Lambda \ge \sup N_1/N_2$. The ratio is continuous, > 1 by (i), and tends to 1 as $t \to 0^+$. It also tends to 1 as $t \to \infty$: after $u = tv$, the normalized densities $\alpha(tv)/(tv)^{M+1} \in [\frac14 \mathrm{B}, \mathrm{B}]$ and $\beta(t(1-v))/(t(1-v))^{M+1} = b(t(1-v)) \in [\frac12, 1)$ converge pointwise to $\mathrm{B} = \mathrm{B}(m+1,n+1)$ and 1 (Lemma 5.5(c)), and dominated convergence gives the limit $\int_0^1 v^{M+2}(1-v)^{M+1}dv / \int_0^1 v^{M+1}(1-v)^{M+2}dv = 1$. So the supremum is a maximum and exceeds 1. □

Remark 5.16. (1) Numerically (script `R2_verify_yhq_remark4.py`): $\Lambda^*_{0,0} \approx 1.11422$, $\Lambda^*_{0,1} \approx 1.08382$, $\Lambda^*_{1,1} \approx 1.07538$, $\Lambda^*_{0,2} \approx 1.06173$, $\Lambda^*_{1,2} \approx 1.06254$, $\Lambda^*_{2,2} \approx 1.05612$ and $\Lambda^*_{2,3} \approx 1.04904$, attained near $t \approx 12$–16. In contrast with Theorem 5.9(ii), the endpoint asymptotics of G_Λ cancel at $\Lambda = 1$ at *both* ends ($\Phi \sim x^{-1}$ at 0 and $\Phi \sim (2x)^{-1}$ at ∞), which is why Λ^* is determined by an interior maximum; no closed form is known to us.

(2) As in Remark 5.10, part (i) at $\Lambda = 1$ can also be deduced by splitting,

$$\begin{aligned} G_1 = {} & |\Phi^{(n)}|\Big(|\Phi^{(M+1)}\Phi^{(m+1)}| - \tfrac{m+1}{M+2}|\Phi^{(M+2)}\Phi^{(m)}|\Big) \\ & + |\Phi^{(m)}|\Big(|\Phi^{(M+1)}\Phi^{(n+1)}| - \tfrac{n+1}{M+2}|\Phi^{(M+2)}\Phi^{(n)}|\Big), \end{aligned}$$

from the Φ-version of Qi's majorization theorem in the form stated in [57, Thm. 17] ("$-Y_{i,j;\ell,m;\Omega}$ is CM iff $\Omega \ge \frac{i!j!}{\ell!m!}$", applied to $(M+2,m) \succ (M+1,m+1)$). We could not consult the original article cited there; the proof above is independent of it.

5.5. **Yin–Han–Qi's Remark 2: an elementary sign guess.**

Problem 5.17 (Yin–Han–Qi [55, Lemma 3 and Remark 2])**.** For $0 < x < y/2$ let

$$F(x,y) = 2\Big(\frac{1}{x} - \frac{1}{y-x}\Big) + \frac12\Big(\frac{2^{y-x}}{y-x} - \frac{2^x}{x}\Big) - \frac{2^{y-x} - 2^x}{(y-x)x}.$$

The authors guess that $F(x,y) > 0$ when $2 < x < y/2$, and $F(x,y) < 0$ when $y > 4$ and $0 < x < 2$, and suggest discussing the sign of F on the whole region $0 < x < y/2$.

Proposition 5.18. *For* $0 < x < y/2$*, with* $E(s) = (2^s - 1)/s$ *(*$E(0) = \log 2$*),*

$$x(y-x)\,F(x,y) = 2(x-2)(y-x-2)\big[E(y-x-2) - E(x-2)\big].$$

Consequently $\operatorname{sign} F(x,y) = \operatorname{sign}\big((x-2)(y-x-2)\big)$*, and* F *vanishes exactly on the lines* $x=2$ *and* $y=x+2$*. Both guesses hold.*

Proof. With $z=y-x$, $u=x-2$, $v=z-2$, clearing denominators gives $xzF = (x-2)2^{z-1} - (z-2)2^{x-1} + 2(z-x) = 2[u(2^v-1) - v(2^u-1)] = 2uv[E(v)-E(u)]$. Since $E(s) = \log 2 \int_0^1 2^{sr}dr$ is strictly increasing and $v-u = z-x > 0$, the bracket is positive. If $2 < x < y/2$ then $u>0$ and $v>u>0$; if $0<x<2<y-2$ then $u<0<v$. □

5.6. **Completely monotonic degrees in the Qi–Agarwal survey.** Let

$$Q_7(x) = 6x^7 + 51x^6 + 181x^5 + 345x^4 + 399x^3 + 283x^2 + 120x + 24,$$
$$Q_6(x) = 24x^6 + 156x^5 + 412x^4 + 621x^3 + 523x^2 + 242x + 48,$$

and

$$H_2 = \psi'(x) - \frac{Q_7(x)}{6x^2(1+x)^4(2+x)^2}, \qquad H_3 = \frac{Q_6(x)}{24x^2(x+1)^4(x+2)} - \psi'(x),$$
$$f_\lambda = H_2 - \lambda g, \qquad g = \frac{4+8x+5x^2}{24x(1+x)^3(2+x)^2}.$$

Then f_λ is the function (5.2) of [56]. It is known that H_2, H_3 are CM, that f_λ is CM iff $\lambda \le 0$, and that $-f_\lambda$ is CM iff $\lambda \ge 4$ (see [56, Sections 3.3, 5.4, 5.5] and the references there).

Problem 5.19 (Qi–Agarwal [56, Sections 5.4 and 5.5])**.** "Can one compute completely monotonic degrees of the functions $H_2(x)$ and $H_3(x)$ on $(0,\infty)$, as one did in [32]?" and "What are completely monotonic degrees of the function (5.2) for $\lambda \le 0$ and its negative for $\lambda \ge 4$ on $(0,\infty)$?"

Partial fractions give the Laplace densities ($\psi' = \mathcal{L}c$):

$$H_2 = \mathcal{L}p_{H_2}, \qquad p_{H_2} = c(t) - t - e^{-t}\Big(\tfrac{43}{6} - \tfrac{7}{2}t + t^2 - \tfrac{t^3}{12}\Big) + e^{-2t}\Big(\tfrac{37}{6} + \tfrac{13}{6}t\Big),$$
$$H_3 = \mathcal{L}p_3, \qquad p_3 = t + \tfrac{13}{24} + e^{-t}\Big(\tfrac{55}{24} - \tfrac{15}{8}t + \tfrac{35}{48}t^2 - \tfrac{t^3}{12}\Big) - \tfrac{11}{6}e^{-2t} - c(t),$$
$$g = \mathcal{L}p_g, \qquad p_g = \tfrac{1}{24} + e^{-t}\Big(-\tfrac{3}{8} + \tfrac{t}{8} - \tfrac{t^2}{48}\Big) + e^{-2t}\Big(\tfrac{1}{3} + \tfrac{t}{6}\Big),$$

with $p_{H_2}(0) = p_3(0) = p_g(0) = 0$. With $T = e^t$ one finds the exponential polynomials

$$\begin{aligned}
\mathcal{E}_3 &:= p_3'(t)(T-1)^2T^2 \\
&= T^3\big(\tfrac{t^3}{12} - \tfrac{47}{48}t^2 + \tfrac{13}{3}t - \tfrac{31}{6}\big) + T^2\big(-\tfrac{t^3}{6} + \tfrac{47}{24}t^2 - \tfrac{20}{3}t + 13\big) \\
&\quad + T\big(\tfrac{t^3}{12} - \tfrac{47}{48}t^2 + \tfrac{10}{3}t - \tfrac{23}{2}\big) + \tfrac{11}{3}, \\
\mathcal{E}_2 &:= p_{H_2}(t)(T-1)^2T^2 \\
&= T^3\big(\tfrac{t^3}{12} - t^2 + \tfrac{9}{2}t - \tfrac{43}{6}\big) + T^2\big(-\tfrac{t^3}{6} + 2t^2 - \tfrac{35}{6}t + \tfrac{41}{2}\big) \\
&\quad + T\big(\tfrac{t^3}{12} - t^2 - \tfrac{5}{6}t - \tfrac{39}{2}\big) + \tfrac{13}{6}t + \tfrac{37}{6}, \\
\mathcal{E}_2' &:= p_{H_2}'(t)(T-1)^2T^2 \\
&= T^3\big(-\tfrac{t^3}{12} + \tfrac{5}{4}t^2 - \tfrac{13}{2}t + \tfrac{35}{3}\big) + T^2\big(\tfrac{t^3}{6} - \tfrac{5}{2}t^2 + \tfrac{20}{3}t - \tfrac{65}{2}\big) \\
&\quad + T\big(-\tfrac{t^3}{12} + \tfrac{5}{4}t^2 + \tfrac{19}{6}t + 31\big) - \tfrac{13}{3}t - \tfrac{61}{6}, \\
\mathcal{E}_4 &:= (4p_g - p_{H_2})'(t)(T-1)^2T^2 \\
&= T^3\big(\tfrac{t^3}{12} - \tfrac{7}{6}t^2 + \tfrac{35}{6}t - \tfrac{29}{3}\big) + T^2\big(-\tfrac{t^3}{6} + \tfrac{7}{3}t^2 - \tfrac{20}{3}t + \tfrac{53}{2}\big) \\
&\quad + T\big(\tfrac{t^3}{12} - \tfrac{7}{6}t^2 - \tfrac{7}{6}t - 25\big) + 3t + \tfrac{49}{6}, \\
p_g'(t)\,T^2 &= T\big(\tfrac{t^2}{48} - \tfrac{t}{6} + \tfrac{1}{2}\big) - \tfrac{t}{3} - \tfrac{1}{2}.
\end{aligned}$$

(The notation $\mathcal{E}_2'$ is a name, not a t-derivative of $\mathcal{E}_2$.)

Lemma 5.20. *The Taylor coefficients $a_N = N![t^N]$ of $\mathcal{E}_3$, $\mathcal{E}_2$, $\mathcal{E}_4$ and of $p_g' T^2$ are all nonnegative; the first nonzero ones are $a_4(\mathcal{E}_3) = \frac{45}{2}$, $a_5(\mathcal{E}_2) = \frac{50}{3}$, $a_5(\mathcal{E}_4) = \frac{82}{3}$, $a_2(p_g' T^2) = \frac{5}{24}$. Hence $p_3' > 0$, $p_{H_2} > 0$, $(4p_g - p_{H_2})' > 0$ and $p_g' > 0$ on $(0,\infty)$.*

Proof. For $p_g' T^2$: $a_0 = a_1 = 0$ and $a_N = (N^2 - 9N + 24)/48 > 0$ for $N \geq 2$ (negative discriminant). For $\mathcal{E} = \sum_{k=0}^{3} Q_k(t) e^{kt}$ one has, for $N \geq 2$, $a_N = 3^{N-3} u_3(N) + 2^{N-3} u_2(N) + u_1(N)$ with

$$\begin{aligned}
\mathcal{E}_3: \quad u_3 &= \tfrac{4N^3 - 153N^2 + 2021N - 6696}{48}, \quad u_2 = -\tfrac{2N^3 - 53N^2 + 371N - 1248}{12}, \\
u_1 &= \tfrac{4N^3 - 59N^2 + 215N - 552}{48}; \\
\mathcal{E}_2: \quad u_3 &= \tfrac{N^3 - 39N^2 + 524N - 2322}{12}, \quad u_2 = -\tfrac{N^3 - 27N^2 + 166N - 984}{6}, \\
u_1 &= \tfrac{N^3 - 15N^2 + 4N - 234}{12}; \\
\mathcal{E}_4: \quad u_3 &= \tfrac{N^3 - 45N^2 + 674N - 3132}{12}, \quad u_2 = -\tfrac{N^3 - 31N^2 + 190N - 1272}{6}, \\
u_1 &= \tfrac{N^3 - 17N^2 + 2N - 300}{12}.
\end{aligned}$$

For $N \geq 90$ each $u_3(N) \geq N^3/275$ (the polynomial $275u_3(90+s) - (90+s)^3$ has nonnegative coefficients), while $|u_2(N)| \leq 7N^3$ and $|u_1(N)| \leq N^3$ for $N \geq 4$ (bound each coefficient using $N \geq 4$). Hence

$$a_N \geq N^3\, 2^{N-3} \big[(3/2)^{N-3}/275 - 8\big] > 0 \qquad (N \geq 90).$$

For $N < 90$ the coefficients are evaluated in exact rational arithmetic (the verification script checks all $N \leq 200$): all are ≥ 0, and the first nonzero ones are as stated. Since $(T-1)^2 T^2 > 0$, the sign claims follow. □

Lemma 5.21 (fractional powers)**.** *Let $p \in C^1[0,\infty)$ with $p(0) = 0$ and $|p'(t)| \leq Ke^{-\kappa t}$ $(K, \kappa > 0)$. For $0 < \varepsilon < 1$ put $D^\varepsilon p(t) = \Gamma(1-\varepsilon)^{-1} \int_0^t (t-u)^{-\varepsilon} p'(u)\,du$. Then $D^\varepsilon p$ is continuous and bounded, and $x^\varepsilon \mathcal{L}p(x) = \mathcal{L}[D^\varepsilon p](x)$ for $x > 0$. If moreover $p > 0$ on $(0,\infty)$, $p' > 0$ on some $(0,\delta)$ and $p(t) \to L > 0$, then $D^\varepsilon p > 0$ on $(0,\infty)$ for all sufficiently small $\varepsilon > 0$; hence $\deg_{\mathrm{cm}} \mathcal{L}p > 0$.*

Proof. The identity is the convolution theorem: $\mathcal{L}[D^\varepsilon p] = x^{\varepsilon-1}\mathcal{L}[p'] = x^{\varepsilon-1}(x\mathcal{L}p - p(0))$. Let $J_\varepsilon(t) = \int_0^t (1-u/t)^{-\varepsilon} p'(u)\,du$, which has the sign of $D^\varepsilon p(t)$. For $t \leq \delta$, $J_\varepsilon(t) > 0$. For $0 < \varepsilon \leq \frac{1}{2}$, splitting at $t/2$ and using $(1-y)^{-\varepsilon} - 1 \leq 2y$ on $[0, \frac{1}{2}]$, we get $|J_\varepsilon(t) - p(t)| \leq \frac{2}{t}\int_0^\infty u|p'(u)|\,du + 4tKe^{-\kappa t/2}$, so there is T with $J_\varepsilon \geq L/2$ on $[T,\infty)$ for all such ε. On $[\delta, T]$, $|J_\varepsilon(t) - p(t)| \leq \max_{[0,T]} |p'|\, T\, \frac{\varepsilon}{1-\varepsilon}$, which is smaller than $\min_{[\delta,T]} p > 0$ for small ε. Lemma 5.2 concludes. □

Theorem 5.22. $\deg_{\mathrm{cm}} H_2 = 0$ *and* $\deg_{\mathrm{cm}} H_3 = 1$.

Proof. By Lemma 5.20, $p_{H_2} > 0$, so H_2 is CM (Lemma 5.2). From the partial fractions, $Q_7/(6x^2(1+x)^4(2+x)^2) = x^{-2} + \frac{37}{24} + O(x)$ and $\psi'(x) = x^{-2} + \zeta(2) + O(x)$, so $H_2(0^+) = \zeta(2) - \frac{37}{24} \in (0,\infty)$. By Lemma 5.1, $x^\alpha H_2$ is not CM for $\alpha > 0$; thus $\deg_{\mathrm{cm}} H_2 = 0$.

Since $p_3(0) = 0$ and $p_3' > 0$ (Lemma 5.20), $p_3 > 0$, so H_3 is CM, and $xH_3(x) = \mathcal{L}[p_3'](x)$ (integration by parts, p_3 bounded) is CM; so $\deg_{\mathrm{cm}} H_3 \geq 1$. As $x \to 0^+$, $Q_6/(24x^2(x+1)^4(x+2)) = x^{-2} + \frac{13}{24}x^{-1} + O(1)$, so $xH_3(x) \to \frac{13}{24}$ and $x^\alpha H_3 \to 0$ for $\alpha > 1$; Lemma 5.1 gives $\deg_{\mathrm{cm}} H_3 = 1$. □

Theorem 5.23. (a) $\deg_{\mathrm{cm}} f_0 = 0$.

(b) *For every $\lambda < 0$, $0 < \deg_{\mathrm{cm}} f_\lambda < 1$; in particular the degree is not an integer. The map $\lambda \mapsto \deg_{\mathrm{cm}} f_\lambda$ is nonincreasing on $(-\infty, 0)$, tends to 1 as $\lambda \to -\infty$ and to 0 as $\lambda \to 0^-$.*

(c) *For every $\mu \geq 4$, $\deg_{\mathrm{cm}}(-f_\mu) = 1$.*

Proof. (a) is Theorem 5.22. By Lemma 5.20, p_g increases from 0 to $\frac{1}{24}$, so g and $xg(x) = \mathcal{L}[p_g'](x)$ are CM; since $xg(x) \to \frac{1}{24}$ at 0^+, $\deg_{\mathrm{cm}} g = 1$.

(b) Let $\lambda<0$. The density $p=p_{H_2}+|\lambda|p_g$ of f_λ is positive, so f_λ is CM; $xf_\lambda(x)\to|\lambda|/24$ at 0^+, so $\deg_{\mathrm{cm}} f_\lambda\le 1$. The leading terms of $\mathcal{E}_2'$ and of $p_g'T^2$ give, as $t\to\infty$,

$$p'_{H_2}(t)=-\tfrac{t^3}{12}e^{-t}\big(1+O(t^{-1})\big),\qquad p_g'(t)=\tfrac{t^2}{48}e^{-t}\big(1+O(t^{-1})\big).$$

Hence $p'(t)<0$ for large t, $xf_\lambda=\mathcal{L}[p']$ is not CM (Lemma 5.2), and $\deg_{\mathrm{cm}} f_\lambda<1$ by Lemma 5.1. The first nonzero Taylor coefficients give $p_{H_2}=\frac{5}{36}t^3+O(t^4)$ and $p_g=\frac{5}{144}t^3+O(t^4)$, so $p'>0$ near 0; moreover $p\to|\lambda|/24>0$ and p' decays exponentially. Lemma 5.21 yields $\deg_{\mathrm{cm}} f_\lambda>0$.

Monotonicity: for $\lambda_1<\lambda_2<0$ and $\alpha\le 1$, $x^\alpha f_{\lambda_1}=x^\alpha f_{\lambda_2}+(\lambda_2-\lambda_1)x^\alpha g$, and $x^\alpha g$ is CM. Limits: fix $\alpha\in(0,1)$. Since $\int_0^\infty p'_{H_2}=0$, splitting the Caputo integral at $t/2$ gives $|D^\alpha p_{H_2}(t)|=O(t^{-\alpha-1})$ as $t\to\infty$, while $D^\alpha p_g(t)\ge t^{-\alpha}\Gamma(1-\alpha)^{-1}\int_0^t p_g'\ge c_\alpha t^{-\alpha}$ for large t; near 0 both are $\asymp t^{3-\alpha}$ with positive constants. Hence $S_\alpha=\sup_t|D^\alpha p_{H_2}|/D^\alpha p_g<\infty$, and for $|\lambda|>S_\alpha$ the density $D^\alpha p=D^\alpha p_{H_2}+|\lambda|D^\alpha p_g$ of $x^\alpha f_\lambda$ (Lemma 5.21) is positive, so $\deg_{\mathrm{cm}} f_\lambda\ge\alpha$. Conversely, $x^\alpha H_2=\mathcal{L}[D^\alpha p_{H_2}]$ is not CM by (a), so $D^\alpha p_{H_2}(t_\alpha)<0$ for some t_α; then $D^\alpha p(t_\alpha)<0$ whenever $|\lambda|<-D^\alpha p_{H_2}(t_\alpha)/D^\alpha p_g(t_\alpha)$, and $\deg_{\mathrm{cm}} f_\lambda<\alpha$.

(c) The density of $-f_\mu$ is $q_\mu=\mu p_g-p_{H_2}$, and $q_\mu'=(4p_g-p_{H_2})'+(\mu-4)p_g'>0$ by Lemma 5.20. With $q_\mu(0)=0$ this gives $q_\mu>0$, so $-f_\mu$ and $-xf_\mu=\mathcal{L}[q_\mu']$ are CM. Since $-xf_\mu(x)\to\mu/24>0$ at 0^+, Lemma 5.1 gives $\deg_{\mathrm{cm}}(-f_\mu)=1$. □

Remark 5.24. (1) Theorem 5.23(b) is the sign-change mechanism of method M9 in a different guise: for $\lambda<0$ the obstruction to degree 1 is not a vanishing moment but the exponentially small negative tail $-\frac{t^3}{12}e^{-t}$ of p'_{H_2}, which beats $|\lambda|\frac{t^2}{48}e^{-t}$ for every fixed λ; numerically the first sign change of p' lies in $(2,2.25)$, $(9,9.25)$ and $(32,32.25)$ for $\lambda=-1,-10,-100$. (2) The related conjecture of [60, Remark 7.4] (degree 4 for h_λ and $-h_\mu$) is false for the trivial reason that $h_\lambda(x)\sim-\lambda/(12x^3)$ at 0^+; this has already been recorded publicly [63] and is not claimed here.

5.7. **A partial answer for Qi's Remark 4.1 (MIA 2021).** Let $\mathcal{G}(x)=x(x\psi'(x)-1)-\frac12=\mathcal{L}w$ with $w(t)=e^t[(t-2)e^t+t+2]/(e^t-1)^3$, $w(0)=\frac16$ (note $w=f''$ with $f=t/(e^t-1)$). For $(i,j)\succ(p,q)$ (same sum, $\max(i,j)>\max(p,q)$, $i,j,p,q\ge0$) Qi [59, Thm. 3.1] proved that $\mathcal{G}_\Lambda=(-1)^{i+j}\mathcal{G}^{(i)}\mathcal{G}^{(j)}-(-1)^{p+q}\Lambda\mathcal{G}^{(p)}\mathcal{G}^{(q)}$ is CM if $\Lambda\le1$, and that $1<\mathcal{G}^{(i)}\mathcal{G}^{(j)}/(\mathcal{G}^{(p)}\mathcal{G}^{(q)})<\frac{i!j!}{p!q!}$. [59, Remark 4.1] asks: "What is the necessary and sufficient condition on $\Lambda_{i,j;p,q}$ such that" $\mathcal{G}_\Lambda$ is CM, and "What is the sharp lower bound of the left hand side inequality in (3.1)?"

Proposition 5.25. *Let* $R(t)=\int_0^1\sigma_{i+1,j+1}(v)K_t(v)dv\big/\int_0^1\sigma_{p+1,q+1}(v)K_t(v)dv$ *with* $K_t(v)=w(tv)w(t(1-v))$. *Then* $\mathcal{G}_\Lambda$ *is CM iff* $\Lambda\le\Lambda_*:=\inf_{t>0}R(t)$, *and*

$$1<\Lambda_*\le\frac{(i+1)!\,(j+1)!}{(p+1)!\,(q+1)!}<\frac{i!\,j!}{p!\,q!}.$$

In particular Qi's sufficient condition $\Lambda\le1$ *is never sharp.*

Proof. The density of $\mathcal{G}_\Lambda$ is $t^{i+j+1}\int_0^1[v^i(1-v)^j-\Lambda v^p(1-v)^q]K_t(v)dv$, so Lemma 5.2 gives the characterization. The pointwise inequality $\sigma_{i+1,j+1}>\sigma_{p+1,q+1}$ on $(0,\frac12)$ (Lemma 5.6) gives $R>1$; $R(0^+)=\frac{i!j!}{p!q!}$, and since $w(y)=(y-2)e^{-y}(1+o(1))$ and $0<w(y)e^y\le C(1+y)$, dominated convergence gives $R(\infty)=\mathrm{B}(i+2,j+2)/\mathrm{B}(p+2,q+2)=\frac{(i+1)!(j+1)!}{(p+1)!(q+1)!}>1$. Hence $\inf R>1$ and $\inf R\le R(\infty)$. The last inequality of the display is $(i+1)(j+1)<(p+1)(q+1)$, i.e. $ij<pq$. □

Numerically R is *not* monotone: it decreases to an interior minimum near $t\approx$ 19–25 and then increases to $R(\infty)$ (script `R2_verify_mia_partial.py`); e.g. $\Lambda_*(2,0;1,1)\approx1.4078<\frac32$, $\Lambda_*(3,0;2,1)\approx1.8157<2$ and $\Lambda_*(3,1;2,2)\approx1.2820<\frac43$. So the sharp constant is not $R(\infty)$,

and no closed form is offered. For the second question the numerics support the answer

$$\inf_{x>0} \frac{\mathcal{G}^{(i)}(x)\mathcal{G}^{(j)}(x)}{\mathcal{G}^{(p)}(x)\mathcal{G}^{(q)}(x)} = \frac{\mu_i\mu_j}{\mu_p\mu_q} \qquad \text{(the value at } x = 0^+\text{)},$$

where $\mu_0 = \frac{1}{2}$, $\mu_1 = 1$ and $\mu_k = k!(k-1)\zeta(k)$ for $k \geq 2$ (e.g. $\pi^2/6$ for $(2,0;1,1)$). *But* the Wronskian density $\int_0^t \alpha(u)\gamma(t-u)(t-2u)du$ changes sign (near $t \approx 50$–60 for $(2,0;1,1)$), so our method does not prove monotonicity of the ratio. This question remains open.

5.8. **Verification and limits.** Independent checks (scripts in the area directory; each prints PASS/FAIL): `R2_verify_kernels.py` (the symbolic identities and the coefficient argument of Lemma 5.5); `R2_verify_remark17.py` (for seven pairs (m,n): signs of $(-1)^j H_1^{(j)}$ and $(-1)^j(-H_{\Omega^\sharp})^{(j)}$ for $j \leq 8$ at eight points in $[0.02, 60]$, with exact polygamma derivatives at 60 digits; failure of CM just outside both thresholds; the Hurwitz form of $-H_{\Omega^\sharp}$; the splitting identity of Remark 5.10; positivity of the density on $[0.05, 120]$); `R2_verify_remark18.py` and `R2_verify_yhq_remark4.py` (the same checks for W_1 and G_1, the monotonicity of h in Lemma 5.6, and the constants Ω_*, Λ^*); `R2_verify_yhq_remark2.py` (the identity of Proposition 5.18 and 20000 random sign tests); `R2_verify_qa2019_degrees.py` (partial fractions, a numerical Laplace check of p_{H_2}, p_3, p_g, the exponential-polynomial identities, the exact Taylor certificates of Lemma 5.20 with the tail bound, the endpoint asymptotics); `R2_verify_mia_partial.py` (Proposition 5.25 and the numerics after it). The finite derivative checks are consistency tests only; the proofs do not rely on them, except for the exact finite coefficient check in Lemma 5.20.

What remains open here: closed forms (if any) for Ω_* and Λ^* in Theorems 5.12(ii) and 5.15(ii); the exact values of $\deg_{\mathrm{cm}} f_\lambda$ for $\lambda < 0$; both questions of [59, Remark 4.1] beyond Proposition 5.25; and the two-variable curvature conjecture of Le Brigant–Preston–Puechmorel recalled in [57, 62], which we did not attempt.

6. Composite integer parts of powers and composite linear recurrences

This section treats two problems posed by other authors in the area of method M10 (trace congruences of powers in residue rings and return of finite-field powers).

- §6.1: the conjecture of Hahn, Ismailescu, Kim and Kim that there are infinitely many biquadratic Pisot numbers all of whose integer parts of powers are $\equiv 3, 4 \pmod 6$ [66, §5]. The literal statement is easy: powers of a single example with small conjugates of total size < 1, such as the authors' own $z^4 - 64z^3 + 8z + 1$, already give infinitely many. The content of Theorem 6.2 is a field-by-field version with infinitely many multiplicatively independent examples in every real biquadratic field, valid for all $n \geq 1$.
- §6.2: Šiurys's conjecture that for every $k > 2$ some sequence obeying the k-step Fibonacci recurrence has coprime positive initial values and no prime terms [67, Conjecture 1]. We obtain a *partial* result (Theorem 6.12): the conjecture holds for all k in nine residue classes modulo 24 and two residue classes modulo 120 (the latter inside the hardest class $k \equiv 11 \pmod{12}$), for all $k \not\equiv 0, 3, 11 \pmod{12}$ outside an explicit set of density zero, and, by computer, for every $3 \leq k \leq 130$. The general conjecture remains open.

§6.3 records the status of further targets that were examined.

6.1. **Biquadratic Pisot numbers with composite integer parts of powers.** A *Pisot number* is a real algebraic integer $\alpha > 1$ whose other conjugates lie in the open unit disc. Following [66], a *biquadratic Pisot number* is a Pisot number α with $\mathbb{Q}(\alpha)$ a biquadratic field $\mathbb{Q}(\sqrt{p}, \sqrt{q})$ ($p, q > 1$ squarefree; in [66] also coprime). Such fields are totally real of degree 4.

Problem 6.1 (Hahn–Ismailescu–Kim–Kim [66, §5, after Table 1])**.** After listing in Table 1 (caption: "Biquadratic Pisot numbers $\alpha < 100$ with the property that $\lfloor \alpha^n \rfloor \equiv 3, 4$ for all $n \geq 3$")

the examples they found, with the remark "For all these entries $\lfloor \alpha^n \rfloor \equiv 3, 4 \pmod 6$ as soon as $n \geq 3$", the authors write: "We conjecture that there are infinitely many biquadratic Pisot numbers with the property described above."

The tool in [66] is their Theorem 6: if the minimal polynomial $z^4 - az^3 + bz^2 - cz + d$ of a biquadratic Pisot number satisfies $a \equiv c \equiv 4 \pmod 6$, $b \equiv 0 \pmod 3$, $d \equiv 1 \pmod 3$, then $\lfloor \alpha^n \rfloor \equiv 3, 4 \pmod 6$ for all sufficiently large n. The difficulty addressed below is to produce infinitely many genuinely different such numbers, with control for *every* n.

Theorem 6.2. *Let K be a real biquadratic field. There is an infinite sequence $\alpha_1, \alpha_2, \ldots$ of Pisot numbers in K, no two of which are multiplicatively dependent, such that for every j:*

(i) $\mathbb{Q}(\alpha_j) = K$;

(ii) *the minimal polynomial P_j of α_j satisfies $P_j(z) \equiv (z-4)^4 \pmod 6$ coefficientwise; in particular $P_j(z) = z^4 - az^3 + bz^2 - cz + d$ with $a \equiv c \equiv d \equiv 4$ and $b \equiv 0 \pmod 6$, so the hypotheses of* [66, Theorem 6] *hold;*

(iii) *for every $n \geq 1$, $\left\lfloor \alpha_j^n \right\rfloor \equiv 3$ or $4 \pmod 6$ and $\left\lfloor \alpha_j^n \right\rfloor$ is composite.*

Corollary 6.3. *The conjecture in Problem 6.1 is true. More precisely, there are infinitely many biquadratic Pisot numbers α with $\lfloor \alpha^n \rfloor \equiv 3, 4 \pmod 6$ for all $n \geq 1$; they can be taken in each of infinitely many fields $\mathbb{Q}(\sqrt{p}, \sqrt{q})$ with p, q coprime squarefree (e.g. $\mathbb{Q}(\sqrt{2}, \sqrt{q})$, q an odd prime), infinitely many in each such field, and pairwise multiplicatively independent (so none is a power of another). As noted in Remark 6.6(b), the literal conjecture already follows from the powers of a single example; the new content is the independence and the statement for all $n \geq 1$ in every field.*

We use two lemmas; the first is a standard consequence of Dirichlet's unit theorem.

Lemma 6.4 (Pisot units)**.** *Let $K \neq \mathbb{Q}$ be a number field with a real embedding σ_1. There is a unit $u \in \mathcal{O}_K^\times$ with $\sigma_1(u) > 1$ and $|\sigma(u)| < 1$ for every embedding $\sigma \neq \sigma_1$ of K into $\mathbb{C}$. Every such u satisfies $\mathbb{Q}(u) = K$, and so does every power u^e, $e \geq 1$.*

Proof. Let $v_1, \ldots, v_s$ be the archimedean places (v_1 induced by σ_1), $n_v \in \{1, 2\}$ the local degrees, and $\ell(x) = (\log |x|_v)_v \in \mathbb{R}^s$. By Dirichlet's theorem $\ell(\mathcal{O}_K^\times)$ is a lattice of full rank $s - 1 \geq 1$ in $H = \{y : \sum_v n_v y_v = 0\}$. The set $C = \{y \in H : y_{v_1} > 0,\ y_v < 0\ (v \neq v_1)\}$ is a nonempty open cone in H (it contains $y_{v_1} = \sum_{v \neq v_1} n_v$, $y_v = -1$), hence contains a ball $B(c, r)$ and therefore all balls $B(tc, tr)$, $t > 0$. For t larger than the covering radius of the lattice divided by r, $B(tc, tr)$ contains some $\ell(u)$. Replacing u by u^2 makes $\sigma_1(u) > 0$. If $\mathbb{Q}(u^e) = F \subsetneq K$, then $[K : F] \geq 2$ embeddings of K restrict to $\sigma_1|_F$, so some $\sigma \neq \sigma_1$ has $\sigma(u^e) = \sigma_1(u^e)$, contradicting $|\sigma(u)| < 1 < \sigma_1(u)$. □

Lemma 6.5 (Trace congruence)**.** *Let K be a number field of degree D, $c \in \mathbb{Z}$, $M \geq 2$, and $\alpha \in \mathcal{O}_K$ with $\alpha \equiv c \pmod{M\mathcal{O}_K}$. Then the characteristic polynomial of α is $\equiv (z-c)^D \pmod M$ coefficientwise, and $\mathrm{Tr}_{K/\mathbb{Q}}(\alpha^n) \equiv Dc^n \pmod M$ for all $n \geq 0$.*

Proof. Write $\alpha = c + M\beta$, $\beta \in \mathcal{O}_K$. With $\beta_1, \ldots, \beta_D$ the conjugates of β, the characteristic polynomial of α is $\prod_i (z - c - M\beta_i) = Q(z - c)$, where $Q(w) = \prod_i (w - M\beta_i) = w^D + \sum_{t \geq 1} (-M)^t e_t(\beta) w^{D-t}$ and the $e_t(\beta) \in \mathbb{Z}$; all non-leading coefficients of Q are divisible by M, hence so are those of $Q(z-c) - (z-c)^D$. Also $\alpha^n - c^n \in M\mathcal{O}_K$ and $\mathrm{Tr}(M\mathcal{O}_K) \subseteq M\mathbb{Z}$. □

Proof of Theorem 6.2. Fix a real embedding σ_1 of K (all four embeddings are real) and take u as in Lemma 6.4. Since u is a unit, its class in the finite group $(\mathcal{O}_K / 6\mathcal{O}_K)^\times$ has some order $m \geq 1$. For $j \geq 1$ put $\alpha_j = 4u^{jm}$, so $\alpha_j \equiv 4 \pmod{6\mathcal{O}_K}$. Let $\rho = \max_{\sigma \neq \sigma_1} |\sigma(u)| < 1$ and choose j_0 with $12\rho^{j_0 m} < 1$; we use $j \geq j_0$ and relabel.

(i) and the Pisot property. $\mathbb{Q}(\alpha_j) = \mathbb{Q}(u^{jm}) = K$ by Lemma 6.4, so the conjugates of α_j are $\sigma(\alpha_j)$, σ running over the four embeddings. Now $\sigma_1(\alpha_j) > 4$ and

$$S_j := \sum_{\sigma \neq \sigma_1} |\sigma(\alpha_j)| \leq 12\rho^{jm} < 1. \tag{15}$$

Thus $\alpha_j = \sigma_1(\alpha_j)$ is a Pisot number of degree 4 in K.

(ii) is Lemma 6.5 with $c = 4$, $M = 6$, $D = 4$; $(z-4)^4 = z^4 - 16z^3 + 96z^2 - 256z + 256$ and $16 \equiv 256 \equiv 4$, $96 \equiv 0 \pmod 6$.

(iii) Let $T_n = \mathrm{Tr}(\alpha_j^n) \in \mathbb{Z}$ and $E_n = \sum_{\sigma \neq \sigma_1} \sigma(\alpha_j)^n \in \mathbb{R}$, so $\alpha_j^n = T_n - E_n$. By Lemma 6.5, $T_n \equiv 4 \cdot 4^n \equiv 4 \pmod 6$ for $n \geq 1$ (as $4^t \equiv 4 \pmod 6$ for $t \geq 1$). By (15), $|E_n| \leq \sum_{\sigma \neq \sigma_1} |\sigma(\alpha_j)|^n \leq S_j^n < 1$, so $\left\lfloor \alpha_j^n \right\rfloor = T_n - \lceil E_n \rceil$ with $\lceil E_n \rceil \in \{0, 1\}$. Hence $\left\lfloor \alpha_j^n \right\rfloor \equiv 4$ or $3 \pmod 6$. Since $\left\lfloor \alpha_j^n \right\rfloor \geq \lfloor \alpha_j \rfloor \geq 4$, it is divisible by 2 (residue 4) or by 3 (residue 3) and exceeds that prime; so it is composite.

Independence. If $\alpha_j^x = \alpha_{j'}^y$ with integers $(x, y) \neq (0, 0)$, taking norms gives $4^{4x} = 4^{4y}$, so $x = y \neq 0$, and then $u^{mx(j-j')} = 1$; as u is not a root of unity, $j = j'$. □

Proof of Corollary 6.3. Apply Theorem 6.2 to $K = \mathbb{Q}(\sqrt{2}, \sqrt{q})$ for each odd prime q; these fields are pairwise distinct (their quadratic subfields differ) and $(2, q)$ are coprime squarefree. Each α_j has $\left\lfloor \alpha_j^n \right\rfloor \equiv 3, 4 \pmod 6$ for all $n \geq 1$, a fortiori for $n \geq 3$. □

Remark 6.6 (Examples and the literal conjecture). (a) Let $K = \mathbb{Q}(\sqrt{2}, \sqrt{5})$ and $u = (1+\sqrt{2})(2+\sqrt{5})(3+\sqrt{10})$, so $u \approx 63.02$; the other conjugates of u have absolute value $\approx 0.092,\ 0.285,\ 0.603$ (a product of quadratic units is a Pisot unit as soon as the three logarithms $x \log \varepsilon_p,\ y \log \varepsilon_q,\ z \log \varepsilon_{pq}$ of the factors satisfy the strict triangle inequalities), its order modulo $6\mathcal{O}_K$ is $m = 4$, and already $\alpha = 4u^4$ has $S = 0.554 < 1$ and minimal polynomial

$$z^4 - 63092752\, z^3 + 34947168\, z^2 - 884992\, z + 256 \equiv (z-4)^4 \pmod 6.$$

Hence every $\alpha_j = 4u^{4j}$ $(j \geq 1)$ has $\left\lfloor \alpha_j^n \right\rfloor \equiv 3, 4 \pmod 6$ for all $n \geq 1$. The script `verify_biquadratic.py` recomputes these data exactly and the same construction in the fields $\mathbb{Q}(\sqrt{p}, \sqrt{q})$ for $(p, q) = (2,3), (2,5), (3,5), (2,7), (5,13), (6,10), (3,7), (11,13)$ (the field $\mathbb{Q}(\sqrt{6}, \sqrt{10})$ has no coprime pair of radicands), checking minimal polynomials, the congruence mod 6, $S < 1$, and $\lfloor \alpha^n \rfloor \bmod 6$ for $1 \leq n \leq 120$.

(b) The literal conjecture also follows from a single example: for the largest root α_* of $z^4 - 64z^3 + 8z + 1$ (Table 1 of [66]) the three small conjugates have $\sum |\cdot| \approx 0.809 < 1$, and the polynomial is $\equiv (z+1)^4 \pmod 2$ and $\equiv (z-1)^4 \pmod 3$, so all reduced roots equal 1, $\mathrm{Tr}(\alpha_*^n) \equiv 4 \pmod 6$, and the argument above gives $\lfloor \alpha_*^n \rfloor \equiv 3, 4 \pmod 6$ for all $n \geq 1$; the powers α_*^e $(e \geq 1)$ are then infinitely many biquadratic Pisot numbers with the property. Theorem 6.2 is the non-degenerate form: infinitely many multiplicatively independent examples in every real biquadratic field. We also checked all ten entries of Table 1 of [66] that are given by explicit polynomials (Galois group V_4, Pisot, congruences, and $\lfloor \alpha^n \rfloor \bmod 6 \in \{3, 4\}$ for $3 \leq n \leq 120$; for three of them $n = 2$, and for one of these also $n = 1$, fall outside $\{3, 4\}$, consistent with the restriction $n \geq 3$ in [66]).

(c) The proof gives more: if $K \neq \mathbb{Q}$ is any number field with a real embedding and $[K : \mathbb{Q}] \not\equiv 0, 2 \pmod 6$, then K contains infinitely many multiplicatively independent Pisot numbers α with $\mathbb{Q}(\alpha) = K$ and all $\lfloor \alpha^n \rfloor$ $(n \geq 1)$ composite: take $\alpha = c\, u^{jm}$ with $c = 4$ if $[K : \mathbb{Q}] \equiv 1 \pmod 3$ (then $\mathrm{Tr}(\alpha^n) \equiv 4$) and $c = 3$ if $[K : \mathbb{Q}]$ is odd (then $\mathrm{Tr}(\alpha^n) \equiv 3 \pmod 6$, and $\lfloor \alpha^n \rfloor \equiv 3$ or $2 \pmod 6$), with $j \geq j_0$ large enough that the conjugates other than α have total size < 1 and $\lfloor \alpha^n \rfloor > 3$.

6.2. **k-step Fibonacci-like sequences without primes.** For $k \geq 2$ and integers $a_0, \ldots, a_{k-1}$ let $S_k(a_0, \ldots, a_{k-1}) = (x_n)_{n\geq 0}$ be given by $x_n = a_n$ $(n < k)$ and $x_n = x_{n-1} + \cdots + x_{n-k}$ $(n \geq k)$. Graham [69] found coprime positive a_0, a_1 with $S_2(a_0, a_1)$ free of primes; Šiurys [67] proved the analogue for $2 \leq k \leq 10$ and for $k \equiv 79 \pmod{120}$, and stated:

Problem 6.7 (Šiurys [67, Conjecture 1]; also [68])**.** "Let $k > 2$ be some fixed positive integer. Then there exist positive integers $a_0, a_1, \ldots, a_{k-1}$ such that $\gcd(a_0, a_1, \ldots, a_{k-1}) = 1$ and the sequence $S_k(a_0, a_1, \ldots, a_{k-1})$ contains no prime numbers."

To our knowledge (searches in October 2026) the only later progress is Lunev's smaller initial values for $k = 3$ [70]. Throughout, $k \geq 3$,

$$f_k(X) = X^k - X^{k-1} - \cdots - X - 1, \quad g_k(X) = (X-1)f_k(X) = X^{k+1} - 2X^k + 1.$$

For a prime p, a *solution mod p* is a sequence $(y_n)_{n\geq 0}$ in $\mathbb{F}_p$ with $y_n = y_{n-1} + \cdots + y_{n-k}$ for $n \geq k$; it is *nonzero* if $(y_0, \ldots, y_{k-1}) \neq 0$. We say that it *vanishes on c mod d* if $y_n = 0$ for all $n \geq 0$ with $n \equiv c \pmod{d}$.

Lemma 6.8 (Assembly)**.** *Let $p_1, \ldots, p_r$ be distinct primes and $y^{(i)}$ a nonzero solution mod p_i $(1 \leq i \leq r)$ such that every $n \geq 0$ has $y_n^{(i)} = 0$ for some i. Then there are positive integers $a_0, \ldots, a_{k-1}$ with $\gcd = 1$ such that every term of $S_k(a_0, \ldots, a_{k-1})$ is composite.*

Proof. Let $\Pi = p_1 \cdots p_r$ and, by the Chinese remainder theorem, $b_j \in [1, \Pi]$ with $b_j \equiv y_j^{(i)} \pmod{p_i}$ for all i, j. Put $a_j = b_j + \Pi$ for $1 \leq j \leq k-1$ and $a_0 = b_0 + \Pi t$ with $t \geq 1$ chosen below. By induction on n, $x_n \equiv y_n^{(i)} \pmod{p_i}$ for all n, i, so every x_n has some p_i as a divisor; moreover $x_n \geq \min_j a_j > \Pi \geq p_i$ (for $n \geq k$ the term x_n is a sum of k earlier positive terms), so x_n is composite. If a prime q divides all a_j, then $q \neq p_i$ (since some $y_j^{(i)} \neq 0$) and $q \mid G := \gcd(a_1, \ldots, a_{k-1})$, a fixed positive integer. Choosing t (Chinese remainder theorem) with $b_0 + \Pi t \not\equiv 0 \pmod{q}$ for the finitely many primes $q \mid G$, $q \nmid \Pi$, gives $\gcd(a_0, \ldots, a_{k-1}) = 1$. □

Lemma 6.9 (The prime 2)**.** *For $0 \leq a < b \leq k$ the sequence with $y_n = 1$ if $n \equiv a$ or $b \pmod{k+1}$ and $y_n = 0$ otherwise is a nonzero solution mod 2.*

Proof. For $n \geq k$, $y_n - \sum_{i=1}^{k} y_{n-i} \equiv \sum_{i=0}^{k} y_{n-i} \pmod 2$, and any $k+1$ consecutive terms of this $(k+1)$-periodic sequence contain exactly two 1's. Also $y_a = 1$ with $a < k$. □

Thus the prime 2 alone annihilates all residues mod $k+1$ except two freely chosen ones. (In fact the solutions mod 2 are exactly the $(k+1)$-periodic sequences of even weight, because $g_k \equiv X^{k+1} + 1 \pmod 2$.) Odd primes are used through pairs of roots of g_k with root-of-unity ratio:

Lemma 6.10 (Pair patterns)**.** *Let p be an odd prime, $d \geq 3$ with $p \nmid d$, $\zeta \in \overline{\mathbb{F}}_p$ of multiplicative order d, and $\theta \in \overline{\mathbb{F}}_p$ with $\theta \neq 1$, $\zeta\theta \neq 1$ and $g_k(\theta) = g_k(\zeta\theta) = 0$. Then for every $c \in \mathbb{Z}$ there is a nonzero solution mod p vanishing on c mod d.*

Proof. Let $F = \mathbb{F}_p(\zeta, \theta)$. Every Galois conjugate of θ or $\zeta\theta$ is a root of g_k different from 1, hence a root of f_k, so $n \mapsto \rho^n$ is a solution (over F) for each such ρ. For $\lambda \in F$ put $y_n(\lambda) = \mathrm{Tr}_{F/\mathbb{F}_p}\big(\lambda\,\theta^n(1 - \zeta^{\,n-c})\big) = \mathrm{Tr}(\lambda\theta^n) - \mathrm{Tr}(\lambda\zeta^{-c}(\zeta\theta)^n)$, a solution mod p with values in $\mathbb{F}_p$. If $n \equiv c \pmod{d}$ then $\zeta^{n-c} = 1$ and $y_n(\lambda) = 0$. Finally, taking the representative $0 \leq c < d$, the element $\theta^{c+1}(1-\zeta)$ is nonzero and the trace form of $F/\mathbb{F}_p$ is nondegenerate, so $y_{c+1}(\lambda) \neq 0$ for some λ; this solution is nonzero. □

For $\zeta \in \overline{\mathbb{Q}}$ of order d with $\zeta^k \neq 1 \neq \zeta^{k+1}$, the identity $\zeta^k(\zeta\theta - 2) = \theta - 2$ for $\theta = 2(\zeta^k - 1)/(\zeta^{k+1} - 1)$ gives $g_k(\zeta\theta) = g_k(\theta)$; reducing modulo primes above p yields pairs as in Lemma 6.10.

For $d \in \{3,4,6\}$ the relevant numbers are rational integers of binomial shape, and Zsigmondy's theorem controls their prime factors:

Lemma 6.11 (Four explicit families)**.** *Let p be an odd prime and $\iota, \omega \in \overline{\mathbb{F}}_p$ roots of X^2+1, X^2+X+1. Put*

$$\begin{aligned} M_k &= 2^{(k+1)/2} - (-1)^{(k-1)/4} && (k \equiv 1 \bmod 4),\\ A_k &= 1 + (-1)^{(k+2)/4}\, 2^{(3k+2)/2} && (k \equiv 2 \bmod 4),\\ B_k &= (-2)^{k+1} - 1 && (k \equiv 1 \bmod 3),\\ C_k &= (-3)^{k/2} + 1 && (k \equiv 2 \bmod 6). \end{aligned}$$

Lemma 6.10 applies in each of the following cases:

(I) $k \equiv 1 \pmod 4$, $p \mid M_k$*:* $d=4$, $\zeta=\iota$, $\theta = 1-\iota$*;*
(II) $k \equiv 2 \pmod 4$, $p \mid A_k$, $p \neq 5$*:* $d=4$, $\zeta=\iota$, $\theta = 2-2\iota$*;*
(III) $k \equiv 1 \pmod 3$, $p \mid B_k$, $p \neq 3$*:* $d=3$, $\zeta=\omega$, $\theta = -2\omega$*;*
(IV) $k \equiv 2 \pmod 6$, $p \mid C_k$, $p \neq 3$*:* $d=6$, $\zeta=-\omega^2$, $\theta = 1-\omega$.

Proof. In each case $\zeta\theta$ is the image of θ under $\iota \mapsto -\iota$ (resp. $\omega \mapsto \omega^2$), namely $1+\iota$, $2+2\iota$, $-2\omega^2$, $1-\omega^2$, and we show that $g_k(\theta)$ equals $\mp M_k$, A_k, $-B_k$, C_k as an identity in $\mathbb{Z}[\iota] = \mathbb{Z}[X]/(X^2+1)$ (resp. $\mathbb{Z}[\omega]$); applying the automorphism, $g_k(\zeta\theta)$ is the same integer, so both vanish mod p. (I) $k = 4t+1$: $(1-\iota)^2 = -2\iota$, $(1-\iota)^4 = -4$, and $g_k(1-\iota) = -(1-\iota)^k(1+\iota)+1 = 1-2(1-\iota)^{4t} = 1-(-1)^t 2^{2t+1}$. (II) $k = 2h$, h odd: $(2-2\iota)^2 = -8\iota$ and $g_k(2-2\iota) = (-8\iota)^h(-2\iota)+1 = 1+(-1)^{(h+1)/2}2^{3h+1} = A_k$. (III) $\omega^k = \omega$ and $\omega(\omega+1) = -1$, so $g_k(-2\omega) = (-2\omega)^k(-2)(\omega+1)+1 = 1-(-2)^{k+1}$. (IV) $k = 2h$, $h \equiv 1 \pmod 3$: $(1-\omega)^2 = -3\omega$ and $1-\omega-2 = \omega^2$, so $g_k(1-\omega) = (-3)^h\omega^h\omega^2 + 1 = (-3)^h + 1$. Side conditions: the orders of $\iota, \omega, -\omega^2$ are $4, 3, 6$ for odd p ($p \neq 3$ for ω and $-\omega^2$); $\theta - 1$ and $\zeta\theta - 1$ are $-\iota, \iota$ in (I) and $-\omega, -\omega^2$ in (IV); in (II) they are $1 \mp 2\iota$, which vanish only if $\iota = \pm 1/2$, i.e. $5 = 0$; in (III) they are $-1-2\omega$, $-1-2\omega^2$, which vanish only if $3 = 0$. Finally $p \nmid d$: p is odd, and $p \neq 3$ in (III), (IV) (note $3 \nmid C_k$). □

We use the theorem of Bang and Zsigmondy [72]: for $a \in \{2,3\}$ and $\delta \geq 2$, $a^\delta - 1$ has a prime divisor q with $\mathrm{ord}_q(a) = \delta$ (a *primitive divisor* $q_\delta(a)$), except for $(a,\delta) = (2,6)$ and $(3,2)$. Primitive divisors for distinct δ are distinct, $q_2(2) = 3$ and $q_4(2) = 5$.

Theorem 6.12 (Šiurys's conjecture: partial resolution)**.** *Let $k \geq 3$. In each of the following cases there are positive integers $a_0, \ldots, a_{k-1}$ with greatest common divisor* 1 *such that* every *term of $S_k(a_0, \ldots, a_{k-1})$ is composite:*

(a) $k \equiv 2, 5, 7, 10, 13, 17, 18, 19, 21 \pmod{24}$*, or* $k \equiv 11, 71 \pmod{120}$*;*
(b) $k \not\equiv 0, 3, 11 \pmod{12}$ *and* $k \notin \mathcal{E} \cup \{5, 7, 10\}$*, where* $\mathcal{E} = \mathcal{E}_1 \cup \cdots \cup \mathcal{E}_5$ *with* $\mathcal{E}_1 = \{k \equiv 1\ (8) : 2^{(k+1)/2} - 1$ *prime*$\}$, $\mathcal{E}_2 = \{k \equiv 14\ (24) :\ k/2$ *prime*$\}$, $\mathcal{E}_3 = \{k \equiv 4, 10\ (12) :\ k+1$ *prime*$\}$, $\mathcal{E}_4 = \{k \equiv 6\ (24) :\ \tau((3k+2)/4) \leq 4\}$, $\mathcal{E}_5 = \{2^j : j \geq 3\}$*;* $\mathcal{E}$ *has natural density* 0*, so the conjecture holds on a set of k of lower density at least* $3/4$*;*
(c) $3 \leq k \leq 130$ *(computer-certified; see Remark 6.13).*

Proof. By Lemma 6.8 it suffices to exhibit coverings by nonzero solutions mod distinct primes. We use four constructions.

Construction I ($k \equiv 1 \pmod 4$, M_k has two distinct prime factors p_1, p_2, necessarily odd). Take the solution mod 2 of Lemma 6.9 with $(a,b) = (1,3)$ (distinct mod $k+1 \geq 6$), and by Lemma 6.11(I) solutions mod p_1, p_2 vanishing on 1 mod 4 and 3 mod 4. Since $k+1$ is even, every $n \equiv 1$ or $3 \pmod{k+1}$ is odd; so even n are covered by 2 and odd n by p_1 or p_2.

Construction II ($k \equiv 2 \pmod 4$, A_k has four distinct prime factors $\neq 5$, necessarily odd): by Lemma 6.11(II), solutions mod these primes vanishing on $0, 1, 2, 3 \bmod 4$.

Construction III ($k \equiv 1 \pmod 3$, B_k has three distinct prime factors $\neq 3$, necessarily odd): solutions vanishing on $0, 1, 2 \bmod 3$ (Lemma 6.11(III)).

Construction IV ($k \equiv 2 \pmod 6$, $k \geq 8$, C_k has two distinct odd prime factors, necessarily ≥ 5 since $3 \nmid C_k$). Here $3 \mid k+1$; take the solution mod 2 with $(a, b) = (0, 3)$ and solutions mod p_1, p_2 vanishing on 0 mod 6 and 3 mod 6. Every $n \equiv 0, 3 \pmod{k+1}$ is $\equiv 0 \pmod 3$, hence covered.

Construction V (class-wide coverings). If y is a d-periodic sequence over $\mathbb{F}_p$ with zero period sum and $y_n = y_{n-1} + \cdots + y_{n-s}$ (indices mod d), then $n \mapsto y_{n \bmod d}$ is a solution mod p for *every* $k \equiv s \pmod d$, because $\sum_{i=1}^{k} y_{n-i} = \frac{k-s}{d} \sum_{t<d} y_t + \sum_{i=1}^{s} y_{n-i}$ (this is the mechanism of [67, Lemma 4.1]). Likewise, if $l \mid k+1$, every l-periodic 0/1 pattern with an even number of ones per period is a solution mod 2 (Lemma 6.9's argument). For $k \equiv 11 \pmod{120}$ we use: the prime 2 with x_n odd exactly for $n \equiv 0, 6 \pmod{12}$; and sequences with $s = 11$ vanishing on $n \equiv 4 \pmod 5$ ($p = 19$, $d = 15$), $n \equiv 0 \pmod 5$ ($p = 29$, $d = 20$), $n \equiv 3 \pmod 5$ ($p = 31$, $d = 30$), $n \equiv 2 \pmod{10}$ ($p = 521$, $d = 40$) and $n \equiv 6 \pmod{10}$ ($p = 49921$, $d = 120$). A term not divisible by 2 has $n \equiv 0, 6 \pmod{12}$, so n is even; it is covered by $29, 31, 19$ when $n \equiv 0, 3, 4 \pmod 5$, and by 521 or 49921 when $n \equiv 2$ or $6 \pmod{10}$. For $k \equiv 71 \pmod{120}$ we use: 2 with x_n odd exactly for $n \equiv 12, 20 \pmod{24}$ (so $n \equiv 4 \pmod 8$); $p = 31$ ($d = 30$, $s = 11$) on $n \equiv 0 \pmod 5$; $p = 181$ ($d = 60$, $s = 11$) on $n \equiv 4 \pmod 5$; and a 40-periodic sequence mod 3 with $s = 31$ vanishing at $n \equiv 12, 28, 36 \pmod{40}$, which are the remaining classes $n \equiv 4 \pmod 8$, $n \equiv 1, 2, 3 \pmod 5$. The explicit periodic sequences are stored in `siurys/fixed_D120_0_119.json` and checked by `verify_siurys_fixed.py` (zero sum, cyclic s-step relation, zeros, covering, and composite initial segments for three values of k in each class).

It remains to count prime factors with the Bang–Zsigmondy theorem.

I, $k \equiv 5 \pmod 8$, $k \geq 13$. $M_k = 2^m + 1$ with $m = (k+1)/2$ odd, $m \geq 7$; $3 \mid M_k$, and $q_{2m}(2) \neq 3$ divides $2^{2m} - 1$ but not $2^m - 1$, hence divides M_k. *I,* $k \equiv 17 \pmod{24}$. $M_k = 2^m - 1$ with $m \equiv 9 \pmod{12}$: $7 = q_3(2)$ and $q_m(2) \neq 7$ divide M_k. *I,* $k \equiv 1 \pmod 8$. $M_k = 2^m - 1$; if it is not prime it has two distinct prime factors, as $2^m - 1$ is not a proper power ($\equiv 3 \pmod 4$ excludes squares; $y^j + 1 = (y+1)(y^{j-1} - \cdots + 1)$ with the second factor odd and > 1 excludes odd $j \geq 3$). The exceptions form $\mathcal{E}_1$.

II, $k \equiv 2 \pmod 8$, $k \geq 18$. Write $k = 8t + 2$, $t \geq 2$; then $A_k = 1 - 2^M$ with $M = 12t + 4 = 4(3t+1)$, $3 \nmid M$. The primes $q_\delta(2)$, $\delta \mid M$, $\delta \notin \{1, 4\}$, are distinct, odd, $\neq 5$, and number $\tau(M) - 2$. If $u = 3t+1$ is odd, $\tau(M) = 3\tau(u) \geq 6$; if $u = 2^b w$ (w odd), $\tau(M) = (b+3)\tau(w) \geq 6$ unless $w = 1$, $b \leq 2$, i.e. $u = 4$, $t = 1$. So $\tau(M) - 2 \geq 4$. *II,* $k \equiv 6 \pmod 8$. $A_k = 2^{2u} + 1$ with $u = (3k+2)/4$ odd; for $\delta' \mid u$, $\delta' > 1$, $q_{4\delta'}(2)$ divides $2^{4u} - 1$ but not $2^{2u} - 1$, hence divides A_k, and $\neq 5$. This gives $\tau(u) - 1 \geq 4$ primes unless $\tau(u) \leq 4$ (the set $\mathcal{E}_4$ for $k \equiv 6 \pmod{24}$).

III, $k \equiv 7 \pmod{12}$, $k \geq 19$. $B_k = 2^N - 1$, $N = k + 1 \equiv 8 \pmod{12}$: the primes $q_4(2) = 5$, $q_{N/2}(2)$, $q_N(2)$ (with $N/2 \geq 10$) are distinct and $\neq 3$. *III,* $k \equiv 4, 10 \pmod{12}$. $N = k + 1$ is odd with $3 \nmid N$ and $-B_k = 2^N + 1$; for $\delta' \mid N$, $\delta' > 1$, $q_{2\delta'}(2) \mid 2^N + 1$ and $\neq 3$, giving $\tau(N) - 1 \geq 3$ primes unless N is prime ($\mathcal{E}_3$); note $N = q^2$ is impossible, since $N \equiv 5, 11 \pmod{12}$ while $q^2 \equiv 1 \pmod{12}$ for primes $q \geq 5$. *IV,* $k \equiv 8 \pmod{12}$. $h = k/2 = 2^v h'$ with $v \geq 1$, h' odd, and $C_k = 3^h + 1$; for $\delta'' \mid h'$ the prime $q_{2^{v+1}\delta''}(3)$ (≥ 5) divides $3^{2h} - 1$ but not $3^h - 1$, so divides C_k; this gives $\tau(h') \geq 2$ primes unless h is a power of 2 ($\mathcal{E}_5$). *IV,* $k \equiv 2 \pmod{12}$. $h = k/2$ is odd and $-C_k = 3^h - 1$; the primes $q_\delta(3)$, $\delta \mid h$, $\delta > 1$, number $\tau(h) - 1 \geq 2$ unless h is prime ($\mathcal{E}_2$ when $k \equiv 14 \pmod{24}$).

(a) By the above, Construction I covers $k \equiv 5, 13, 21, 17 \pmod{24}$ ($k \geq 13$), II covers $k \equiv 2, 10, 18 \pmod{24}$ ($k \geq 18$), III covers $k \equiv 7, 19 \pmod{24}$ ($k \geq 19$); the remaining $k \in \{5, 7, 10\}$ are in (c) (and in [67]); Construction V covers $k \equiv 11, 71 \pmod{120}$.

(b) Residues mod 24 not in (a) and not $\equiv 0, 3, 11 \pmod{12}$: $k \equiv 1, 9$ (I, exceptions $\mathcal{E}_1$), $k \equiv 14$ (IV, exceptions $\mathcal{E}_2$), $k \equiv 4, 16, 22$ (III, exceptions in $\mathcal{E}_3$), $k \equiv 6$ (II, exceptions $\mathcal{E}_4$), $k \equiv 8, 20$ (IV, exceptions $\mathcal{E}_5$). Each $\mathcal{E}_i$ has density 0: $\mathcal{E}_1, \mathcal{E}_2, \mathcal{E}_3$ lie in images of the set of primes under affine

maps; for $\mathcal{E}_4$, $\tau(u) \geq 2^{\omega(u)}$ and $\omega(u) \to \infty$ on a set of density one (Hardy–Ramanujan [77]), and $k \mapsto (3k+2)/4$ maps the class 6 mod 24 bijectively onto 5 mod 18. (The decay for $\mathcal{E}_4$ is slow: integers with $\tau(u) \leq 4$ have counting function $\asymp x \log\log x / \log x$, and about 64% of the $u = 18j+5$, $j < 10^5$, still have $\tau(u) \leq 4$; the other $\mathcal{E}_i$ are very sparse.)

(c) For each k in (c) an explicit covering was found by computer and is certified as in Remark 6.13. □

Remark 6.13 (Certificates for (c)). For each k in Theorem 6.12(c) the certificate file for k (directory `siurys/certs` of the R6 area) lists positive integers $a_0, \dots, a_{k-1}$ and covering data: optionally the prime 2 with b (pattern of Lemma 6.9 with $a = 0$), and odd primes p with either one class c mod d (type P) or several classes mod d (type R). The verifier script (`verify_siurys_certs.py`) trusts only the integers a_i and checks: $\gcd(a_i) = 1$; $\min a_i >$ every covering prime; for type P that $x_{c+dj} \equiv 0 \pmod p$ for $0 \leq j < k$, which forces $x_{c+dj} \equiv 0$ for all $j \geq 0$ because $j \mapsto x_{c+dj} \bmod p$ satisfies the order-k recurrence given by the characteristic polynomial of C^d (C the companion matrix; Cayley–Hamilton); for type R that $x_{n+d} \equiv x_n \pmod p$ for $n < k$ (hence for all n) and $x_c \equiv 0$ at the listed classes; for 2 the $(k+1)$-periodicity and the parity pattern; and that the union of the annihilated classes is $\mathbb{Z}/L\mathbb{Z}$. The primes come from two sources: pair patterns of Lemma 6.10 (prime factors below 10^7 of $\mathrm{N}_{\mathbb{Q}(\zeta_d)/\mathbb{Q}}\big(2^{k+1}(\zeta^s-1)^k\zeta^s(1-\zeta) + (\zeta^{s+1}-1)^{k+1}\big)$, $s = k \bmod d$, the numerator of $g_k(\theta)$, for $d \leq 96$, and for $d = (k+1)m$ with $2 \leq m \leq 28$ when $k+1$ is prime), and patterns from several roots of f_k that are d-th roots of unity mod p (type R); the coverings were found by depth-first search and integer programming. Of the k in (c), the cases $k \leq 10$ were known [67]. The coverings use between 3 and 33 primes (e.g. $k = 13$: the primes $2, 3, 43$; $k = 12$: eleven primes with moduli dividing 780; $k = 108$: 33 odd primes with moduli dividing 1680). The hardest k are those with $k+1$ prime and k highly divisible. Then the prime 2 leaves two classes modulo $k+1$; such a class meets every residue class modulo any d coprime to $k+1$, so the odd primes must either cover all of $\mathbb{Z}$ by themselves or use moduli divisible by $k+1$; and by Remark 6.14(i) no odd prime can use a modulus dividing k. For $k = 72, 108$ coverings by odd primes alone were found; for $k = 60, 96, 100$ the coverings use pair patterns with moduli divisible by $k+1$ (for $k = 60$, where every modulus dividing 60 is unavailable to odd primes: 23 primes, moduli $7, 8, 9, 14, 16, 18, 24, 28, 36, 63, 84$ and $122, 488, 976$, $L = 61 \cdot 1008$).

Remark 6.14 (Why $k \equiv 0, 3, 11 \pmod{12}$ remain open). (i) *For an odd prime p and a divisor d of k or of $k+1$, no nonzero solution mod p vanishes on a residue class modulo d.* Let C be the companion matrix of f_k over $\overline{\mathbb{F}}_p$, so a solution is $y_n = e_1 C^n v$ and vanishing on $c \bmod d$ means $e_1 C^c (C^d)^j v = 0$ for all $j \geq 0$, i.e. v lies in the largest C^d-invariant subspace U of $\ker(e_1 C^c)$. Every root θ of f_k satisfies $\theta^k = 1/(2-\theta)$ and $\theta^{k+1} = \theta/(2-\theta)$ (from $g_k(\theta) = 0$, and $g_k(2) = 1$), so $\theta \mapsto \theta^d$ is injective on the roots. If $p \mid d$ then $p \mid k(k+1)$ and $g_k' \equiv X^k$ resp. $2X^{k-1}$ shows that the roots are simple, so C^d is diagonalizable with distinct eigenvalues; if $p \nmid d$, each Jordan block $J_m(\theta)$ of the non-derogatory matrix C has $J_m(\theta)^d \sim J_m(\theta^d)$. In both cases C^d is non-derogatory, so C (which commutes with C^d) is a polynomial in C^d and U is C-invariant; then $e_1 C^{c+i} U = 0$ for all $i \geq 0$, i.e. the solutions with initial vector in U vanish from index c on, whence $U = 0$ (C is invertible). In particular modulus 2 is available only to the prime 2, and for $k \equiv 0$ or $11 \pmod{12}$ none of the moduli $3, 4, 6, 12$ is available to odd primes, so none of the families of Lemma 6.11 applies. (ii) For $k \equiv 3 \pmod{12}$ the moduli $3 \mid k$ and $4 \mid k+1$ are excluded by (i), so among the moduli $d \in \{3, 4, 6\}$ (those with θ quadratic) only $d = 6$ remains: $\zeta = -\omega^2$, $\theta = 4(1-\omega)/3$, for which $g_k(\theta) = 1 - 2^{2k+1}/3^{(k+1)/2}$ is rational (and the norm computed by our search equals $(2^{2k+1} - 3^{(k+1)/2})^2$ for all $k \equiv 3 \pmod{12}$, $k \leq 63$). The integer $2^{2k+1} - 3^{(k+1)/2}$ is not a difference of like powers, and we have no primitive-divisor input. (iii) For other moduli the relevant integers are norms of $2^{k+1}(\zeta^k - 1)^k \zeta^k (1-\zeta) + (\zeta^{k+1} - 1)^{k+1}$ from fields $\mathbb{Q}(\zeta_d)$ with units of infinite order, for which no Zsigmondy-type theorem is available. For example, the pair patterns with ratio of

order dividing $2(k+1)$ but not $k+1$ (then $\theta = 1+\zeta^{-1}$, $\zeta\theta = 1+\zeta$) come exactly from the prime factors of $R_k = \mathrm{Res}(f_k, (X-1)^{k+1}+1)$; for $3 \le k \le 60$ one finds $R_k = \pm W_k^2$ with $W_k \in \mathbb{Z}$, and W_k is prime exactly for $k = 3, 4, 7, 12, 16$ in this range (e.g. $W_3 = 7$, $W_4 = 29$, $W_7 = 4241$), so for these k this source supplies a single prime. Thus the full conjecture needs either new uniform families for $k \equiv 0, 3, 11 \pmod{12}$ or control of prime factors of such norms; the exceptional sets $\mathcal{E}_i$ would also have to be handled.

6.3. **Further targets examined.**

Remark 6.15 (A conjecture that is already a classical theorem). Ismailescu, Ko, Lee and Park [71, Conjecture 6.2] state: "Let $(u_n)_{n\ge 0}$ be the Lucas sequence associated to some $|a| \ge 3$ and $b = -1$. Then for any two different primes p and q we have that u_p and u_q are relatively prime." Here $u_0 = 0$, $u_1 = 1$, $u_{n+1} = au_n - u_{n-1}$, i.e. $u_n = U_n(a, 1)$ in Lucas's notation. Since $\gcd(a, 1) = 1$, (U_n) is a strong divisibility sequence, $\gcd(U_m, U_n) = |U_{\gcd(m,n)}|$ (Lucas [73]), so $\gcd(u_p, u_q) = |u_1| = 1$ for all distinct primes p, q (indeed for all coprime indices, and for every integer a). We record this only to note that the statement is not open; no new mathematics is involved.

Remark 6.16 (Varona's lacunary prime-representing series). Varona [74, §1] discusses $\beta = \sum_{k\ge 1} p_k 10^{-2^k}$ and $\beta' = \sum_{k\ge 1} p_k r^{-k^2}$ (p_k the k-th prime) and writes that "it is possible that these β are transcendental numbers", the defining series being unsuitable for Liouville's or Roth's theorems. For β this follows at once from Ridout's theorem [75] (for algebraic ξ and a finite set S of primes, $|\xi - P/Q| < Q^{-1-\varepsilon}$ has finitely many solutions with Q an S-unit): with $Q_K = 10^{2^K}$ and $P_K = Q_K \sum_{k\le K} p_k 10^{-2^k} \in \mathbb{Z}$, using $p_k \le 2^k$,

$$0 < \beta - \frac{P_K}{Q_K} \le \sum_{k>K} 2^k 10^{-2^k} \le 2^{K+2} Q_K^{-2} < Q_K^{-3/2} \qquad (K \ge 2),$$

infinitely many distinct approximations, while $\beta \notin \mathbb{Q}$. So β is transcendental. The second number β' is not covered (the approximation exponent tends to 1); its transcendence remains open. We list this as a minor, partial item.

Remark 6.17 (Open questions not closed here). (1) [66, §7] asks: "Can one find a rational number or a transcendental number that satisfies this condition?" (that $\lfloor \alpha^n \rfloor$ is composite for all but finitely many n). The existence of uncountably many such α (hence transcendental ones) is attributed in [66] to Baker and Harman, without an explicit example; an explicit example is not produced here. (2) The second question of [66, §7] (an $\alpha > 1$ with $\lfloor \alpha^n \rfloor$ prime for all large n), the totally real case of Saito's Problem 1.1 [76], and the irrationality of $\xi(2^k)$ were not attacked; the latter two are partly treated in [7, 13] and the first is far beyond the methods available.

7. Floquet isospectrality to the zero potential: conjugate-symmetric potentials and period lattices

This section treats the two items left open in the paper of Faust, Friedman, O'Malley, Ramos and Sharma [78] after the results of the earlier manuscript [10]. Part XII of [10] completed the classification of rectangular periods. Part XIII proved Conjecture 1, and resolved Conjecture 3 by proving its multiplicity formula and disproving its singularity patterns. The two remaining items are Conjecture 2 (existence of conjugate-reflection-symmetric potentials, open for odd periods) and Problem 1 (classification of all period lattices). The tools are those of method M4 (coordinate averaging, induction on the dimension, eigenvalue perturbation), together with a degeneration argument on diagonal fibres, interval-Newton certificates and exact computer algebra. *Neither target is closed here; both results are partial.*

For Conjecture 2:

- we give exact explicit solutions for $q = 5$ and $q = 9$;

- we give computer-assisted (interval-arithmetic) proofs for every odd period $5 \le q \le 69$;
- it follows that Conjecture 2 holds for every period that is even (already known) or has a divisor in $[5, 69]$.

Primes larger than 69 remain open.

For Problem 1:

- we prove a first-order structure theorem, valid for every lattice in every dimension;
- we deduce rigidity for an infinite class of non-rectangular lattices. It contains the families treated in [78] and the $(3\mathbb{Z})^d$ theorem of [10];
- we reduce the expected classification to a single "core statement";
- using the structure theorem and exact computer algebra, we verify the expected planar classification for all lattices of index at most 12. The previously reported check was index at most 8.

7.1. **Setting and the plane-wave form of the Floquet matrices.** Let $\Gamma \subset \mathbb{Z}^d$ be a full-rank sublattice, $G = \mathbb{Z}^d/\Gamma$, $N = |G|$, and let $V : \mathbb{Z}^d \to \mathbb{C}$ be Γ-periodic. Put $(\Delta f)(n) = \sum_{\|m-n\|_1=1} f(m)$. For $z \in \mathbb{T}^d$ let $L_V(z)$ be the Floquet matrix of $\Delta + V$ (acting on functions on a fundamental domain with the Bloch conditions $f(n+\gamma^{(i)}) = z_i f(n)$ for a basis $\gamma^{(i)}$ of Γ). As in [78, 79], V is *Floquet isospectral to* 0 if $\det(\lambda - L_V(z)) = \det(\lambda - L_0(z))$ for all $z \in \mathbb{T}^d$ and $\lambda \in \mathbb{C}$.

For $\xi \in (\mathbb{C}^*)^d$ and $n \in \mathbb{Z}^d$ write $\xi^n = \prod_j \xi_j^{n_j}$. Let

$$H = \{\xi \in (\mathbb{C}^*)^d : \ \xi^\gamma = 1 \text{ for all } \gamma \in \Gamma\}, \qquad b_h = \frac{1}{N} \sum_{n \in G} V(n) h^{-n} \quad (h \in H),$$

so that $|H| = N$, H consists of roots of unity, and $V(n) = \sum_{h \in H} b_h h^n$. Let $g_j = g_j(\Gamma) \ge 1$ be the positive generator of the j-th coordinate projection of Γ, and define the subsets

$$K_j = \{h \in H : h_i = 1 \ (i \ne j)\} \cong \mu_{g_j}, \qquad H^\circ = \{h \in H : h_j \ne 1 \text{ for all } j\},$$
$$H_D^\varepsilon = \{h \in H : \ h_1^{\varepsilon_1} = \cdots = h_d^{\varepsilon_d}\}$$

for $\varepsilon \in \{\pm 1\}^d$; H_D^ε is a subgroup.

Lemma 7.1 (plane-wave form)**.** *For $x \in (\mathbb{C}^*)^d$ put $D_\xi(x) = \sum_{j=1}^d \big(x_j \xi_j + (x_j \xi_j)^{-1}\big)$, $D(x) = \operatorname{diag}(D_\xi(x))_{\xi \in H}$ and $B = (b_{\zeta\xi^{-1}})_{\zeta,\xi \in H}$. Then V is Floquet isospectral to 0 if and only if $\det(\lambda - D(x) - B) = \det(\lambda - D(x))$ for all $x \in (\mathbb{C}^*)^d$, $\lambda \in \mathbb{C}$.*

Proof. Fix x and let W_x be the space of $f : \mathbb{Z}^d \to \mathbb{C}$ with $f(n+\gamma) = x^\gamma f(n)$ ($\gamma \in \Gamma$). The functions $e_\xi(n) = (x\xi)^n$, $\xi \in H$, lie in W_x and are linearly independent (distinct characters), hence form a basis ($\dim W_x = N$). One has $\Delta e_\xi = D_\xi(x) e_\xi$ and $V e_\xi = \sum_h b_h e_{\xi h}$, so $\Delta + V$ acts on W_x by $D(x) + B$. Restriction to a fundamental domain identifies W_x with the domain of $L_V(z)$ for $z_i = x^{\gamma^{(i)}}$. The homomorphism $x \mapsto (x^{\gamma^{(i)}})_i$ of $(\mathbb{C}^*)^d$ is surjective, because the exponent matrix is nonsingular. Both characteristic polynomials are Laurent polynomials in z, so equality on $\mathbb{T}^d$ is equivalent to equality on $(\mathbb{C}^*)^d$. □

Translations of V multiply b_h by h^{n_0}, and $\Gamma \subset \Gamma'$ with V Γ'-periodic makes $D(x) + B$ block diagonal along the cosets of $\Gamma'^\perp \subset H$; hence translates of isospectral potentials are isospectral, and isospectrality for Γ' implies isospectrality for every sublattice $\Gamma \subset \Gamma'$.

7.2. **Target 1: conjugate-reflection-symmetric potentials.**

Problem 7.2 ([78, Section 4, Conjecture 2])**.** "There exist nonzero $q\mathbb{Z}$-periodic potentials V of the form $(v_1, v_2, \ldots, \overline{v_2}, \overline{v_1})$ for $q > 3$ such that Δ and $\Delta + V$ are Floquet isospectral."

The authors prove the conjecture for even q, using the support-four potentials of [80]. Table 1 of [78] lists numerical counts of solutions of this form for $q \le 7$: 4 for $q = 5$ and 120 for $q = 7$. The odd case was open, and is not covered by [10]. For odd $q = 2m + 1$ we index sites by $\mathbb{Z}/q$ and place the middle entry v_{m+1} at 0. A cyclic shift, which preserves isospectrality, turns the required form into

$$V_{-j} = \overline{V_j} \qquad (j \in \mathbb{Z}/q), \qquad \text{in particular } V_0 \in \mathbb{R}. \tag{16}$$

Lemma 7.3 (one-dimensional reduction)**.** *Let $q \ge 3$ and let C_q be the adjacency matrix of the q-cycle. A q-periodic V is Floquet isospectral to 0 if and only if $\det(\lambda - C_q - \operatorname{diag} V) = \det(\lambda - C_q)$.*

Proof. In $L_V(z)$ the parameter enters only through the two corner entries $z^{\pm 1}$ (up to the choice of convention). In the permutation expansion of $\det(\lambda - L_V(z))$, a term containing a corner entry but not its transpose must use all q edges of the cycle in one direction. These are the two cyclic permutations, which contain no diagonal entry and are therefore independent of V. Terms using a corner entry together with its transpose contain $z \cdot z^{-1} = 1$. Hence $\det(\lambda - L_V(z)) - \det(\lambda - L_0(z))$ does not depend on z, and it suffices to take $z = 1$. □

Lemma 7.4 (repetition)**.** *Let $q_0 \mid q$ and let $W : \mathbb{Z}/q_0 \to \mathbb{C}$ satisfy $W_{-j} = \overline{W_j}$ and be Floquet isospectral to 0 (period q_0). Then $V_j := W_{j \bmod q_0}$ ($j \in \mathbb{Z}/q$) satisfies* (16) *and is Floquet isospectral to 0 for the period q.*

Proof. Reduction modulo q_0 commutes with $j \mapsto -j$, which gives (16). The potential is $q_0\mathbb{Z}$-periodic and $q_0\mathbb{Z}$-isospectral to 0, hence $q\mathbb{Z}$-isospectral by the sublattice remark after Lemma 7.1. □

Theorem 7.5 (explicit odd solutions)**.** (a) *Let $q = 5$, $\omega = e^{2\pi i/5}$, $t = 24^{1/10} > 0$, $u = 5^{1/4} > 0$, and*

$$b_1 = -\frac{t^4}{2u^3}, \quad b_{-1} = -\frac{t^6}{2u^3}, \quad b_2 = \frac{6}{u^3 t^2}, \quad b_{-2} = -\frac{t^2}{u^3},$$
$$V_j = \sum_{h \in \{\pm 1, \pm 2\}} b_h \, \omega^{hj} \quad (j \in \mathbb{Z}/5).$$

Then $V \neq 0$ satisfies (16) *and is Floquet isospectral to 0. Numerically,*

$$V_0 = -1.1541016\ldots, \qquad V_{\pm 1} = -0.7878062\ldots \pm 1.3408501\ldots i,$$
$$V_{\pm 2} = 1.3648570\ldots \mp 1.1625702\ldots i.$$

(b) *Let $q = 9$ and $V_0 = V_{\pm 4} = 0$, $V_{\pm 1} = e^{\mp i\pi/6}$, $V_{\pm 2} = -e^{\mp i\pi/6}$, $V_{\pm 3} = \mp i$. Then $V \neq 0$ satisfies* (16) *and is Floquet isospectral to 0.*

Proof. (a) All b_h are real, so $\overline{V_j} = \sum_h b_h \omega^{-hj} = V_{-j}$. By Lemma 7.3 it suffices to prove $\det(\lambda - C_5 - \operatorname{diag} V) = \det(\lambda - C_5)$. Multiplying by a power of t and u clears denominators, and the difference becomes a polynomial in $\mathbb{Q}[t, u, \omega, \lambda]$. It lies in the ideal generated by $t^{10} - 24$, $\Phi_5(\omega)$ and $u^2 - (1 + 2\omega + 2\omega^4)$; we verified this by reducing it to zero with an exact lex Gröbner basis. All three generators vanish at $t = 24^{1/10}$, $\omega = e^{2\pi i/5}$, $u = 5^{1/4}$, because $1 + 2(\omega + \omega^4) = 1 + 4\cos\frac{2\pi}{5} = \sqrt{5} = u^2$. Hence the identity holds at the stated values.

(b) Symmetry (16) is immediate. All entries lie in $\mathbb{Q}(\zeta_{12})$ with $\zeta_{12} = e^{i\pi/6}$. The difference $\det(\lambda - C_9 - \operatorname{diag} V) - \det(\lambda - C_9)$, computed exactly as a polynomial in ζ_{12} and λ, is divisible by $\Phi_{12}(\zeta_{12}) = \zeta_{12}^4 - \zeta_{12}^2 + 1$. It therefore vanishes, and Lemma 7.3 applies. □

Remark 7.6. The potential in (a) was found numerically and then identified exactly. The products $b_h b_{-h} = \pm 6/(5\sqrt{5})$ and the ratios $(b_1/b_{-1})^5 = 1/24$, $(b_2/b_{-2})^5 = -27/2$ were recognised by integer-relation detection. In the normalisation $b_h = t^{-h} s_h$ one has $(s_{-2}, s_{-1}, s_1, s_2) =$

$5^{-3/4}(-1,-\sqrt{6},-\sqrt{6},6)$. The solution in (b) has support six and unimodular entries. A search for sparse solutions of form (16) found no analogue for $q = 7, 11, 13$.

Theorem 7.7 (computer-assisted, odd periods)**.** *For every odd q with $5 \le q \le 69$ there is a nonzero $V : \mathbb{Z}/q \to \mathbb{C}$ satisfying* (16) *that is Floquet isospectral to* 0.

Proof. Write $V_0 = r$, $V_{\pm j} = x_j \pm iy_j$ $(1 \le j \le m)$, and let $F : \mathbb{R}^q \to \mathbb{R}^q$ be the vector of real parts of the coefficients of $\lambda^{q-1}, \dots, \lambda^0$ in $\det(\lambda - C_q - \operatorname{diag} V) - \det(\lambda - C_q)$. These coefficients are real on the slice (16): if R is the permutation matrix of $j \mapsto -j$, then $RC_qR = C_q$ and $R \operatorname{diag}(V)R = \operatorname{diag}(\overline{V})$, so $R(C_q + \operatorname{diag} V)R = \overline{C_q + \operatorname{diag} V}$, and the characteristic polynomial of $C_q + \operatorname{diag} V$ equals its own complex conjugate. Hence a zero of F is a zero of all q complex coefficients, and by Lemma 7.3 a zero of F with $(r, x, y) \ne 0$ is the required potential. For $q \ge 29$ the coefficient system is badly conditioned in floating point, so we use an equivalent system instead. Put $A = C_q + \operatorname{diag} V$, $G(\lambda) = \det(\lambda - A)/\det(\lambda - C_q) - 1$, $z_k = e^{(1+2\pi ik)/q}$ and $\lambda_k = z_k + z_k^{-1}$ $(0 \le k < q)$. These are q distinct points on a Bernstein ellipse around $[-2, 2]$, and $\det(\lambda_k - C_q) = z_k^q + z_k^{-q} - 2 = e + e^{-1} - 2 \ne 0$. The choice $|z_k|^q = e$ balances the Chebyshev coefficients of the difference polynomial and makes the system well conditioned. Let

$$E(w) = \big(\operatorname{Re} G(\lambda_0),\ \operatorname{Re} G(\lambda_k),\ \operatorname{Im} G(\lambda_k)\ (1 \le k \le m)\big) \in \mathbb{R}^q.$$

The polynomial $\det(\lambda - A) - \det(\lambda - C_q)$ has degree $< q$ and, on the slice, real coefficients. Hence $G(\lambda_0)$ is real, the values at $\lambda_{q-k} = \overline{\lambda_k}$ are conjugate, and $E(w) = 0$ forces the polynomial to vanish at all q points λ_k, hence identically. So the zeros of E and F on the slice coincide.

For each q the script `R3_conj2_krawczyk.py` finds a floating-point zero by damped Newton iteration from random starts. It refines the zero by Newton steps at 400-bit precision to a centre w_0 with exactly representable coordinates, and fixes $\rho = 10^{-30}$. In Arb ball arithmetic [85] it then evaluates $F(w_0)$ (resp. $E(w_0)$) and an enclosure $J(X)$ of the Jacobian over the box $X = w_0 + [-\rho, \rho]^q$. The Jacobian uses $\partial_{V_j} \det(\lambda - A) = -\det(\lambda - A_{\hat{j}\hat{j}})$ (resp. $\partial_{V_j} G = -(G+1)[(\lambda - A)^{-1}]_{jj}$) and the chain rule. With Y the midpoint matrix of the inverse of the midpoint Jacobian, it checks, with rigorous ball comparisons, the Krawczyk inclusion $K(X) = w_0 - YF(w_0) + (I - YJ(X))(X - w_0) \subset w_0 + (-0.99\rho, 0.99\rho)^q \subset \operatorname{int} X$ (with E in place of F when $q \ge 29$). By the Krawczyk–Moore theorem [83, 84], the system then has a unique zero in X. Since $|w_0| - \sqrt{q}\,\rho > 0$, this zero is nonzero. □

As a consistency check, for $q = 5$ the certified zero coincides, to the printed digits, with the reflection $V_j \mapsto V_{-j}$ of the exact solution of Theorem 7.5(a). The residuals at the centres are below 10^{-100}. For $29 \le q \le 45$ the log also records the residual of the coefficient system F at the same centre (below 10^{-90}; this is not needed for the proof).

period q	certificate
5, 9	exact (Theorem 7.5); also Krawczyk with F
odd $7 \le q \le 27$	Krawczyk with F, $\rho = 10^{-30}$, 400-bit balls
odd $29 \le q \le 69$	Krawczyk with E, $\rho = 10^{-30}$, 400-bit balls

TABLE 2. Certificates for Theorem 7.7 (logs `R3_conj2_krawczyk.log` for $q \le 27$ and `R3_conj2_krawczyk_ellipse.log` for $q \ge 29$).

Corollary 7.8 (partial resolution of Problem 7.2)**.** *Conjecture 2 of* [78] *holds for every $q > 3$ that is even or has a divisor in $[5, 69]$. In particular it holds for every $q > 3$ divisible by* 5, 7 *or* 9*, hence for every power of* 3 *that is at least* 9*. The only odd $q > 3$ not covered are $q = n$ and $q = 3n$ with $n > 1$ odd and all prime factors of n larger than* 69*, for instance every prime $q > 69$.*

Proof. Even q: [78]. Otherwise apply Lemma 7.4 to the solution for a divisor $q_0 \in [5, 69]$ given by Theorems 7.5 and 7.7. If an odd $q > 3$ has no divisor in $[5, 69]$, then $9 \nmid q$ and no prime $5 \le p \le 69$ divides q, which gives the stated form. □

Remark 7.9 (why degree theory does not settle all odd q). Let $q = 2m + 1$ and F be as in the proof of Theorem 7.7. The leading forms of F are $\pm e_k(V)$, with $V = V(w)$. Their only common real zero is 0, so F is proper. The degree of F on large spheres equals that of the leading map, which is 0. Indeed, on the slice the roots of $\prod_j(\lambda - V_j)$ are V_0 together with pairs $V_j, \overline{V_j}$, so this polynomial has at most one simple real root. The leading map therefore misses the nonempty open set of coefficient vectors of polynomials with two simple real roots. The real zero set of F is finite (the complex system has finitely many solutions), so the indices of the real zeros sum to 0. The bipartite-type gauge $V \mapsto -V$ (valid for odd q together with $z \mapsto -z$) gives $F(-w) = \Sigma F(w)$ with $\Sigma = \mathrm{diag}((-1)^k)_{k=1}^q$. Comparing degrees on a small sphere around 0 gives $\mathrm{ind}_0 F = (-1)^m \mathrm{ind}_0 F$. Thus for $q \equiv 3 \pmod 4$ the local index at the degenerate zero 0 vanishes and no Brouwer-degree count can detect the nonzero real solutions, although Table 1 of [78] reports 120 of them for $q = 7$. For $q \equiv 1 \pmod 4$ the symmetry imposes nothing, and $\mathrm{ind}_0 F \neq 0$ would force nonzero real zeros. For $q = 5$ a certified real-root isolation of the full system (msolve [87]) finds exactly four nonzero real zeros. They form one orbit under $V \mapsto V(-\cdot)$ and $V \mapsto -V$, both of which preserve the local index when $q = 5$, and they are nondegenerate. Hence $|\mathrm{ind}_0 F| = 4$. We do not know how to compute $\mathrm{ind}_0 F$ in general. A proof of Conjecture 2 for all odd q therefore seems to require an explicit infinite family, a computation of the local index for $q \equiv 1 \pmod 4$, or an equivariant refinement of the degree for $q \equiv 3 \pmod 4$.

7.3. **Target 2: classification of period lattices.**

Problem 7.10 ([78, Section 6, Problem 1])**.** "For the $\mathbb{Z}^d$-periodic grid graph, classify all periods Γ such that there exist nonzero Γ-periodic potentials that are Floquet isospectral to the discrete Laplacian." The authors add: "Problem 1 has been solved in the case where $d = 1$ [CFK], but it is still open for $d \ge 2$." For $d = 2$ they observe that the undecided lattices (those not contained in some $q\mathbb{Z} \oplus \mathbb{Z}$ or $\mathbb{Z} \oplus q\mathbb{Z}$ with $q > 3$) have Hermite normal form $(a, 0)\mathbb{Z} \oplus (b, c)\mathbb{Z}$ with $0 \le b < a$ and $\gcd(a, b), c \in \{1, 2, 3\}$. They write: "We symbolically checked all possible combinations of a, b, c where $ac \le 8$" and "found no examples with nonzero potentials isospectral to $\mathbf{0}$".

The rectangular case is settled. Existence when some period exceeds 3, and rigidity when all periods are at most 3 and at most two equal 3, are [79, Theorem C]. The remaining cases, with at least three periods equal to 3, are covered by [10, Part XII], through the $(3\mathbb{Z})^d$ theorem and a reduction. The natural expected answer is the following.

Conjecture 7.11 (expected classification)**.** *Let $\Gamma \subset \mathbb{Z}^d$ be a full-rank sublattice. A nonzero Γ-periodic potential Floquet isospectral to 0 exists if and only if $\max_j g_j(\Gamma) \ge 4$.*

Proposition 7.12 (the "if" direction)**.** *If $g_j(\Gamma) \ge 4$ for some j, a nonzero Γ-periodic potential Floquet isospectral to 0 exists.*

Proof. By [79, Theorem 4.3] (see also [80] for even periods) there is a nonzero $W : \mathbb{Z}/g_j \to \mathbb{C}$ that is Floquet isospectral to 0. Put $V(n) = W(n_j)$. Then V is periodic for $\Gamma' = \{n : n_j \in g_j\mathbb{Z}\} \supset \Gamma$. For Γ' the operator separates: in Lemma 7.1 the dual group is μ_{g_j} in the j-th coordinate, and $D(x) + B$ is the 1-dimensional Floquet matrix plus scalars. So V is Γ'-isospectral to 0, and hence Γ-isospectral by the sublattice remark. □

Theorem 7.13 (first-order structure, all lattices)**.** *Let $d \ge 2$, let $\Gamma \subset \mathbb{Z}^d$ be any full-rank sublattice, and let V be Γ-periodic and Floquet isospectral to 0. Then:*

(i) $b_1 = 0$;

(ii) (averaging) *for each j, the potential on $\mathbb{Z}^{d-1}/p_j(\Gamma)$ (p_j deletes the j-th coordinate) whose Fourier coefficients are $b|_{\{h\in H:h_j=1\}}$ (equivalently: the average of V over the fibres of $\mathbb{Z}^d/\Gamma \to \mathbb{Z}^{d-1}/p_j(\Gamma)$) is $p_j(\Gamma)$-Floquet isospectral to 0;*

(iii) (diagonal fibres) $b_h = 0$ *for every $h \in H_D^\varepsilon$ and every $\varepsilon \in \{\pm 1\}^d$;*

(iv) (pair products) $b_h b_{h^{-1}} = 0$ *for every $h \in H^\circ$.*

Proof. (i) is the trace of Lemma 7.1 at order λ^{N-1}.

(ii) Take $j = d$. Fix $x' = (x_1, \dots, x_{d-1})$ and let $x_d = s \to \infty$. Group H by the value r of ξ_d. Each group P_r is a coset $\xi^0 H^{(d)}$ of $H^{(d)} = \{h : h_d = 1\}$, and $H^{(d)}$ is the dual group of $p_d(\Gamma)$. For $\xi \in P_{r'}$, $D_\xi = sr' + O(1)$. Inside P_r, the $O(1)$ part of $D_{\xi^0 h}$ is $\sum_{i<d}\big((x_i\xi_i^0)h_i + ((x_i\xi_i^0)h_i)^{-1}\big)$, and the coupling is $b_{hh'^{-1}}$ with $hh'^{-1} \in H^{(d)}$. This block is the plane-wave Floquet matrix $M'(x'\xi^0)$ of the averaged potential for the lattice $p_d(\Gamma)$. Substituting $\lambda = sr + \mu$ and taking the Schur complement with respect to the other groups, whose diagonal entries are $s(r - r') + O(1)$ and whose couplings are $O(1)$, gives

$$\det(\lambda - D - B) = \prod_{r'\neq r} (s(r-r'))^{|P_{r'}|}\big(\det(\mu - M'(x'\xi^0)) + O(s^{-1})\big),$$

and the same with $B = 0$. Equality of the left sides for all s forces $\det(\mu - M'(y)) = \det(\mu - M_0'(y))$ for all $y \in (\mathbb{C}^*)^{d-1}$, $\mu \in \mathbb{C}$.

(iii) Fix ε and $\xi^0 \in H$. Since $d \geq 2$, choose $a \in (\mathbb{C}^*)^d$ with $\sum_j a_j(\xi_j^0)^{\varepsilon_j} = 0$ and otherwise generic. Then, for $\xi \in H$, $\sum_j a_j \xi_j^{\varepsilon_j} = 0$ holds exactly when $(\xi_j^{\varepsilon_j})_j \in \mathbb{C} \cdot ((\xi_j^0)^{\varepsilon_j})_j$, i.e. when $\xi \in P := \xi^0 H_D^\varepsilon$. Indeed, for ξ outside P the condition cuts out a proper subspace of the hyperplane, and there are finitely many such ξ. Put $x_j = sa_j$ if $\varepsilon_j = 1$ and $x_j = (sa_j)^{-1}$ if $\varepsilon_j = -1$. Then $D_\xi = s\alpha_\xi + s^{-1}\beta_\xi$ with $\alpha_\xi = \sum_j a_j \xi_j^{\varepsilon_j}$ and $\beta_\xi = \sum_j a_j^{-1}\xi_j^{-\varepsilon_j}$. Thus $\alpha = 0$ on P and $\alpha \neq 0$ off P. For bounded μ and large s, the Schur complement of $\mu - D - B$ onto P is $\mu - B_{PP} + O(s^{-1})$. Hence $D + B$ has exactly $|P|$ eigenvalues in a fixed disc containing the spectrum of B_{PP}, and they tend to the eigenvalues of B_{PP}. By isospectrality these eigenvalues are $s^{-1}\beta_\xi \to 0$ ($\xi \in P$). So B_{PP} is nilpotent. But B_{PP} is the convolution matrix of $b|_{H_D^\varepsilon}$ on the finite abelian group H_D^ε, whose eigenvalues are the Fourier transforms $\sum_{h\in H_D^\varepsilon} b_h\chi(h)$. All of them vanish, so $b|_{H_D^\varepsilon} = 0$.

(iv) Take $x = sa$ with generic a, so $D = s\,\mathrm{diag}(\alpha) + s^{-1}\,\mathrm{diag}(\beta)$ with pairwise distinct $\alpha_\xi = \sum_j a_j\xi_j$. With $\epsilon = 1/s$, the matrix $(D + B)/s = \mathrm{diag}(\alpha) + \epsilon B + \epsilon^2\,\mathrm{diag}(\beta)$ has simple eigenvalue branches analytic in ϵ [88]. The branch through α_1 must equal $\alpha_1 + \epsilon^2\beta_1$, by isospectrality and continuity at $\epsilon = 0$. Comparing the ϵ^2 Rayleigh–Schrödinger coefficients and using (i) gives

$$\sum_{h\neq 1} \frac{b_h b_{h^{-1}}}{L_h(a)} = 0, \qquad L_h(a) = \sum_j a_j(1 - h_j), \tag{17}$$

identically in a. For $h, h' \in H^\circ$ the vectors $(1 - h_j)_j$ and $(1 - h'_j)_j$ are proportional only if $h = h'$ or both h, h' are constant vectors. In two coordinates, write $h_k = e^{i\theta_k}$ with $\theta_k \in (0, 2\pi)$; then $(1-h_2)/(1-h_1) = e^{i(\theta_2-\theta_1)/2}\sin(\theta_2/2)/\sin(\theta_1/2)$ determines $\theta_2 - \theta_1 \in (-2\pi, 2\pi)$. When $\theta_2 \neq \theta_1$ it determines θ_1 as well, by strict monotonicity of $\cot(\theta_1/2)$. Apply this to every pair of coordinates. Hence for non-constant $h \in H^\circ$ the hyperplane $L_h = 0$ is a simple pole of (17) carried by the single term h. It is not the pole of any h' with a coordinate equal to 1, since $L_{h'}$ then omits that variable. The residue gives $b_h b_{h^{-1}} = 0$. A constant $h \in H^\circ$ lies in $H_D^{(1,\dots,1)}$, so $b_h = 0$ by (iii). □

For $h \in H$ let $J(h) = \{j : h_j \neq 1\}$. Call h *sign-constant* if there are signs ε_j ($j \in J(h)$) with $h_j^{\varepsilon_j}$ independent of $j \in J(h)$.

Corollary 7.14 (a rigidity criterion)**.** *Suppose $g_j(\Gamma) \le 3$ for all j and every $h \in H$ is sign-constant. Then* 0 *is the only* Γ*-periodic potential Floquet isospectral to* 0*. This applies in particular to:*

(a) *every $\Gamma \supset 3\mathbb{Z}^d$ or $\Gamma \supset 2\mathbb{Z}^d$. This includes the theorem for $(3\mathbb{Z})^d$ in all dimensions (*[10, Part XII]*), and gives a proof of it that does not use the cubic perturbation identity employed there;*

(b) *for every $d \ge 2$, $a \ge 1$ and $\varepsilon \in \{\pm 1\}^d$, the lattice $\Gamma_{a,\varepsilon} = \{n \in \mathbb{Z}^d : \sum_j \varepsilon_j n_j \equiv 0 \pmod a\}$. This family contains the lattices of* [78, Theorem 6.1] *and the planar family $(a,0)\mathbb{Z} \oplus (1,\pm 1)\mathbb{Z}$.*

Proof. Let $h \in H$ and $J = J(h)$. Iterating Theorem 7.13(ii) over the coordinates outside J shows the following: the $|J|$-dimensional potential for the projected lattice $p_J(\Gamma) \subset \mathbb{Z}^J$, with Fourier coefficients $b|_{H_J}$ where $H_J = \{h' : h'_i = 1 \ (i \notin J)\}$, is Floquet isospectral to 0. If $|J| = 0$, then $b_h = 0$ by (i). If $J = \{j\}$, this is a one-dimensional problem of period $g = g_j \le 3$, which is rigid ([79, Theorem C]; directly, for $g \le 3$ the characteristic polynomial at $z = 1$ yields $e_1(V) = \cdots = e_g(V) = 0$, hence $V = 0$). So $b|_{K_j} = 0$. If $|J| \ge 2$, then h lies in the ε-diagonal subgroup of the $|J|$-dimensional problem, and part (iii) applied there gives $b_h = 0$.

For (a), every coordinate of $h \in H \subset \mu_3^d$ (resp. μ_2^d) different from 1 is $\omega^{\pm 1}$ (resp. -1), and $g_j \mid 3$ (resp. 2).

For (b), $H = \{(\zeta^{\varepsilon_1}, \dots, \zeta^{\varepsilon_d}) : \zeta^a = 1\}$, and $e_1 - \varepsilon_1\varepsilon_k e_k \in \Gamma_{a,\varepsilon}$ gives $g_j = 1$. □

Proposition 7.15 (reduction to a core statement)**.** *For $d \ge 1$ let* CS(d) *be the statement:* for every full-rank $\Gamma \subset \mathbb{Z}^d$ with all $g_j \le 3$ and every Γ-periodic V that is Floquet isospectral to 0 and satisfies $\operatorname{supp} b \subset H^\circ$, one has $V = 0$. *If* CS(d') *holds for all $d' \le d$, then Conjecture 7.11 holds in all dimensions $\le d$.*

Proof. The "if" direction is Proposition 7.12. For "only if", argue by induction on d; the case $d = 1$ is periods ≤ 3. Let all $g_j \le 3$ and $h \notin H^\circ$, so $J(h) \subsetneq \{1, \dots, d\}$. As in the proof of Corollary 7.14, the projected $|J(h)|$-dimensional potential is isospectral to 0. Its lattice has g-values $g_j \le 3$ ($j \in J(h)$), so by the induction hypothesis it vanishes, and $b_h = 0$. Thus $\operatorname{supp} b \subset H^\circ$ and CS(d) gives $V = 0$. □

Remark 7.16 (a tool for the core statement). Let k be the least size of a nonzero principal minor of B; by (i) and (iv), $k \ge 3$ when $\operatorname{supp} b \subset H^\circ$. Expanding $\det(\lambda - D - B) = \sum_M \det(-B_{MM}) \prod_{\xi \notin M} (\lambda - D_\xi)$, dividing by $\prod_\xi (\lambda - D_\xi)$ and inserting x as in (iii) with $\lambda = s\mu$ shows the following. For every ε, the functions $\sum_{|M| = k'} \det(B_{MM}) \prod_{\xi \in M} (\mu - \alpha_\xi^\varepsilon(a))^{-1}$, $k' \in \{k, k+1\}$, vanish identically in (μ, a); the next corrections occur at relative order s^{-2}. When $\operatorname{supp} b = \{h\}$ is a single element this gives $b_h^{\operatorname{ord} h} = 0$. A general argument would need to control the cancellations among these minors.

Proposition 7.17 (computer-assisted, $d = 2$; partial)**.** (a) (exact.) *Let $\Gamma = (a,0)\mathbb{Z} \oplus (b,c)\mathbb{Z}$ with $0 \le b < a$, $\gcd(a,b) \le 3$, $c \le 3$ and $ac \le 12$. Then* 0 *is the only* Γ*-periodic potential Floquet isospectral to* 0*. This extends the symbolic check $ac \le 8$ reported in* [78].

(b) (modular evidence, not a proof.) *For the same lattices with $13 \le ac \le 14$, every branch ideal described below has the origin as its only zero over $\overline{\mathbb{F}}_p$ for one prime $p \equiv 1 \pmod{ac}$ with $p \approx 10^6$.*

Method. The symmetries of $\mathbb{Z}^2$ preserve Δ, isospectrality and the conditions $g_1, g_2 \le 3$, so it suffices to treat one Hermite normal form in each symmetry class. Fix Γ. By Theorem 7.13(i)–(iii) and the rigidity of the one-dimensional periods $g_1 = \gcd(a,b) \le 3$ and $g_2 = c \le 3$, $b_h = 0$ for every h outside $A = H^\circ \setminus (H_D^{(1,1)} \cup H_D^{(1,-1)})$. If $A = \emptyset$, Γ is rigid. Otherwise A is a disjoint union of inverse pairs $\{h, h^{-1}\}$. By (iv), V lies in one of the $2^{\#\text{pairs}}$ *branch* subspaces, on which one chosen

member of every pair has $b = 0$; let S be the set of surviving members. For each branch, the identity of Lemma 7.1 at $ac + 8$ random rational points x gives an ideal $I_S \subset \mathbb{Q}(\zeta_{ac})[b_h : h \in S]$, and it suffices to show that its only zero is the origin. Two symmetries reduce the work. If σ_k is the automorphism $\zeta_{ac} \mapsto \zeta_{ac}^k$, then $\sigma_k(I_S) = I_{S^k}$ after renaming $b_h \mapsto b_{h^k}$, because the points x are rational; so I_S and I_{S^k} have the origin as only zero simultaneously. A lattice symmetry T (a signed permutation with $T\Gamma = \Gamma$) maps isospectral potentials of branch S to isospectral potentials of branch $T(S)$. Hence one branch per orbit suffices; the orbits include $V \mapsto V(-\cdot)$, i.e. $k = -1$. For each representative, SINGULAR [86] computes an exact standard basis of I_S over $\mathbb{Q}(\zeta_{ac})$. It checks that $\mathbb{Q}(\zeta_{ac})[b]/I_S$ is finite-dimensional and that every variable is nilpotent modulo I_S. For index ≤ 10 it also checks that the total length equals the local length at the origin, computed with a local ordering. Hence the origin is the only zero over $\mathbb{C}$, and $V = 0$. As controls, the same code finds the known nonzero solutions for $4\mathbb{Z} \oplus \mathbb{Z}$ and $5\mathbb{Z} \oplus \mathbb{Z}$ when the constraints on K_1 are dropped (lengths 24 vs. 16 and 120 vs. 60), and none for $3\mathbb{Z} \oplus \mathbb{Z}$. For (b) the same branch ideals, built from $ac + 8$ random points, are treated over $\mathbb{F}_p$, with $\zeta_{ac} \in \mathbb{F}_p$. A solution over $\mathbb{C}$ could reduce to the origin modulo an unlucky prime, so (b) is evidence only. The scripts are `R3_problem1_exact_branches.py` and `explore/branch2d_fast.py`. □

$N = ac$	rigid by Cor. 7.14	exact branch computation: $[a, b, c]$ (orbits/branches; max. length)
1–3	all	–
4	$[2, 0, 2]$, $[4, 1, 1]$	$[2, 1, 2]$ (1/2; 4)
5	$[5, 1, 1]$	$[5, 2, 1]$ (1/4; 12)
6	$[6, 1, 1]$	$[2, 0, 3]$ (1/2; 6), $[2, 1, 3]$ (1/2; 6), $[3, 1, 2]$ (1/2; 6)
7	$[7, 1, 1]$	$[7, 2, 1]$ (2/8; 44)
8	$[4, 2, 2]$, $[8, 1, 1]$	$[4, 1, 2]$ (3/8; 43), $[8, 3, 1]$ (2/4; 32)
9	$[3, 0, 3]$, $[9, 1, 1]$	$[3, 1, 3]$ (2/8; 61), $[9, 2, 1]$ (2/8; 61)
10	$[10, 1, 1]$	$[5, 1, 2]$ (4/16; 118), $[5, 2, 2]$ (1/4; 29), $[10, 3, 1]$ (4/16; 119)
11	$[11, 1, 1]$	$[11, 2, 1]$ (4/32; 332), $[11, 3, 1]$ (4/32; 330)
12	$[12, 1, 1]$	$[4, 1, 3]$ (3/8; 98), $[4, 2, 3]$ (4/16; 164), $[6, 1, 2]$ (4/16; 167), $[6, 2, 2]$ (2/4; 28), $[12, 5, 1]$ (2/4; 40)

TABLE 3. Exact verification of Proposition 7.17(a). Here $[a, b, c]$ is the Hermite normal form $(a, 0)\mathbb{Z} \oplus (b, c)\mathbb{Z}$, and one representative is listed per class under the symmetries of $\mathbb{Z}^2$. There are 37 classes: 17 are rigid by Corollary 7.14 (the allowed set A is empty) and 20 are settled by the computation. Lengths are $\dim_{\mathbb{Q}(\zeta_{ac})}$ of the branch quotient ring at $ac + 8$ sample points; for $N \leq 10$ they also equal the local length at the origin. Logs: `R3_problem1_exact_branches_1_4.log` and `R3_problem1_exact_branches.log`.

7.4. **Status of the two problems.**

- [78, Conjecture 2]: **partial**. Proved for every even $q > 3$ (known) and every q with a divisor in $[5, 69]$ (Corollary 7.8). It is exact for $q = 5, 9$ and computer-assisted (Krawczyk) for the other odd $q \leq 69$. Open for $q = n$ and $q = 3n$ with all prime factors of n larger than 69, in particular for every prime $q > 69$.
- [78, Problem 1]: **partial**. The structure theorem holds for all lattices and all d; rigidity is proved for the class of Corollary 7.14; the classification reduces to the core statement CS(d) (Proposition 7.15). The planar case is checked exactly for index ≤ 12 and modulo a prime for index 13 and 14. The core statement itself remains open.

Remark 7.18 (candidates not pursued). Some neighbouring questions were considered and set aside. Liu asked whether Fermi isospectrality is rigid for two-dimensional discrete periodic operators (see also [81]); this was answered negatively by a numerically certified counterexample [82], which also disproves a conjecture of Gieseker, Knörrer and Trubowitz. Conjectures 1 and 3 of [78], and the $(3\mathbb{Z})^d$ case of the rectangular problem, are already treated in [10].

8. Spectral volumes of weighted Laplacians under edge addition and suspension

Summary.

- Problem 8.1 (EJT Question 3): *partial*. It is proved for all connected graphs on at most six vertices, and in every order for two infinite families.
- Problem 8.2 (EJT Question 4): *partial*. It is proved for all graphs on at most four vertices, together with a uniform multiplicative bound.
- EJT Question 5 and Fallat–Gupta–Lin's boundary-linearity question were already answered in [3], and are not treated here.

8.1. **Setting, the two target questions, and summary.** Let G be a simple graph on $[n] = \{1, \dots, n\}$, $n \geq 2$. For an edge $f = ab$ put $b_f = e_a - e_b$ and $B_f = b_f b_f^{\mathsf{T}}$. For $w \in \mathbb{R}_{\geq 0}^{E(G)}$ let $L_G(w) = \sum_{f \in E(G)} w_f B_f$. Every such matrix satisfies $L_G(w)\mathbf{1} = 0$; we list its eigenvalues as $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$ and call $(\lambda_2, \dots, \lambda_n)$ its *reduced spectrum* (zeros other than λ_1 are allowed). Write

$$C_m = \Big\{ x \in \mathbb{R}^m : 0 \leq x_1 \leq \cdots \leq x_m,\ \textstyle\sum_i x_i = 1 \Big\}, \qquad m = n - 1,$$

and define the (relaxed, trace-normalized, ordered) spectral region

$$\mathcal{R}(G) = \big\{ (\lambda_2, \dots, \lambda_n)(L_G(w)) : w \in \mathbb{R}_{\geq 0}^{E(G)},\ \operatorname{tr} L_G(w) = 1 \big\} \subseteq C_{n-1}.$$

It is compact (continuous image of a compact simplex). Following Erchenko–Jakobson–Tsypin [89, Section 8], $P(G)$ is the proportion of the standard simplex $\{x \in \mathbb{R}_{\geq 0}^{n-1} : \sum x_i = 1\}$ occupied by realizable (unordered) reduced spectra; since that set is the union of the $(n-1)!$ coordinate permutations of $\mathcal{R}(G)$, which overlap only in a null set,

$$P(G) = \frac{\operatorname{vol} \mathcal{R}(G)}{\operatorname{vol} C_{n-1}} = \Pr\Big[\tfrac{\operatorname{sort}(X)}{X_1 + \cdots + X_{n-1}} \in \mathcal{R}(G) \Big], \qquad X_1, \dots, X_{n-1} \text{ i.i.d. } \operatorname{Exp}(1), \tag{18}$$

the last equality being the classical exponential representation of the uniform distribution on a simplex. If G is disconnected then $\lambda_2 \equiv 0$ on $\mathcal{R}(G)$ and $P(G) = 0$.

Problem 8.1 (Erchenko–Jakobson–Tsypin [89, Section 8, Question 3])**.** "Consider a non-complete graph G. Let G' be a graph obtained from G by adding an edge. Is $P(G') > P(G)$?"

Problem 8.2 (Erchenko–Jakobson–Tsypin [89, Section 8, Question 4])**.** "Let G be a non-complete graph and G' be the suspension of G. Is $P(G') \geq P(G)$?" (The suspension adds one vertex joined to every vertex of G, [89, Definition 7.2].)

Taken literally, Problem 8.1 has a trivial negative answer (add an edge to the empty graph on $n \geq 3$ vertices: both volumes are 0), as already observed in [3]; the meaningful question concerns connected G. For $n \leq 4$ the connected version follows from the complete census of four-vertex volumes [89, 90, 3]; for every fixed $n \geq 5$ it was open (the concluding section of [3] lists the positivity of the fixed-order gain c_n^{conn}, $n \geq 5$, as open). Problem 8.2 is trivial for disconnected G, known along the family $K_n - e$ [3], and was open in general (EJT Question 5, on iterated suspensions, is answered affirmatively in [3]).

Results of this section.

(a) A reduction: for connected G and a non-edge e, $P(G+e) > P(G)$ *if and only if* $\mathcal{R}(G+e) \neq \mathcal{R}(G)$ (Proposition 8.5); together with an exact, finitely checkable linear (von Neumann) certificate of non-realizability (Lemma 8.6).

(b) Problem 8.1 has a positive answer, in every order n, for every connected G whose complement minus the added edge is a matching (possibly empty) (Theorem 8.8), and for every connected G for which adding e increases the Hadwiger number (Theorem 8.11; e.g. all trees).

(c) Problem 8.1 has a positive answer for *every* connected non-complete graph with at most 6 vertices (Theorem 8.13); this is computer-assisted, with exact rational certificates and rigorous ball arithmetic. In particular the fixed-order constants c_5^{conn} and c_6^{conn} are positive. **Partial**: the general case $n \geq 7$ remains open.

(d) For Problem 8.2, we obtain the following.
- An exact "lift identity" for the uniform join (Proposition 8.16).
- The unconditional bound $P(G') \geq \gamma_n P(G)$, with an explicit constant $\gamma_n \geq \frac{n+1}{2n}$ (Theorem 8.17).
- A positive answer for every graph with at most four vertices (Theorem 8.19; computer-assisted: exact rational box certificates for the lower bounds, based on the exact four-vertex regions of [89]; the volumes of the graphs themselves are the closed forms of [3], re-checked by quadrature, and the case $K_4 - e$ uses the $K_5 - e$ region of [3]).
- A reduction of the full question to a purely probabilistic "cone-lifting inequality" (Conjecture 8.21), supported by computation.

Partial: Problem 8.2 remains open for $n \geq 5$ (except along $K_n - e$, [3]).

8.2. **Regular points and the reduction of Question 3.** For $w \in \mathbb{R}^{E(G)}$ write $\det(tI + L_G(w)) = \sum_{k=0}^{n} q_k(w) t^{n-k}$, so $q_k = e_k(\lambda_2, \ldots, \lambda_n)$ and $q_n = 0$. By the all-minors matrix-tree theorem [92],

$$q_k(w) = \sum_{F} \Big(\prod_{C \in \pi(F)} |C| \Big) \prod_{f \in F} w_f, \tag{19}$$

the sum running over k-edge forests $F \subseteq E(G)$ and $\pi(F)$ being the set of components of the spanning forest $([n], F)$.

Lemma 8.3 (A regular weighting)**.** *Let H be a connected graph on $n \geq 2$ vertices. There is $w \in \mathbb{R}_{>0}^{E(H)}$ such that $L_H(w)$ has $n-1$ distinct non-zero eigenvalues and the Jacobian matrix $\big(\partial \lambda_i / \partial w_f\big)_{2 \leq i \leq n,\, f \in E(H)}$ has rank $n-1$.*

Proof. This is a weak form of the universal lacunary cone of [4] (its tree lacunary chart and regular spanning-subgraph lift). That result gives, for every connected graph, an open cone of strict realizations at regular points. We include the short argument only to keep the section self-contained. Choose a spanning tree T and order its edges $f_1, \ldots, f_{n-1}$ so that every prefix $\{f_1, \ldots, f_k\}$ spans a subtree on $k+1$ vertices (list the vertices in breadth-first order from a root and let f_k join the $(k+1)$st vertex to its parent). For $0 < s < 1$ put $w_{f_j} = s^j$ and $w_f = 0$ for $f \notin E(T)$. Every k-subset of $E(T)$ is a forest, and the prefix forest is the unique k-subset minimizing $\sum_{f_j \in F} j$; its coefficient in (19) is $k+1$. Hence $q_k = (k+1) s^{k(k+1)/2} (1 + O(s))$, and in logarithmic coordinates

$$\frac{\partial \log q_k}{\partial \log w_{f_j}} = \begin{cases} 1 + O(s), & j \leq k, \\ O(s), & j > k, \end{cases}$$

because every term of q_k containing f_j with $j > k$ is $O(s)$ times the leading term. The matrix $(\mathbf{1}[j \leq k])_{k,j}$ is unitriangular, so for small s the map $(w_{f_j})_j \mapsto (q_k)_k$ is a local diffeomorphism

(in the original coordinates too, since $\partial q_k/\partial w_j = (q_k/w_j)\,\partial \log q_k/\partial \log w_j$). Moreover $q_k/q_{k-1} = \frac{k+1}{k}s^k(1+O(s))$, so by the Newton polygon the reduced eigenvalues are $\frac{k+1}{k}s^k(1+O(s))$, $1 \le k \le n-1$, pairwise distinct for small s; near a polynomial with simple roots, roots and coefficients are local diffeomorphic coordinates. Thus the $(n-1)\times(n-1)$ minor of the eigenvalue Jacobian indexed by $E(T)$ is non-singular. Now give every edge of $E(H)\setminus E(T)$ the weight $\varepsilon > 0$; for ε small, simplicity of the spectrum and non-singularity of that minor persist by continuity (eigenvalues and eigenvectors of a simple spectrum depend smoothly on the matrix, and $\partial\lambda_i/\partial w_f = (b_f^{\mathsf T}x_i)^2$ for unit eigenvectors x_i). All weights are then positive. □

Lemma 8.4 (Regions are closures of their interiors)**.** *If H is connected, then $\mathcal{R}(H)$ is the closure of its interior relative to the affine hull of C_{n-1}.*

Proof. Let $\Lambda(w) = (\lambda_2,\dots,\lambda_n)(L_H(w))$ and let $Z \subset \mathbb{R}^{E(H)}$ be the set where the reduced spectrum is not simple or the coefficient map $w \mapsto (q_1,\dots,q_{n-1})$ has rank $< n-1$. It is the zero set of the product of the discriminant with the sum of squares of the $(n-1)$-minors of the coefficient Jacobian, a polynomial, and it is proper by Lemma 8.3. It is also a cone (q_k is homogeneous of degree k), so it cannot contain the hyperplane $\{\sum_f w_f = 1/2\}$ (otherwise it would contain the open set $\{\sum_f w_f > 0\}$); hence the weights $w \in \Delta(H) = \{w \ge 0,\ \sum w_f = 1/2\}$ (i.e. $\operatorname{tr} L = 1$) with all $w_f > 0$ and $w \notin Z$ are dense in $\Delta(H)$. At such w the spectrum is simple and positive (H is connected) and $d\Lambda$ is onto $\mathbb{R}^{n-1}$. Since $\sum_i \lambda_i = \operatorname{tr} L_H(w) = 2\sum_f w_f$, the differential of Λ maps the tangent space $\{\sum_f \dot w_f = 0\}$ of $\{\sum_f w_f = 1/2\}$ into $\{\sum_i \dot x_i = 0\}$, and it maps the remaining (radial) direction w to $\Lambda(w)$, which has coordinate sum 1; hence the restriction of Λ to $\{w > 0,\ \sum_f w_f = 1/2\}$ is a submersion onto the affine hyperplane $\{\sum_i x_i = 1\}$ near $\Lambda(w)$. As the image point has strictly increasing positive coordinates, it is an interior point of $\mathcal{R}(H)$ relative to the affine hull of C_{n-1}; every point of $\mathcal{R}(H)$ is a limit of such points. □

Proposition 8.5 (Reduction)**.** *Let G be a graph on $[n]$ and $e \notin E(G)$ such that $G+e$ is connected. Then $\mathcal{R}(G) \subseteq \mathcal{R}(G+e)$, and*

$$P(G+e) > P(G) \iff \mathcal{R}(G+e) \neq \mathcal{R}(G).$$

Proof. Inclusion: give e weight zero. If the two compact sets are equal the volumes agree. Conversely let $p \in \mathcal{R}(G+e)\setminus\mathcal{R}(G)$. As $\mathcal{R}(G)$ is closed, some relatively open ball $U \ni p$ misses $\mathcal{R}(G)$; by Lemma 8.4, U contains an interior point p' of $\mathcal{R}(G+e)$, hence a small ball around p' lies in $\mathcal{R}(G+e)\setminus\mathcal{R}(G)$, which therefore has positive volume. □

Note that p may be any point of $\mathcal{R}(G+e)$, including degenerate spectra with extra zeros.

Lemma 8.6 (Linear certificate of non-realizability)**.** *Let Y be a real symmetric $n\times n$ matrix with $b_f^{\mathsf T}Y b_f \le 0$ for every $f \in E(G)$, and let $\theta_1 \le \cdots \le \theta_{n-1}$ be the eigenvalues of HYH on $\mathbf{1}^\perp$, $H = I - \frac{1}{n}J$. If $\sigma \in \mathbb{R}^{n-1}_{\ge 0}$ satisfies*

$$\sum_{i=1}^{n-1} \theta_i\,\sigma_i^{\downarrow} > 0 \tag{20}$$

($\sigma^\downarrow$ = decreasing rearrangement), then no $L \in \{L_G(w) : w \ge 0\}$ has reduced spectrum σ.

Proof. If $L = L_G(w)$, $w \ge 0$, has reduced spectrum σ, then, since $L = HLH$ and $L\mathbf{1} = 0$, von Neumann's trace inequality on $\mathbf{1}^\perp$ [91, Ch. III] gives $\operatorname{tr}(YL) = \operatorname{tr}\big((HYH)L\big) \ge \sum_i \theta_i\sigma_i^\downarrow > 0$, while $\operatorname{tr}(YL) = \sum_f w_f\, b_f^{\mathsf T}Y b_f \le 0$. □

The certificates of Lemma 8.6 detect exactly the spectra that do not majorize any spectrum of $\mathcal{R}(G)$: by convexity of $\{L_G(w) : w \ge 0\}$ and of the orbitope $\operatorname{conv}\{QD_\sigma Q^{\mathsf T}\}$ and the separating hyperplane theorem, (20) is solvable iff σ does not majorize the reduced spectrum of any $L_G(w)$ of

the same trace. Equivalently they are dual to maximizing a concave spectral functional $\sum_i c_i\lambda_{i+1}$ ($c_1 \geq \cdots \geq c_{n-1}$) over trace-one weightings.

8.3. **An all-order family: complements that become matchings.**

Lemma 8.7 (Missing-edge inequality and its equality case)**.** *Let $L = L_G(w)$, $w \geq 0$, with $\lambda_2 > 0$, and let uv be a pair with $L_{uv} = 0$, $n \geq 3$. Then $\lambda_n \geq \frac{n}{n-2}\lambda_2$. If equality holds, then $e_u - e_v$ is a λ_2-eigenvector and $L_{uk} = L_{vk} = -\lambda_n/n < 0$ for every $k \notin \{u, v\}$.*

Proof. (The inequality and its equality form are proved in [3] (the missing-edge inequality and the block form at equality); we include a short argument for completeness.) Put $m = \frac{\lambda_2+\lambda_n}{2}$, $\delta = \frac{\lambda_n-\lambda_2}{2}$, $T = L - mH$, $p = He_u$, $q = He_v$. On $\mathbf{1}^\perp$, $\|T\| \leq \delta$; $|p| = |q| = \sqrt{(n-1)/n}$ and $p^\mathsf{T}Tq = L_{uv} - mH_{uv} = m/n$. Cauchy–Schwarz gives $m/n \leq \delta\frac{n-1}{n}$, i.e. $n\lambda_2 \leq (n-2)\lambda_n$. At equality $Tq = \delta p$; writing $q = q_+ + q_-$ along the $\pm\delta$ eigenspaces of T (forced by $|Tq| = \delta|q|$) we get $p = q_+ - q_-$ and $Tp = \delta q$. Hence $L(p-q) = \lambda_2(p-q)$ and $L(p+q) = \lambda_n(p+q)$ with $p - q = e_u - e_v$, $p + q = e_u + e_v - \frac{2}{n}\mathbf{1}$. Reading coordinate $k \notin \{u, v\}$ of both identities gives $L_{ku} = L_{kv}$ and $L_{ku} + L_{kv} = -2\lambda_n/n$. □

Theorem 8.8 (Positive answer when the new complement is a matching)**.** *Let G be connected and non-complete on $n \geq 3$ vertices, let $F = E(\overline{G})$ and $e \in F$. If $F \setminus \{e\}$ is a matching (possibly empty), then $P(G + e) > P(G)$.*

Proof. By Proposition 8.5 it suffices to exhibit a point of $\mathcal{R}(G + e) \setminus \mathcal{R}(G)$.

Case $F = \{e\}$. The uniformly weighted $K_n = G + e$ has reduced spectrum $(\frac{1}{n-1}, \ldots, \frac{1}{n-1})$. A relaxed G-realization would have $\lambda_2 = \lambda_n > 0$ and a zero entry at e, contradicting Lemma 8.7.

Case $F\setminus\{e\} = M$, a matching with $r \geq 1$ edges. Give every edge of $G+e = K_n - M$ weight $\alpha > 0$. Then $L = \alpha\big(nH - \sum_{xy\in M} B_{xy}\big)$, whose reduced spectrum is $\sigma_r = \{((n-2)\alpha)^{(r)}, (n\alpha)^{(n-1-r)}\}$, attaining equality in Lemma 8.7. Suppose $L' = L_G(w')$, $w' \geq 0$, had reduced spectrum σ_r, and let S be its positive support (a spanning subgraph of G, connected as $\lambda_2 > 0$). Every non-edge of S is a pair with $L'_{xy} = 0$ and equality holds, so by Lemma 8.7 every vertex outside a non-edge xy of S is joined to both x and y; thus the non-edges of S form a matching M', and the vectors $e_x - e_y$ ($xy \in M'$) are pairwise orthogonal λ_2-eigenvectors, so $\mathrm{mult}(\lambda_2) \geq |M'|$. Now $M' \supseteq F$. If F is not a matching this is impossible. If F is a matching then $|M'| \geq |F| = r + 1 > r = \mathrm{mult}_{\sigma_r}((n-2)\alpha)$, a contradiction. □

Remark 8.9 (Relation to [3])*.* The case $F = \{e\}$ is the last-edge gap computed exactly in [3] ($P(K_n) - P(K_n - e) = \binom{(n-2)(n+1)/2}{n-2}^{-1}$). The general case is a short consequence of the equality analysis of [3] (its "support rigidity on the universal facet": equality in Lemma 8.7 forces a matching complement and the facet spectra σ_r of $K_n - M_r$), combined with the multiplicity count in the proof above; [3] does not draw this edge-addition consequence, so we record it. The theorem covers all graphs $K_n - F$ in which F is a matching plus one edge, for every n.

8.4. **An all-order family on the sparse side: growth of the Hadwiger number.** For a graph H let $\mathrm{had}(H)$ be the largest h for which $V(H)$ can be partitioned into h sets, each inducing a connected subgraph, that are pairwise joined by at least one edge (for connected H this is the Hadwiger number).

Lemma 8.10 (Singular limits)**.** *Let H be a graph on $[n]$ and $w^{(k)} \in \mathbb{R}^{E(H)}_{\geq 0}$ with $w^{(k)}_f \to w^0_f \in [0, \infty]$ for every f. Let $\Gamma = \{f : w^0_f = \infty\}$, let $V_1, \ldots, V_h$ be the vertex sets of the components of $([n], \Gamma)$ (singletons allowed), $M = \mathrm{diag}(|V_i|)$, and let L_Q be the Laplacian on $[h]$ with the finite weights $W_{ij} = \sum_{f\in E_H(V_i,V_j)} w^0_f$. Let $\lambda^{(k)}_1 \leq \cdots \leq \lambda^{(k)}_n$ be the eigenvalues of $L_H(w^{(k)})$ and $\xi_1 \leq \cdots \leq \xi_h$ those of the pencil $L_Q y = \xi M y$. Then $\lambda^{(k)}_j \to \infty$ for $j > h$ and $\lambda^{(k)}_j \to \xi_j$ for $j \leq h$.*

Proof. Let P be the space of vectors constant on every V_i and $\iota : \mathbb{R}^h \to P$ the obvious isomorphism; $|\iota y|^2 = y^{\mathsf{T}} M y$, and for $x = \iota y$ the edges inside a V_i contribute nothing, so $x^{\mathsf{T}} L_H(w^{(k)}) x \to y^{\mathsf{T}} L_Q y$ (edges between different V_i are not in Γ). Courant–Fischer with the test spaces ι(first j pencil eigenvectors) gives $\limsup \lambda_j^{(k)} \le \xi_j$ for $j \le h$. Let L_Γ be the unweighted Laplacian of Γ; its kernel is P, so $x^{\mathsf{T}} L_\Gamma x \ge \kappa |x|^2$ on $P^\perp$ for some $\kappa > 0$. Every $(h+1)$-dimensional subspace contains a unit $x \perp P$, and $x^{\mathsf{T}} L_H(w^{(k)}) x \ge (\min_{f \in \Gamma} w_f^{(k)})\kappa \to \infty$, whence $\lambda_{h+1}^{(k)} \to \infty$. For the lower bound take orthonormal eigenvectors $x_1^{(k)}, \dots, x_h^{(k)}$ and a subsequence along which they converge to $x_1, \dots, x_h$. Boundedness of $\lambda_j^{(k)} \ge (\min_\Gamma w^{(k)})\, x_j^{(k)\mathsf{T}} L_\Gamma x_j^{(k)}$ forces $x_j \in P$. For a unit $x = \sum_{i \le j} c_i x_i$ put $x^{(k)} = \sum_{i \le j} c_i x_i^{(k)}$; then $\lambda_j^{(k)} \ge x^{(k)\mathsf{T}} L_H(w^{(k)}) x^{(k)} \ge \sum_{f \notin \Gamma} w_f^{(k)} (x_a^{(k)} - x_b^{(k)})^2 \to y^{\mathsf{T}} L_Q y$ with $y = \iota^{-1} x$, so $\liminf \lambda_j^{(k)}$ is at least the maximum of the pencil Rayleigh quotient on the j-dimensional space $\iota^{-1}\operatorname{span}(x_1, \dots, x_j)$, which is $\ge \xi_j$. As every subsequence has a further subsequence with these limits, the whole sequence converges. □

Theorem 8.11 (Hadwiger growth forces a strict volume gain)**.** *Let G be connected and $e \notin E(G)$. If $\operatorname{had}(G+e) > \operatorname{had}(G)$, then $P(G+e) > P(G)$.*

Proof. Let $h = \operatorname{had}(G+e)$ and fix a partition $V_1, \dots, V_h$ of $[n]$ witnessing it in $G+e$, spanning trees T_i of $(G+e)[V_i]$, and for $i < j$ an edge $f_{ij} \in E(G+e)$ between V_i and V_j. Put $m_i = |V_i|$. For $t > 0$ weight every edge of $\bigcup_i T_i$ by t, every f_{ij} by $m_i m_j / n$, and all other edges by 0. By Lemma 8.10 (as $t \to \infty$) the quotient pencil is $L_Q = M - m m^{\mathsf{T}}/n$, $M^{-1/2} L_Q M^{-1/2}$ is the orthogonal projection onto $(M^{1/2}\mathbf{1})^\perp$, and so the reduced spectrum $\sigma(t)$ has $h-1$ entries tending to 1 and $n-h$ entries tending to ∞.

Suppose $\sigma(t_k)$ were the reduced spectrum of $L_G(w^{(k)})$, $w^{(k)} \ge 0$, for a sequence $t_k \to \infty$. Passing to a subsequence with $w^{(k)} \to w^0 \in [0,\infty]^{E(G)}$ and applying Lemma 8.10 to G: exactly h eigenvalues stay bounded, so $([n], \Gamma)$ has exactly h components $V_1', \dots, V_h'$, each spanning a connected subgraph of G, and the quotient pencil $(L_{Q'}, M')$ has eigenvalues $0, 1^{(h-1)}$. Then $M'^{-1/2} L_{Q'} M'^{-1/2}$ is the projection onto $(M'^{1/2}\mathbf{1})^\perp$, so $L_{Q'} = M' - m' m'^{\mathsf{T}}/n$ has all off-diagonal entries non-zero: every two of the V_i' are joined by an edge of G. Thus $\operatorname{had}(G) \ge h$, a contradiction. Hence for some t the normalized $\sigma(t)$ lies in $\mathcal{R}(G+e) \setminus \mathcal{R}(G)$, and Proposition 8.5 applies. □

Remark 8.12. Theorem 8.11 contains, for instance, every tree G (adding any edge creates a cycle, i.e. a K_3 minor), and every graph with no K_h minor to which e adds one. The same argument shows more generally that $\mathcal{R}(G+e) = \mathcal{R}(G)$ would force, for every h, the union of the quotient ("mass-minor") regions of all h-part contractions of $G+e$ to coincide with that of G; we use only the equal-spectrum point of complete quotients. The two families of Theorems 8.8 and 8.11 together cover 21 of the 42 non-isomorphic pairs (G, e) (G connected, e up to $\operatorname{Aut}(G)$) with $n = 5$, 131 of the 402 with $n = 6$, and 1348 of the 5381 with $n = 7$; for $n \le 6$ the remaining pairs are handled by the certificates of Theorem 8.13, while the 4033 remaining pairs with $n = 7$ have not been certified.

8.5. **All connected graphs on at most six vertices.**

Theorem 8.13 (Question 3 for $n \le 6$; computer-assisted)**.** *Let G be a connected non-complete graph on $n \le 6$ vertices and $e \notin E(G)$. Then $P(G+e) > P(G)$. Consequently, for $n \le 6$ the fixed-order gain $c_n^{\mathrm{conn}} = \min\{P(G+e) - P(G)\}$, the minimum over connected non-complete G with $|V(G)| = n$ and $e \notin E(G)$, is positive.*

Proof. For $n = 3$ the only pair is $P_3 \subset K_3$, and for $n = 4$ the six connected graphs (5 non-complete, 6 pairs (G, e) up to isomorphism) have the known volumes $P(K_{1,3}) \approx 0.180$, $P(P_4) \approx 0.384$,

$P(\text{paw}) = 9/13$, $P(C_4) = 3/4$, $P(K_4 - e) = 9/10$, $P(K_4) = 1$ (the four-vertex census of [3], based on [89, 90]), so every edge addition is strict. (Our certificates below re-prove $n = 3, 4$ as well.)

For $n \in \{5, 6\}$, for each of the connected non-complete graphs on n vertices (20 for $n = 5$, 111 for $n = 6$, taken from the complete atlas of graphs on at most seven vertices [95]) and each orbit of non-edges under $\mathrm{Aut}(G)$ (42 resp. 402 pairs), the file of certificates contains a pair (w, Y) with: w a vector of non-negative rationals on $E(G) \cup \{e\}$, so that the reduced spectrum σ of $L_{G+e}(w)$ (normalized) lies in $\mathcal{R}(G + e)$; and Y a rational symmetric matrix with $b_f^{\mathsf{T}} Y b_f \le 0$ for all $f \in E(G)$ (checked exactly) and (20) (checked rigorously: the characteristic polynomials of $L_{G+e}(w)$ and HYH are computed exactly over $\mathbb{Q}$, divided by t, their roots enclosed in certified balls (FLINT/Arb [96]), and a lower bound for $\min_\pi \sum_i \theta_i \sigma_{\pi(i)} = \sum_i \theta_i \sigma_i^{\downarrow}$ is obtained from the ball lower bounds of all products, minimized over all permutations). By Lemma 8.6, $\sigma \notin \mathcal{R}(G)$, and Proposition 8.5 gives $P(G + e) > P(G)$. Isomorphic pairs have equal volumes, so all pairs are covered. Finitely many pairs give a positive minimum. □

Remark 8.14 (The two classical extremal problems do not suffice). A natural sufficient condition for $P(G + e) > P(G)$ is that the trace-normalized absolute algebraic connectivity $\max \lambda_2 / \operatorname{tr}$ increases, or the trace-normalized minimal spectral radius $\min \lambda_n / \operatorname{tr}$ decreases, when e is added (each is a certificate of the type in Lemma 8.6). Numerically (to 10^{-5}) this holds for all 42 pairs with $n = 5$ and for 401 of the 402 pairs with $n = 6$. It fails, provably, for the remaining pair: let $D = P_2 \times P_3$ be the domino, with rows 1 0 5 and 2 3 4 (edges $01, 05, 23, 34$ and rungs $12, 03, 54$), and $e = 14$ the long diagonal. The weights $\frac{1}{22}$ on the three rungs and $\frac{1}{11}$ on the four row edges give $\operatorname{tr} L_D = 1$ and spectrum $\{0, \frac{1}{11}, \frac{1}{11}, \frac{2}{11}, \frac{3}{11}, \frac{4}{11}\}$, while the vector $x = (1, -1, -3, -1, 1, 3)$ (indexed by $0, \dots, 5$) satisfies $\sum x_i = 0$, $|x|^2 = 22$ and $|x_a - x_b| = 2$ on all eight edges of $D + e$, so $\lambda_2 \le \frac{x^{\mathsf{T}} L x}{|x|^2} = \frac{1}{11}$ for every trace-one weighting of $D + e$. Similarly the weights $\frac{1}{9}$ on the outer rungs and $\frac{1}{18}$ on the other five edges give $\lambda_n(L_D) = \frac{1}{3}$, while the bipartition vector ($+1$ on $\{0, 2, 4\}$, -1 on $\{1, 3, 5\}$; e joins the two classes) gives $\lambda_n \ge \frac{1}{3}$ for every trace-one weighting of $D + e$. So both extremal values coincide for D and $D + e$, yet $P(D + e) > P(D)$ by Theorem 8.13; its certificate excludes a $(D+e)$-spectrum with $\lambda_2 / \operatorname{tr} \approx \frac{1}{11}$ (within 10^{-8}) but $\lambda_3 / \operatorname{tr} \approx \frac{2}{11}$. (All of this is checked in exact arithmetic by `R4_domino.py`.) Thus any general proof of Problem 8.1 must use spectral information beyond λ_2 and λ_n.

8.6. **Suspension: a lift identity and a partial answer to Question 4.**

Lemma 8.15 (Uniform join [89, Lemma 7.3])**.** *If L is a weighted Laplacian on G (vertices $1, \dots, n$) with reduced spectrum $\mu = (\mu_2, \dots, \mu_n)$ and the new vertex 0 is joined to every vertex with weight $t \ge 0$, the resulting Laplacian on $G' = G \vee K_1$ has reduced spectrum $\{\mu_i + t\}_{i=2}^n \cup \{(n+1)t\}$.*

Proof. $L'(0, x) = (0, (L + tI)x)$ for $x \perp \mathbf{1}$, and $L'(n, -\mathbf{1}) = (n + 1)t\,(n, -\mathbf{1})$. □

For $\nu \in \mathbb{R}^n_{\ge 0}$ call an index j *admissible* if $\nu_j \le (n + 1) \min_i \nu_i$, and put

$$\mu^{(j)}(\nu) = \big(\nu_i - \tfrac{\nu_j}{n+1}\big)_{i \ne j} \in \mathbb{R}^{n-1}_{\ge 0}.$$

Admissibility is exactly $\mu^{(j)} \ge 0$, and $\nu = \{\mu_i^{(j)} + t\} \cup \{(n + 1)t\}$ with $t = \nu_j/(n + 1)$. For a symmetric cone $A \subseteq \mathbb{R}^{n-1}_{\ge 0}$ let $J(A) = \{\nu : \ \mu^{(j)}(\nu) \in A \text{ for some admissible } j\}$ and let $U_m(A)$ denote the probability that m i.i.d. Exp(1) variables lie in A. By Lemma 8.15 and (18),

$$P(G \vee K_1) \ \ge \ U_n\big(J(\widehat{\mathcal{R}}(G))\big), \qquad P(G) = U_{n-1}\big(\widehat{\mathcal{R}}(G)\big), \tag{21}$$

where $\widehat{\mathcal{R}}(G)$ is the cone over all permutations of $\mathcal{R}(G)$.

Proposition 8.16 (Lift identity)**.** *Let $\nu_1, \dots, \nu_n$ be i.i.d.* Exp(1)*, $N(\nu)$ the number of admissible indices and $N_A(\nu)$ the number of admissible j with $\mu^{(j)} \in A$. Then N is uniform on $\{1, \dots, n\}$ and, for every measurable symmetric A (more generally for every measurable symmetric $f \geq 0$),*

$$\mathbb{E}\Big[\sum_{j \text{ adm}} f(\mu^{(j)})\Big] = \frac{n+1}{2}\,\mathbb{E}f(X), \qquad X \sim \mathrm{Exp}(1)^{\otimes(n-1)};$$

in particular $\mathbb{E}N_A = \frac{n+1}{2}U_{n-1}(A)$. More precisely, for each admissible j the map $(\nu) \mapsto (\mu^{(j)}, t)$, $t = \nu_j/(n+1)$, has Jacobian $n+1$ and transports the density $e^{-\sum\nu}$ to $(n+1)e^{-\sum\mu}e^{-2nt}$.

Proof. By symmetry the left side is $n\,\mathbb{E}[f(\mu^{(n)})\mathbf{1}\{n \text{ adm}\}]$. Substitute $\nu_i = \mu_i + t$ $(i < n)$, $\nu_n = (n+1)t$; the region $\{n \text{ admissible}\}$ becomes $\{\mu \geq 0, t \geq 0\}$, the Jacobian is $n+1$ and $\sum\nu = \sum\mu + 2nt$. Integrating t gives the factor $\frac{n+1}{2n}$. For the law of N: with $a = \min\nu$ (so e^{-na} is uniform on $(0,1)$) the other $n-1$ variables exceed a by i.i.d. Exp(1) amounts (Rényi's representation [94]), so $N-1$ is binomial$(n-1, 1-e^{-na})$ mixed over a uniform parameter, i.e. uniform on $\{0, \dots, n-1\}$. □

Because the transported density factorizes, a *fixed* index is exact: conditionally on j being admissible (probability $\frac{n+1}{2n}$), $\mu^{(j)}$ is exactly $\mathrm{Exp}(1)^{\otimes(n-1)}$ and independent of t. Geometrically, on the simplex of directions $\Delta_{n-1} = \{\omega \geq 0,\ \sum\omega_i = 1\}$ the set where j is admissible is the pyramid with base the facet $\{\omega_j = 0\}$ and apex $p_j^* = \frac{1}{2n}(1, \dots, 1, n+1, 1, \dots, 1)$ ($n+1$ in position j), and the direction of $\mu^{(j)}$ is the central projection from p_j^* onto that facet, which pushes the uniform measure of the pyramid to the uniform measure of the facet; the n pyramids cover Δ_{n-1} with average multiplicity $\frac{n+1}{2}$.

Theorem 8.17 (Partial answer to Question 4)**.** *For every graph G on $n \geq 2$ vertices,*

$$P(G \vee K_1) \;\geq\; \gamma_n\,P(G), \qquad \gamma_n := \inf_{\theta}\ \frac{n+1}{2}\,\mathbb{E}_{r\sim\rho_n}\Big[\frac{1}{N(\theta,r)}\Big] \;\geq\; \frac{n+1}{2n},$$

where θ ranges over the open simplex (directions of $\mu \in \mathbb{R}^{n-1}_{>0}$), $\rho_n(r) = 2(n-1)(1+2r)^{-n}$ on $[0,\infty)$, and $N(\theta,r)$ is the number of admissible indices of the vector $\nu(\theta,r) = \{\theta_i + r/n\} \cup \{(n+1)r/n\}$, namely $N = 1 + \#\{i : \theta_i \leq (n+2)r\}$ if $r < \min\theta$ and $N = 1 + \#\{i : \theta_i \leq (n+1)\min\theta + r\}$ otherwise.

Proof. Select, for each ν, one admissible index uniformly at random. Then $\Pr[\nu \in J(A)] \geq \mathbb{E}\big[\sum_{j \text{ adm}} N^{-1}\mathbf{1}_A(\mu^{(j)})\big]$. By Proposition 8.16 (with $f = \mathbf{1}_A \cdot N^{-1}$ read along the fibre) this equals $\frac{n+1}{2}\mathbb{E}[\mathbf{1}_A(X)\,\mathbb{E}_\tau N^{-1}]$ with $\tau = nt \sim \mathrm{Exp}(2)$ independent of X; writing $X = S\theta$ ($S \sim \mathrm{Gamma}(n-1)$, θ uniform and independent) and $r = \tau/S$, whose density is ρ_n, the inner expectation depends only on (θ, r), and the formula for N follows from $\min\nu = \min(\min\mu + \tau/n, (n+1)\tau/n)$. Use (21) with $A = \widehat{\mathcal{R}}(G)$. The lower bound $\frac{n+1}{2n}$ is $N \leq n$. □

Remark 8.18 (Size of γ_n; numerical). The integrand is piecewise explicit, since $\rho_n([a,\infty)) = (1+2a)^{-(n-1)}$. Numerically (`R4_q4_numerics.py`) $\gamma_3 \approx 0.934$ (numerically the minimum is at $\theta = (\frac{1}{6}, \frac{5}{6})$, where the integrand equals $\frac{239}{256}$), $\gamma_4 \approx 0.826$, $\gamma_5 \approx 0.748$, $\gamma_6 \approx 0.700$, $\gamma_7 \approx 0.667$, $\gamma_8 \approx 0.643$; for $4 \leq n \leq 8$ the infimum is approached at a vertex of the simplex, where the integrand tends to $\frac{n+1}{2}\big(\frac{1-3^{1-n}}{n-1} + \frac{3^{1-n}}{n}\big)$, so presumably $\gamma_n \to \frac{1}{2}$. Only $\gamma_n \geq \frac{n+1}{2n}$ is proved. Since the θ-average of the integrand is 1 (Proposition 8.16), $\gamma_n \leq 1$ in any case: a uniformly random choice of the removed index cannot answer Problem 8.2. Nor can any rule that chooses the removed entry by its rank among the admissible ones: a linear program over such rules gives a best guaranteed density of only about 0.997, 0.984, 0.979, 0.977 for $n = 3, 4, 5, 6$.

For small n the bound (21) can be evaluated directly, because the four-vertex regions are known exactly.

Theorem 8.19 (Question 4 for at most four vertices; computer-assisted)**.** *Let G be a non-complete graph on $n \le 4$ vertices. Then $P(G \vee K_1) \ge P(G)$, with strict inequality when G is connected. The case $G = K_4 - e$ uses the exact region of $K_5 - e$ from* [3].

Proof. If G is disconnected, then $P(G) = 0$. The connected non-complete graphs with $n \le 4$ are P_3, $K_{1,3}$, P_4, the paw, C_4 and $K_4 - e$. Their suspensions satisfy $P_3 \vee K_1 = K_4 - e$ and $(K_4 - e) \vee K_1 = K_5 - e$. For P_3: the paw is $K_4 - e$ minus an edge, and setting that weight to 0 shows that the region of the paw lies in that of $K_4 - e$, so $P(K_4 - e) \ge P(\text{paw}) = \frac{9}{13} > \frac{1}{2} = P(P_3)$ by [89, Theorem 6.1]; here $P(P_3) = \frac{1}{2}$ is a direct computation, since $\lambda_2\lambda_3 = 3ab$ and $\lambda_2 + \lambda_3 = 2(a + b)$ give exactly $\lambda_2 \le \lambda_3/3$. For $K_4 - e$: the exact regions of $K_4 - e$ and $K_5 - e$ in [3] give $P(K_5 - e) = 1 - \binom{9}{3}^{-1} = \frac{83}{84} > \frac{9}{10} = P(K_4 - e)$.

For $G \in \{K_{1,3}, P_4, \text{paw}, C_4\}$ we use $P(G \vee K_1) \ge U_4\big(J(\widehat{\mathcal{R}}(G))\big)$ from (21). The regions are known exactly for nonnegative weights [89, Theorems 3.1, 4.1, 5.1, 6.1]. For an ordered triple $0 < a \le b \le c$ we use the following sufficient conditions for membership.

- $K_{1,3}$: $F_a > 0 > F_b$, where $F_v = 6v^2 - 2vs_v + p_v$, with s_v and p_v the sum and the product of the other two weights. EJT's condition is $F_aF_bF_c \le 0$, and $F_c \ge 6c^2 - 4c^2 + ab > 0$.
- P_4: $b > 2a$ and $g = c(b-2a) - b(b-a) > 0$. EJT's conditions are $c \ge \frac{a+b+c}{2}$ together with a sextic inequality lhs $\le$ rhs. Sympy factors the difference as $\text{rhs} - \text{lhs} = -256 f_a f_b f_c$, where $f_v = v^2 - vs_v + 2p_v$ (notation as above). Here $f_b = -g < 0$, and $f_a \ge (a - b)^2 + b^2 > 0$. Also $g > 0$ forces $c > \frac{b(b-a)}{b-2a} > a + b$, which gives $c \ge \frac{a+b+c}{2}$ and $f_c = c(c - a - b) + 2ab > 0$.
- paw: $c > \kappa a$ with a rational $\kappa > 2 + \sqrt{3}$. This gives $\sqrt{3}(c - a) \ge a + c$.
- C_4: $c > a + b$.

Write the order statistics of $\nu \sim \text{Exp}(1)^{\otimes 4}$ through Rényi's representation [94]: $s_1 = \frac{E_1}{4}$, $s_2 - s_1 = \frac{E_2}{3}$, $s_3 - s_2 = \frac{E_3}{2}$, $s_4 - s_3 = E_4$, with E_i i.i.d. Exp(1). The event $\nu \in J(\widehat{\mathcal{R}}(G))$ is invariant under scaling, so it is an event for the point $E/\sum_i E_i$, which is uniform on the 3-simplex. In the coordinates (e_2, e_3, e_4) of that point, the four candidates $\mu^{(k)}$ (remove s_k and subtract $s_k/5$) and their admissibility conditions $s_k < 5s_1$ are affine.

We cover the simplex by dyadic boxes. A box is counted only if, for some k, exact rational interval arithmetic proves on the whole box that k is admissible and that $\mu^{(k)}$ satisfies the strict conditions above. Its mass $6\,\text{vol}(\text{box} \cap \text{simplex})$ is computed exactly by inclusion–exclusion. The total counted mass is therefore an exact rational lower bound ℓ_G for $U_4(J(\widehat{\mathcal{R}}(G)))$.

With boxes of side 2^{-8} (`R4_q4_small.py`), the bound ℓ_G exceeds the volume of G in each case:

- $\ell_{K_{1,3}} = 0.24663\ldots > 0.17997\ldots = P(K_{1,3})$;
- $\ell_{P_4} = 0.52378\ldots > 0.38350\ldots = P(P_4)$;
- $\ell_{\text{paw}} = 0.77563\ldots > \frac{9}{13}$;
- $\ell_{C_4} = 0.83521\ldots > \frac{3}{4}$.

The volumes of G are the closed forms of the four-vertex census [3] (for the paw and C_4 also [89]), which we re-computed independently as $P(G) = 12 \iint_\Omega (1 + p + q)^{-3}\, dp\, dq$ by quadrature; the margins exceed 0.06 in every case. □

Remark 8.20. Monte Carlo estimates of $U_4(J(\widehat{\mathcal{R}}(G)))$ are about 0.290, 0.567, 0.785 and 0.847 for $K_{1,3}$, P_4, paw and C_4. So the uniform join alone exceeds $P(G)$ by 0.09–0.18. The same method gives nothing for $n \ge 5$, because the five-vertex regions are not known.

Conjecture 8.21 (Cone-lifting inequality)**.** *For every $n \ge 3$ and every closed symmetric cone $A \subseteq \mathbb{R}^{n-1}_{\ge 0}$, $U_n(J(A)) \ge U_{n-1}(A)$.*

By (21), Conjecture 8.21 implies a positive answer to Problem 8.2 for every G. By Strassen's theorem [93] it is equivalent, up to closedness issues at the apices where some $\mu^{(j)}$ vanishes, to the

existence of a coupling of a uniform direction θ' in the $(n-1)$-simplex and a uniform direction θ in the $(n-2)$-simplex (up to permutation) with $\theta' \in J(\mathbb{R}_{>0}\theta)$ almost surely, i.e. to a randomized choice, for each point of Δ_{n-1}, of one pyramid containing it such that the central projections described after Proposition 8.16 produce exactly the uniform law on the facet; by Proposition 8.16 the "multiplicity budget" is exactly $\frac{n+1}{2}$ per target point, so the question is whether the overlaps can be arranged without loss. Two facts are provable: (i) choosing the minimal coordinate as the removed index gives, by Proposition 8.16, $\Pr[\mu^{(\arg\min)} \in A] = \frac{n+1}{2}\mathbb{E}[\mathbf{1}_A(X)(1-e^{-2\min X})]$, and since $\mu^{(\arg\min)} = X' + \frac{n\min\nu}{n+1}\mathbf{1}$ with X' an exact $\mathrm{Exp}(1)^{\otimes(n-1)}$ sample, the conjecture holds for every cone that is closed under $x \mapsto x+c\mathbf{1}$ ($c \geq 0$); (ii) the conjecture holds for A equal to the whole orthant (with equality). Regions of non-complete graphs are, however, never closed under adding constants: by Lemma 8.7 they omit a neighbourhood of the equal-spectrum ray. So neither fact settles Problem 8.2. The cone-lifting inequality has been tested numerically (`R4_q4_numerics.py`). Monte Carlo estimates of both sides for sixteen families of cones and $n \in \{3,4,5,6,8\}$ give ratios ≥ 1 up to sampling error; the tightest cases are the graph-like cones $\{\max/\min \geq \kappa\}$. A discretized Hall/max-flow test shows no deficit for $n = 3,4,5$. What remains open is exactly Conjecture 8.21, or any graph-specific substitute for it.

8.7. **Computations accompanying this section.** All scripts are run with `python3 -I`, and each prints what it checks.

Used in proofs.

- `q3_verify.py`: the independent verifier for Theorem 8.13. It works in exact $\mathbb{Q}$-arithmetic with Arb balls. For $N = 3,\dots,6$ it checks the file `q3_certs_n`N`.json`, reporting 1, 6, 42 and 402 valid certificates, with no pair uncovered. The smallest certified margins are 0.316, $1.08 \cdot 10^{-3}$, $1.84 \cdot 10^{-2}$ and $9.9 \cdot 10^{-6}$.
- `R4_q4_small.py`: Theorem 8.19, using exact rational interval arithmetic and exact box masses.
- `R4_domino.py`: Remark 8.14, in exact arithmetic.

Not used in proofs.

- The certificate generators `q3_certify*.py`.
- The sanity checks `R4_checks.py` and `R4_regions_check.py`.
- The numerical evidence `R4_q4_numerics.py` for Remark 8.18 and Conjecture 8.21.

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